Testing a Candidate Dependency Architecture:

An Adversarial Experimental Programme for Maintenance, Heredity, Learning, Agency and Self-Modelling

Ken Craggs
Independent Researcher, United Kingdom

20 September 2026

Working Paper 3, Theory of Everything Research Programme

DOI: 10.5281/zenodo.22857230

Abstract

This paper develops an adversarial experimental programme for testing candidate dependencies identified by the companion Minimal Principles and Mathematics of Constraint papers. It does not assume that organisation, evolution, intelligence and reflective self-modelling form a necessary hierarchy, nor that artificial construction establishes natural necessity or biological equivalence. Instead, five experimental blocks isolate distinct functional and causal questions and subject proposed dependencies to matched controls, targeted intervention, rescue, bypass architectures and held-out tests.

Block I distinguishes persistent organisation from internally generated maintenance using driven dynamical baselines, passive and externally maintained controls, and causal maintenance networks. Block II establishes reproduction, heredity, variation and differential reproductive success before natural-selection-like population change is inferred. Block III separates policy-accessible retained state, within-episode feedback and between-episode learning through a factorial intervention design. Block IV tests adaptive agency through structural generalisation, causal novelty, model-based control and counterfactual rollout while allowing serious model-free alternatives to succeed. Block V distinguishes accurate self-monitoring from causal use of self-related representations through misinformation, correction and rescue experiments. An auxiliary conflict-representation test remains outside the principal architecture and receives no privileged status.

Measurements are organised through a typed framework rather than a universal constraint score or cross-scale vector. Dynamical, functional, temporal or historical, and organisational quantities retain their own baselines, representations, explananda and causal status. Every principal hypothesis is assigned one of five preregistered outcomes: Retain, Restrict, Reduce, Suspend or Reject. Failure of one dependency does not terminate logically independent later tests, and successful bypasses count as evidence against necessity rather than as experimental anomalies.

The programme is designed so that the final dependency structure need not resemble the starting proposal. It may form a chain, a branching or convergent network, a collection of

partly independent mechanisms, or no coherent hierarchy at all. Complete experimental provenance, including machine-readable manifests, raw trajectories, interventions, bypasses, negative results and decision records, is treated as part of the evidential architecture.

No result produced by the proposed Paper III experiments, by itself, is interpreted as evidence of phenomenal consciousness. The purpose of the experimental programme is not to preserve a Theory of Everything architecture by construction, but to discover which proposed relations remain after controlled attempts have been made to break them.

Keywords: candidate dependency architecture; adversarial experimentation; causal intervention; self-maintenance; heredity; natural selection; causally consequential retention; feedback control; experience-dependent learning; structural generalisation; world models; reflective self-modelling; bypass architectures; preregistration; open science; Theory of Everything

1 Introduction

A proposed hierarchy of reality becomes scientifically useful only when its transitions can be placed at risk. The companion Minimal Principles Paper concluded that the route from fundamental dynamics to life, intelligence and consciousness cannot presently be represented as a simple succession of universal generative principles. It instead identified framework conditions, enabling conditions, effective mechanisms, unresolved transition problems and an open problem concerning phenomenal consciousness. The present paper asks which of those distinctions can be tested within deliberately constructed artificial systems.

Originally, the Artificial Hierarchy Experiment was motivated by the possibility that increasingly complex artificial environments might reproduce a sequence from dynamical persistence through adaptive organisation towards increasingly sophisticated intelligence. That formulation is now too permissive. If successive experimental phases are designed so that each contains more architecture, memory, feedback and computational capacity than the last, improved performance would be almost guaranteed and would reveal little about which mechanism produced which transition.

Consequently, the experiment is reformulated as a programme of adversarial dependency tests. Each candidate transition is treated as a hypothesis of the form

$$A \xrightarrow{M} B, \tag{1}$$

where A is an experimentally established capacity, B a proposed higher-order capacity and M the mechanism hypothesised to mediate the transition.

Crucially, three different claims must be kept separate:

$$\begin{gathered} M \text{ is sufficient to help produce } B, \\ M \text{ is necessary for } B, \\ M \text{ is the mechanism by which natural systems produce } B. \end{gathered} \tag{2}$$

An artificial experiment may provide evidence for the first, occasionally test the second within a specified architecture class, and still establish nothing about the third.

Successful construction of a capacity in an artificial system does not demonstrate that biology uses the same route. Aeroplanes demonstrate that heavier-than-air flight can be engineered without reproducing avian anatomy. In the same way, an artificial agent could instantiate a functional dependency relevant to learning or control without thereby recapitulating the evolutionary history, material organisation or phenomenology of an organism.

Failure to produce a capacity in one artificial architecture does not establish that the capacity is impossible or that natural systems cannot realise it differently. Negative evidence concerns the specified mechanism, implementation and experimental class unless a stronger impossibility result can be derived.

The paper therefore distinguishes *synthetic sufficiency* from *natural necessity*. If architecture $A + M$ reliably produces capability B while matched architecture A does not, the result supports the claim that M can contribute to B under the tested conditions. It does not establish that M is universally necessary.

The first major target is causally consequential retention. A system should not be credited with memory merely because information about the past can be decoded from its present state. The retained distinction must alter relevant future behaviour. History-dependent environments should therefore be paired with matched tasks that can be solved completely from the present observable state.

Feedback must likewise be distinguished from generic action capacity. Closed-loop performance should be compared not only with an inactive system but with capacity-matched open-loop control, yoked replay of previously successful action sequences and perturbations of the observation–action channel. A specifically feedback-dependent advantage should appear primarily where unpredictable disturbances make online correction useful.

Learning must also be separated from feedback. A fixed controller can respond successfully to changing observations without modifying the policy that generates those responses. Learning requires experience-dependent alteration of future behaviour or policy, together with evidence that the modification transfers appropriately beyond the exact training history.

Superior performance on familiar tasks is insufficient to establish the transition from adaptive regulation to intelligent agency. The stronger target concerns adaptation to structural novelty: changed causal relations, unfamiliar compositions of known elements, altered task families or situations in which previously successful policies cease to work.

Model-based control should be distinguished from successful reactive behaviour. If an agent allegedly uses an internal model, perturbing or misleading the relevant internal representation should produce selective behavioural consequences that cannot be explained by loss of generic capacity alone.

Self-modelling requires a still stronger test. A system that can produce statements about itself

does not thereby use a self-model in control. Evidence should require that internally represented information about the system's own capacities, uncertainty, state or limitations has causal access to subsequent action selection or verification.

None of these experiments is interpreted as a test for phenomenal experience. Memory, feedback, learning, generalisation, metacognition, self-report and recursive self-representation can all be studied behaviourally and mechanistically without assuming subjective awareness.

Paper II also changes the measurement strategy. The proposed cross-scale constraint vector did not survive adversarial analysis as a vector of independent coordinates. The present experiment therefore does not compute one total constraint score.

Instead, where relevant, measurements are organised through the typed comparative profile

$$\mathfrak{C} = [\mathfrak{D}, \mathfrak{F}, \mathfrak{T}, \mathfrak{O}], \tag{3}$$

distinguishing dynamical divergence, functional effects, temporal or historical descriptors and organisational relations.

Even this profile is not treated as an intrinsic property of an agent. Its interpretation depends upon the comparison being made:

$$\mathfrak{C}\,[\mathcal{M}_1 \| \mathcal{M}_0; \mathfrak{R}, \Psi, T, \mathcal{K}], \tag{4}$$

where $\mathcal{M}_1$ and $\mathcal{M}_0$ are comparison and baseline conditions, $\mathfrak{R}$ is the representation, Ψ the explanandum, T the evaluation horizon and $\mathcal{K}$ the causal or structural status of the claim.

The strongest evidence sought in Paper III is therefore not a monotonic increase in one complexity or constraint variable across successive experimental blocks. It is selective dependence:

$$\boxed{\text{remove or disrupt hypothesised mechanism } M \Longrightarrow \text{selective loss of capacity } B.} \tag{5}$$

An even stronger result occurs when restoring the relevant function rescues the capacity, while irrelevant or capacity-matched manipulations do not. Such intervention patterns provide substantially more information than correlations between architecture size and performance.

Bypass architectures are especially important. If system S acquires capacity B without candidate precursor A, then the proposed dependency

$$A \longrightarrow B \tag{6}$$

cannot be treated as universally necessary. Artificial counterexamples may therefore be more theoretically informative than artificial successes.

Ultimately, Paper III asks not whether an artificial hierarchy can be made to look plausible, but

whether the candidate dependency architecture survives controlled attempts to break it:

$$\boxed{\begin{array}{c}\text{candidate transition} \longrightarrow \text{matched controls} \longrightarrow \text{causal ablation} \\ \longrightarrow \text{counterexample search} \longrightarrow \text{retain, restrict, reduce, suspend or reject.}\end{array}} \tag{7}$$

2 Methodological Commitments

The experiment adopts the same evidential asymmetry as the two companion papers: evidence may support a claim only at the level directly tested. Behavioural success does not automatically establish mechanism, mechanism does not establish universality, and artificial implementation does not establish natural equivalence.

2.1 The hierarchy is a hypothesis, not the experimental result

The ordering of experimental blocks is treated as a candidate dependency structure rather than as a developmental law.

If the programme proposes

$$A \longrightarrow B \longrightarrow C, \tag{8}$$

then the experiment must actively search for:

$$A \not\Rightarrow B, \tag{9}$$

$$B \not\Rightarrow C, \tag{10}$$

and possible bypass

$$A \longrightarrow C. \tag{11}$$

Discovery of a bypass is evidence against a strict hierarchy, not an inconvenient anomaly to be engineered away.

2.2 No upward propagation of evidential status

Success at one level does not validate claims at higher levels.

For example,

$$\text{history-sensitive behaviour} \tag{12}$$

does not establish

$$\text{learning}, \tag{13}$$

and successful learning does not establish

$$\text{intelligent agency}. \tag{14}$$

Similarly,

$$\text{self-model use} \tag{15}$$

does not establish

$$\text{phenomenal consciousness.} \tag{16}$$

Each transition requires its own evidence.

2.3 Mechanism labels must be operational

Terms such as "memory", "feedback", "learning", "model", "agency" and "self-model" are not accepted as architectural labels without intervention-sensitive definitions.

A component counts as relevant to mechanism M only if:

1. its state or operation can be specified;
2. its hypothesised causal contribution is stated before testing;
3. perturbing it produces the predicted selective effect;
4. simpler loss-of-capacity explanations are controlled.

Naming a module "memory" or "world model" therefore supplies no evidence that it performs the hypothesised function.

2.4 Matched controls outrank inactive controls

Each principal architecture should face a control possessing comparable computational opportunity wherever practicable.

Relevant comparisons include:

$$\begin{gathered}\text{retention-enabled} \quad \text{vs} \quad \text{capacity-matched retention-ablated,} \\ \text{closed-loop} \quad \text{vs} \quad \text{open-loop} \quad \text{vs} \quad \text{yoked replay,} \\ \text{learning} \quad \text{vs} \quad \text{fixed-policy regulation,} \\ \text{model-based} \quad \text{vs} \quad \text{capacity-matched reactive control.}\end{gathered} \tag{17}$$

An architecture should not receive mechanistic credit merely for outperforming a deliberately weak baseline.

2.5 Task pairs must isolate the claimed function

Every candidate mechanism should be tested on at least two classes of task:

1. tasks in which the proposed mechanism is theoretically relevant;
2. matched tasks in which the same generic computational capacity is available but the proposed mechanism should not be required.

For causally consequential retention, this means comparing history-dependent environments with matched present-state-solvable environments.

For feedback, it means comparing unpredictable disturbances with matched predictable conditions.

For explicit models, it means comparing tasks requiring latent-structure inference or counterfactual prediction with matched reactive tasks.

Selective impairment provides stronger evidence than generic performance loss.

2.6 Ablation must be selective rather than destructive

Deleting a component can produce uninterpretable failure if the intervention destroys generic computational capacity.

A useful ablation should therefore preserve as much of the architecture as possible while removing or scrambling the hypothesised causal function.

Where intervention is expressed using do(·) notation, the notation is used in the interventionist causal sense associated with structural causal models [1].

Where direct deletion is destructive, alternatives include:

1. channel randomisation;
2. temporal delay;
3. state permutation;
4. matched replacement;
5. information masking;
6. targeted freezing of parameter updates.

The aim is to identify mechanism-specific deficits rather than demonstrate that damaged systems perform badly.

2.7 Restoration and rescue strengthen causal claims

Causal attribution becomes stronger when an ablated function can be selectively restored.

Suppose intervention

$$\mathrm{do}(M = 0) \tag{18}$$

reduces target capacity B.

If supplying a functionally equivalent replacement

$$M' \tag{19}$$

restores performance while irrelevant replacements do not, then the evidence supports the functional role more strongly than ablation alone.

Such rescue experiments should be used wherever implementation permits.

2.8 Architecture capacity must be controlled

Larger architectures may outperform smaller ones for reasons unrelated to the hypothesised mechanism.

Comparisons should therefore match, as far as practicable:

$$\begin{gathered}\text{parameter count}, \quad \text{training budget}, \quad \text{observation bandwidth}, \\ \text{action space}, \quad \text{optimisation opportunity}, \quad \text{random-seed distribution}.\end{gathered} \tag{20}$$

Where exact capacity matching is impossible, the residual difference must be measured and included as a limitation rather than ignored.

2.9 Training success and experimental success are different

An agent reaching training criterion does not count as confirmation of the research hypothesis.

Training merely produces a candidate system for testing.

Confirmatory evidence comes from fixed held-out evaluations, causal interventions and preregistered comparisons performed after training.

Therefore,

$$\boxed{\text{training performance} \neq \text{experimental confirmation.}} \tag{21}$$

2.10 Generalisation must involve structural novelty

Performance on new samples from the same narrow distribution is a weak test of generalisation.

The programme distinguishes at least:

1. interpolation within familiar distributions;

2. parameter variation within familiar causal structure;

3. unfamiliar composition of known components;

4. altered causal relations;

5. previously unseen problem families.

Claims of intelligent agency should rely primarily upon the latter categories rather than ordinary held-out sampling alone.

2.11 Representation dependence must be tested

Measured effects can depend upon state variables, temporal resolution, system boundary and coarse-graining.

Where a claimed mechanism survives only one arbitrary representation, its interpretation should be restricted.

At minimum, decisive experiments should test sensitivity to:

$$\text{state encoding}, \qquad \text{observation granularity}, \qquad \text{sampling interval}, \qquad \text{evaluation horizon}. \tag{22}$$

A robust result need not be numerically invariant, but its transformation should be intelligible.

2.12 No universal constraint score is permitted

Paper II rejected the proposed cross-scale constraint vector as a vector of commensurable independent coordinates.

Paper III therefore does not construct

$$\Gamma_{\text{total}} \tag{23}$$

by summing dynamical, functional, temporal and organisational quantities.

When relevant, the experiment reports the typed profile

$$\mathfrak{C} = [\mathfrak{D}, \mathfrak{F}, \mathfrak{T}, \mathfrak{O}], \tag{24}$$

with each descriptor interpreted separately.

Mechanisms such as feedback and selection-like filtering should be treated as causes of profile changes, not additional scalar coordinates.

2.13 Artificial success does not establish cross-scale identity

An artificial system can instantiate the same formal relation as a biological system without sharing its material mechanism.

Therefore,

$$\text{same functional relation} \neq \text{same ontology}. \tag{25}$$

Paper III's primary confirmatory domain is artificial systems themselves.

Mappings to biological, chemical or physical systems remain exploratory until they pass the validation protocol established in Paper II.

2.14 No claim to artificial life is built into the experiment

Persistence, viability, self-maintenance or closure-like organisation in an artificial environment does not automatically constitute life.

The experiment may test formal properties associated with living systems, including:

$$\text{maintenance}, \qquad \text{heredity}, \qquad \text{selection-like dynamics}, \tag{26}$$

without deciding the classificatory question

$$\text{Is this system alive?} \tag{27}$$

That question requires criteria beyond successful task construction.

2.15 No consciousness inference is permitted

No behavioural or architectural result in this paper is sufficient evidence of phenomenal consciousness. Contemporary consciousness science contains multiple competing theoretical frameworks and no settled inference rule from the functional capacities tested here to phenomenal experience [2, 3].

Specifically:

$$\begin{gathered} \text{recursive processing} \not\Rightarrow \text{consciousness}, \\ \text{self-report} \not\Rightarrow \text{consciousness}, \\ \text{self-modelling} \not\Rightarrow \text{consciousness}, \\ \text{conflict monitoring} \not\Rightarrow \text{consciousness}. \end{gathered} \tag{28}$$

The physical basis of phenomenality remains Open Problem 1 in Paper I and is not assigned an artificial proxy here.

2.16 Pre-registration precedes decisive evaluation

Preregistration is used to preserve the distinction between exploratory hypothesis generation and confirmatory testing before decisive outcomes are known [4].

The following should be frozen before final evaluation:

1. hypotheses;
2. environments;
3. architectures;
4. seeds or seed-generation procedure;
5. training budgets;
6. ablations;
7. primary metrics;
8. exclusion rules;
9. statistical tests;
10. break conditions.

Exploratory analyses remain permitted but must be labelled as exploratory and re-tested independently before being treated as confirmation.

2.17 Null models must be capable of winning

Every major hypothesis should face at least one serious alternative capable of reproducing the observed behaviour without the proposed mechanism.

Nulls may include:

$$\begin{gathered}\text{simpler reactive policies,} \qquad \text{larger but mechanism-free architectures,} \\ \text{randomised historical state,} \qquad \text{yoked action sequences,} \qquad \text{domain-specific heuristic solutions.}\end{gathered} \tag{29}$$

A null model that is designed to fail does not constitute a meaningful adversarial test.

2.18 Bypass tests are mandatory

Every proposed dependency

$$A \longrightarrow B \tag{30}$$

should face an explicit search for architectures capable of producing B without A.

If such architecture exists and performs comparably under the decisive tests, A cannot be treated as universally necessary for B.

Bypass discovery should therefore trigger revision of the dependency map rather than reinterpretation designed to preserve it.

2.19 Failure is classified rather than hidden

Each candidate mechanism or transition should finish in one of five states:

1. **Retain**: survives the relevant causal and robustness tests within the stated scope.
2. **Restrict**: survives only for specified architectures, environments, representations or scales.
3. **Reduce**: behaviour is explained equally well by a simpler established mechanism.
4. **Suspend**: evidence is insufficient because a decisive test remains unavailable or underpowered.
5. **Reject**: the predicted selective dependency repeatedly fails or a robust bypass defeats the proposed necessity claim.

These outcomes should be decided from preregistered criteria rather than from whether the result makes the wider Theory of Everything architecture look more unified.

2.20 Progression between blocks is not automatic

Successful completion of one experimental block does not require promotion of its hypothesis into the interpretation of the next block.

A candidate dependency may instead receive:

$$\text{Retain}, \quad \text{Restrict}, \quad \text{Reduce}, \quad \text{Suspend}, \quad \text{Reject}. \tag{31}$$

Later blocks can still be run when they test logically separable questions, but their interpretation must not presuppose a failed dependency.

This prevents the experimental design from becoming a one-way ladder in which every block is forced to confirm the preceding one.

2.21 Raw trajectories must be preserved

Summary statistics are insufficient for a programme concerned with transition structure, history dependence and causal intervention.

The experiment should preserve:

$$\begin{gathered}\text{observations}, \quad \text{actions}, \quad \text{internal states where accessible}, \\ \text{rewards or viability variables}, \quad \text{intervention labels}, \quad \text{environment states}.\end{gathered} \tag{32}$$

Derived figures and tables should be regenerable from these raw trajectories.

2.22 The experimental architecture itself remains falsifiable

Ultimately, Paper III is not protected from the possibility that no artificial hierarchy survives.

The programme would count as scientifically successful if it established any of the following:

1. several proposed transitions are genuinely dependent;
2. some transitions are contingent rather than necessary;
3. apparently higher capacities bypass proposed lower ones;
4. several mechanisms reduce to simpler dynamics;
5. no coherent hierarchy appears at all.

The governing rule is therefore:

$$\boxed{\begin{array}{c}\text{the experiment is not required to discover a hierarchy;} \\ \text{it is required to make the hierarchy fail if it is wrong.}\end{array}} \tag{33}$$

3 Experimental Architecture and Scope

The experiment is organised as a sequence of controlled dependency tests rather than as a ladder whose later stages are assumed to emerge from earlier ones. The five experimental blocks correspond to questions left open by the revised Minimal Principles architecture, but their ordering is methodological rather than ontological. Each block isolates a different candidate capacity, identifies mechanisms hypothesised to contribute to it, constructs matched alternatives, and searches explicitly for counterexamples to the proposed dependency.

The programme therefore tests the schematic architecture

$$\boxed{\begin{array}{c}\text{Block I: Dynamical Persistence and Maintenance}\\ \downarrow ?\\ \text{Block II: Heredity, Reproduction and Population Change}\\ \downarrow ?\\ \text{Block III: Retention, Feedback and Learning}\\ \downarrow ?\\ \text{Block IV: Adaptive Agency and Structural Generalisation}\\ \downarrow ?\\ \text{Block V: Reflective Self-Modelling.}\end{array}} \tag{34}$$

The question marks are part of the experimental architecture. They indicate candidate dependencies to be tested rather than transitions already accepted by the Theory of Everything programme.

3.1 The five experimental blocks

The five blocks separate capacities that the earlier three-phase design risked combining prematurely.

Table 1: Revised experimental architecture of Paper III.

Block	**Primary target**	**Central question**	**Principal adversarial test**
I	Dynamical persistence and maintenance	Can continued operation be attributed to internally generated maintenance rather than passive stability or external support?	Passive, external, feedforward and reciprocal-maintenance architectures are compared under matched disturbances, ablations and rescue tests.
II	Heredity, reproduction and population change	What reproduces, what is transmitted, and under what conditions does differential reproduction constitute natural-selection-like population change?	Reproduction and heredity are established before selection is inferred; neutral, non-heritable and alternative-unit controls test whether selection language is warranted.

Block	Primary target	Central question	Principal adversarial test
III	Retention, feedback and learning	Do historical retention, closed-loop regulation and experience-dependent learning make distinct causal contributions?	History-dependent tasks are paired with matched Markov tasks; closed-loop control faces open-loop and yoked controls; learning is compared with fixed policies.
IV	Adaptive agency and structural generalisation	Can the system adapt to changed causal structure, unfamiliar task composition and novel problem families rather than merely interpolate?	Reactive, model-free and model-based alternatives face held-out structural novelty and intervention on candidate internal representations.
V	Reflective self-modelling	Does information about the system's own state, competence, uncertainty or limitations causally alter subsequent behaviour?	Self-description is separated from causal self-model use through perturbation, misinformation, calibration and bypass tests.

3.2 Block ordering does not imply necessary dependence

Later blocks may require technical capabilities that resemble capacities studied earlier, but implementation dependence must not be confused with theoretical necessity.

Suppose Block IV system S possesses internal retention because its software architecture requires recurrent state. That fact does not establish the universal dependency

$$\text{causally consequential retention} \longrightarrow \text{intelligent agency}. \tag{35}$$

Necessity would require evidence that serious alternative architectures fail to realise the higher capacity without the proposed precursor.

Accordingly,

$$\boxed{\text{experimental implementation order} \neq \text{natural dependency order}.} \tag{36}$$

The experiment therefore permits later blocks to be run even when an earlier dependency has been restricted or rejected, provided the later question can be formulated independently.

3.3 Each arrow is a separate empirical hypothesis

A sequence such as

$$A \longrightarrow B \longrightarrow C \tag{37}$$

contains at least two dependency claims:

$$A \longrightarrow B \tag{38}$$

and

$$B \longrightarrow C. \tag{39}$$

Evidence for the first does not propagate automatically to the second.

Nor does observing

$$A, \qquad B, \qquad C \tag{40}$$

within one architecture demonstrate that the arrows are causal.

Each proposed transition must therefore possess its own:

1. operational definition;
2. candidate mechanism;
3. matched control;
4. causal perturbation;
5. bypass search;
6. break condition.

3.4 Bypass architectures are first-class experimental outcomes

The strongest challenge to a proposed hierarchy is a system that realises a higher capacity while lacking a supposedly necessary lower one.

For candidate dependency

$$A \longrightarrow B, \tag{41}$$

a bypass architecture S^* satisfies

$$A(S^*) = 0 \tag{42}$$

while

$$B(S^*) = 1 \tag{43}$$

under the preregistered operational definitions.

If the bypass survives capacity matching, held-out testing and robustness analysis, then A cannot be retained as universally necessary for B.

Hence,

$$\boxed{\text{successful bypass} = \text{evidence against strict hierarchical necessity.}} \tag{44}$$

The experiment should therefore allocate computational effort to constructing counterexamples rather than using every available resource to optimise the preferred architecture.

3.5 Synthetic sufficiency and natural necessity remain distinct

Successful artificial construction answers a possibility question.

If architecture

$$A + M \tag{45}$$

reliably produces capacity

$$B, \tag{46}$$

the result supports:

$$M \text{ is sufficient to contribute to } B \tag{47}$$

within the stated architecture and environment.

It does not establish:

$$M \text{ is universally necessary for } B, \tag{48}$$

nor:

$$M \text{ is the mechanism used by natural systems.} \tag{49}$$

Consequently,

$$\boxed{\text{synthetic sufficiency} \neq \text{natural necessity} \neq \text{cross-domain identity.}} \tag{50}$$

This distinction governs interpretation throughout all five blocks.

3.6 Measurements are typed rather than common

The experiment does not require every block to emit the same set of numerical quantities.

Where appropriate, results can be organised through the Paper II profile

$$\mathfrak{C} = [\mathfrak{D}, \mathfrak{F}, \mathfrak{T}, \mathfrak{O}], \tag{51}$$

but individual blocks use only the categories relevant to their explananda.

For example, Block I may require:

$$\mathfrak{D}, \qquad \mathfrak{F}, \qquad \mathfrak{T}, \qquad \mathfrak{O}, \tag{52}$$

because dynamics, viability, timescale and maintenance topology are all relevant.

Block IV may instead rely primarily upon:

$$\text{generalisation performance}, \qquad \text{causal model-use tests}, \qquad \text{trajectory adaptation}, \tag{53}$$

without inventing an organisational-closure score simply for numerical symmetry.

Therefore,

$$\boxed{\text{shared experiment} \neq \text{shared scalar metric.}} \tag{54}$$

3.7 Common experimental infrastructure

Several procedural elements should nevertheless remain shared across blocks.

Where applicable, every block should use:

1. preregistered environments;
2. capacity-matched architectures;
3. multiple random seeds;
4. fixed training budgets;
5. held-out evaluation conditions;
6. raw trajectory preservation;
7. predetermined ablations;
8. serious null models;
9. robustness checks;
10. explicit decision criteria.

This shared infrastructure provides methodological comparability without pretending that the substantive quantities measured in each block are identical.

3.8 Training and evaluation remain separated

Optimisation is treated as preparation for experiment rather than as the decisive experiment itself.

Each architecture may undergo an exploratory or preregistered training phase:

$$\theta_0 \longrightarrow \theta_{\text{trained}}. \tag{55}$$

Once training is complete, the decisive evaluation begins on fixed held-out conditions.

Architectural changes, metric redefinitions or new stopping rules introduced after inspecting decisive results should generate a new experimental version rather than being folded silently into the original analysis.

Thus,

$$\boxed{\text{optimisation produces the test subject;held-out intervention tests the hypothesis.}} \tag{56}$$

3.9 Progression between blocks is conditional

No block automatically "passes" an agent to the next level of an artificial hierarchy.

Instead, each candidate dependency receives one of:

$$\text{Retain}, \qquad \text{Restrict}, \qquad \text{Reduce}, \qquad \text{Suspend}, \qquad \text{Reject}. \tag{57}$$

A later block may still be conducted after an earlier rejection if its experimental question remains logically meaningful.

For example, rejection of strong organisational closure as necessary for artificial self-maintenance would not prevent investigation of learning or agency in systems constructed through another route.

The experimental architecture is therefore a network of questions rather than a mandatory staircase.

3.10 The five blocks should be interpreted modularly

Each block has two roles.

First, it tests one local dependency:

$$A \xrightarrow{M} B. \tag{58}$$

Second, its results can modify the wider dependency map.

A failure in Block II may alter the proposed relationship between self-maintenance and Darwinian populations without invalidating Block I.

Likewise, a bypass discovered in Block IV may weaken one route to intelligent agency without invalidating the experimentally demonstrated existence of feedback or learning.

Accordingly,

$$\boxed{\text{local failure} \neq \text{global programme failure.}} \tag{59}$$

The architecture is deliberately decomposable so that failed hypotheses can be removed without protecting them through dependence of later experiments upon their truth.

3.11 Cross-domain claims are deferred

Paper III tests artificial systems first.

If an artificial mechanism exhibits a reproducible dependency, comparison with biological, chemical or physical systems becomes a separate research step governed by the validation protocol of Paper II.

A cross-domain claim therefore requires:

$$\begin{gathered}\text{within-artificial-system validation} \\ \downarrow \\ \text{independently validated natural-domain quantity} \\ \downarrow \\ \text{preregistered mapping} \\ \downarrow \\ \text{held-out cross-domain prediction.}\end{gathered} \tag{60}$$

Superficial functional resemblance is insufficient.

In particular,

$$\text{artificial maintenance} \neq \text{biological life}, \tag{61}$$

and

$$\text{artificial self-modelling} \neq \text{phenomenal consciousness}. \tag{62}$$

3.12 Scope boundary

The experiment ends at reflective self-modelling. It does not introduce a sixth block for phenomenal consciousness. The absence is deliberate because the current research architecture contains no experimentally validated bridge from self-modelled intelligence to phenomenal experience.

Behavioural sophistication, recursive processing, conflict representation, metacognitive reporting and self-description may all be investigated without being treated as consciousness indicators.

Accordingly,

$$\boxed{\begin{array}{c}\text{Paper III tests functional and causal architecture;}\\ \text{it does not manufacture a proxy for phenomenality.}\end{array}} \tag{63}$$

3.13 Overall experimental logic

The complete programme can be summarised as

$$\boxed{\begin{array}{c}\text{define candidate capacity}\\ \downarrow\\ \text{identify proposed mechanism}\\ \downarrow\\ \text{construct matched alternatives}\\ \downarrow\\ \text{train without using decisive test conditions}\\ \downarrow\\ \text{evaluate on held-out tasks}\\ \downarrow\\ \text{ablate, perturb and rescue}\\ \downarrow\\ \text{search for bypass architectures}\\ \downarrow\\ \text{retain, restrict, reduce, suspend or reject.}\end{array}} \tag{64}$$

The architecture therefore does not define success as completion of all five blocks in their proposed order.

Its success criterion is stricter:

$$\boxed{\begin{array}{c}\text{each proposed dependency must survive}\\ \text{a serious opportunity to fail.}\end{array}} \tag{65}$$

4 Experimental Block I: Dynamical Persistence and Maintenance

The first experimental block addresses the earliest unresolved transition in the revised dependency architecture:

$$\text{persistent dynamical organisation} \longrightarrow \boxed{?} \longrightarrow \text{internally maintained organisation.} \tag{66}$$

The experiment does not assume that persistence implies self-maintenance. Instead, it asks which causal structures, if any, distinguish systems that merely remain stable from systems whose own internal processes contribute to preserving the conditions required for their continued operation.

Four forms of persistence must be separated:

1. passive persistence produced by intrinsically slow degradation;
2. persistence produced by continuous external support;
3. persistence produced by internal but predominantly feed-forward maintenance;
4. persistence produced by reciprocal internal maintenance among organisational components.

Only the latter two are candidates for internal self-maintenance, and even reciprocal maintenance is not assumed to constitute life [5].

4.1 Primary experimental question

Block I asks:

> **Can an artificial system be shown to preserve its own operational organisation through causally identifiable internal maintenance, rather than merely persisting because it is intrinsically stable or externally stabilised?**

The candidate transition can be written

$$P \xrightarrow{M} S, \tag{67}$$

where P denotes persistent organisation, S denotes internally maintained organisation and M is a candidate maintenance architecture.

The proposed dependency should fail if equivalent persistence and recovery can be obtained from passive stability or external maintenance without the proposed internal architecture.

4.2 Driven pattern-forming baseline

Block I should include a deliberately simpler physical baseline in which persistent organisation can arise without any claim of internal self-maintenance. A driven reaction–diffusion system provides a useful test because sustained pattern formation can be produced transparently through local dynamics and external non-equilibrium conditions.

A Gray–Scott-type system is suitable [6]:

$$\frac{\partial u}{\partial t} = D_u \nabla^2 u - uv^2 + f(1-u), \tag{68}$$

$$\frac{\partial v}{\partial t} = D_v \nabla^2 v + uv^2 - (f+k)v, \tag{69}$$

where u and v are field concentrations, D_u and D_v are diffusion coefficients, and f and k control feed and removal.

Crucially, persistent spatial structure in this model is not classified as self-maintenance merely because it survives for a long period. The system is included precisely to test the weaker relation

$$\boxed{\begin{array}{c}\text{sustained non-equilibrium driving} \\ \longrightarrow \text{persistent organised pattern,}\end{array}} \tag{70}$$

without assuming

$$\text{persistent organised pattern} \longrightarrow \text{internally maintained organisation.} \tag{71}$$

4.2.1 Baseline manipulations

The reaction–diffusion baseline should be tested under:

1. sustained driving versus matched removal of driving;
2. open versus closed boundary conditions;
3. spatial barriers or channel restrictions;
4. parameter sweeps through pattern-forming and non-pattern-forming regimes;
5. standardised perturbations after pattern formation.

These manipulations test whether persistence depends upon external driving, boundary structure and dynamical regime.

4.2.2 Baseline measurements

Relevant descriptive measurements include:

1. persistence time after perturbation;
2. recovery time after local damage;
3. spatial mutual information;
4. spectral descriptors of spatial structure;

5. state-space occupancy;

6. transition or path divergence under matched interventions.

Spatial mutual information and spectral structure are retained only as descriptive pattern measures. They are not treated as scalar measures of organisation, viability or self-maintenance.

Likewise, reduced or altered state-space occupancy does not by itself imply greater organisation.

4.2.3 Baseline prediction and break condition

The preregistered prediction is that sustained driving should produce persistent non-trivial organisation within reproducible parameter regions.

The corresponding break condition is:

> **If apparent organisation depends primarily upon visually selected patterns, fails preregistered quantitative descriptions, or cannot be distinguished reproducibly from matched non-pattern-forming regimes, the reaction–diffusion baseline supplies no robust evidence even for driven pattern persistence.**

Importantly, successful confirmation of this baseline establishes only that driven physical dynamics can sustain persistent organisation. It does not establish self-maintenance, organisational closure, life or any subsequent stage of the experimental hierarchy.

4.3 Artificial system

Formally, let the system contain n organisational components

$$C_1, C_2, \ldots, C_n. \tag{72}$$

Each component has a bounded functional state

$$q_i(t) \in [0, 1], \tag{73}$$

where larger q_i represents greater preservation of the component's specified operational function.

Environmental disturbances are represented by

$$\xi_i(t), \tag{74}$$

and uncontrolled degradation by rate

$$d_i \geq 0. \tag{75}$$

A general discrete-time update is

$$q_i(t+1) = \mathrm{clip}_{[0,1]}\Big[q_i(t) - d_i - \xi_i(t) + m_i^{\mathrm{int}}(t) + m_i^{\mathrm{ext}}(t)\Big], \tag{76}$$

where m_i^{int} denotes internally generated maintenance and m_i^{ext} externally supplied maintenance.

Crucially, the same degradation process, disturbance distribution and total available maintenance budget should be used across matched architectures wherever practicable.

4.4 Viability criterion

The system requires an independently specified operational region

$$\mathcal{V} = \{\mathbf{q} : q_i \geq \theta_i \text{ for all essential } i\}, \tag{77}$$

where the thresholds θ_i are fixed before the decisive experiments.

For horizon T, path viability is

$$V^{\mathrm{path}}(T) = P\left(\mathbf{q}(t) \in \mathcal{V} \;\; \forall t \leq T\right). \tag{78}$$

Terminal viability is recorded separately:

$$V^{\mathrm{term}}(T) = P\left(\mathbf{q}(T) \in \mathcal{V}\right). \tag{79}$$

Importantly, neither quantity is interpreted as a universal measure of organisation. They specify whether the artificial system remains within the predefined operational conditions relevant to this experiment.

4.5 Experimental architectures

Comparatively, at least four capacity-matched system classes should be tested.

Architecture A: Passive persistence. Here

$$m_i^{\mathrm{int}}(t) = m_i^{\mathrm{ext}}(t) = 0, \tag{80}$$

while degradation rates are chosen sufficiently low that the system can persist for substantial periods without active maintenance.

This architecture is a critical null because long persistence alone should not be mistaken for self-maintenance.

Architecture B: External maintenance. Here internal maintenance is absent,

$$m_i^{\text{int}}(t) = 0, \tag{81}$$

while an external controller supplies

$$m_i^{\text{ext}}(t) > 0 \tag{82}$$

in response to component degradation.

This architecture can display excellent viability and recovery while possessing no endogenous maintenance autonomy.

Architecture C: Internal feed-forward maintenance. Here maintenance is generated internally, but the dependency graph is not reciprocally closed.

A representative structure is

$$C_1 \to C_2, \qquad C_1 \to C_3, \tag{83}$$

without a maintenance path returning from C_2 or C_3 to C_1.

This architecture tests whether internality alone explains any apparent self-maintenance advantage.

Architecture D: Reciprocal internal maintenance. Here the components form at least one preregistered strongly connected maintenance structure, for example

$$C_1 \to C_2 \to C_3 \to C_1. \tag{84}$$

The arrows represent hypothesised maintenance contributions and must later be validated interventionally rather than inferred merely from the architecture diagram.

4.6 Maintenance graph

Structurally, define directed maintenance graph

$$G_{\mathcal{C}} = (V, E). \tag{85}$$

For candidate edge

$$C_i \to C_j, \tag{86}$$

define intervention-sensitive maintenance effect

$$\begin{aligned} w_{ij}(T) = P\left(C_j \text{ remains functional at } T \mid \text{do}(C_i = \text{active})\right) \\ - P\left(C_j \text{ remains functional at } T \mid \text{do}(C_i = \text{disabled})\right). \end{aligned} \tag{87}$$

An edge is retained only if

$$w_{ij}(T) > \epsilon_w, \tag{88}$$

where ϵ_w is preregistered.

Causally, architectural connectivity is therefore not sufficient. The maintenance relations themselves must survive perturbation.

4.7 Internal maintenance fraction

Let

$$u_{aj}(T) \tag{89}$$

denote the maintenance contribution of external support E_a to component C_j.

The exploratory internal maintenance fraction is

$$\chi_{\text{int}}(T) = \frac{\sum_{i,j}[w_{ij}(T)]_+}{\sum_{i,j}[w_{ij}(T)]_+ + \sum_{a,j}[u_{aj}(T)]_+}, \tag{90}$$

where

$$[z]_+ = \max(z, 0). \tag{91}$$

However,

$$\boxed{\chi_{\text{int}} \neq \text{organisational closure.}} \tag{92}$$

A feed-forward architecture can possess high internal maintenance while lacking reciprocal causal dependence.

4.8 Disturbance regimes

Systems should be tested under several independently specified disturbance classes:

1. low-amplitude stationary degradation;
2. stochastic component-specific shocks;
3. temporary resource deprivation;
4. repeated subcritical disturbance;
5. rare severe perturbation;
6. altered disturbance statistics not encountered during calibration.

The final category is important because a maintenance architecture that performs only under its calibration distribution may represent tuning rather than robust self-maintenance.

4.9 Primary outcome measures

Block I should report several distinct outcomes rather than one total organisation or constraint score.

The principal functional quantities are:

$$V^{\mathrm{path}}(T), \qquad V^{\mathrm{term}}(T), \tag{93}$$

together with mean time to failure:

$$\tau_{\mathrm{fail}} = \mathbb{E}[T_{\mathrm{fail}}]. \tag{94}$$

When a single experimental condition is being described, the condition index is suppressed for readability. Comparative analyses should restore an explicit architecture or condition index, for example

$$V_a^{\mathrm{path}}(T), \qquad V_a^{\mathrm{term}}(T), \tag{95}$$

for condition a.

Recovery after standardised perturbation can be measured by

$$\tau_{\mathrm{rec}}, \tag{96}$$

the time required to return to a preregistered recovery region.

Dynamical divergence relative to a matched baseline can be reported using Γ_{trans}, or using path-space divergence Γ_{path} where appropriate.

Organisational quantities should include:

$$G_{\mathcal{C}}, \qquad \chi_{\mathrm{int}}, \tag{97}$$

rather than collapsing them into one scalar.

4.10 Typed profile for Block I

The Block I comparison can therefore use only those portions of the Paper II typed profile that match the present explanandum.

$$\mathfrak{C}_{\mathrm{Block\,I}} = [\mathfrak{D}, \mathfrak{F}, \mathfrak{T}, \mathfrak{O}]. \tag{98}$$

Here:

$$\mathfrak{D} = \{\Gamma_{\text{trans}}, \Gamma_{\text{path}}\}, \tag{99}$$

$$\mathfrak{F} = \{V^{\text{path}}, V^{\text{term}}, \tau_{\text{fail}}, \tau_{\text{rec}}\}, \tag{100}$$

$$\mathfrak{T} = \{\phi_K\}, \tag{101}$$

$$\mathfrak{O} = \{G_{\mathcal{C}}, \chi_{\text{int}}\}. \tag{102}$$

No norm

$$\|\mathfrak{C}_{\text{Block I}}\| \tag{103}$$

is defined.

4.11 Test 1: Persistence versus active maintenance

Test 1 compares passive Architecture A with actively maintained Architectures B–D.

If Architecture A matches active architectures in

$$V^{\text{path}}(T), \qquad \tau_{\text{fail}}, \qquad \tau_{\text{rec}}, \tag{104}$$

across the decisive disturbance regimes, then active maintenance has not demonstrated independent functional value under those conditions.

Conversely, active architectures should outperform passive persistence specifically when disturbances require ongoing correction rather than merely slow degradation.

An operational prediction associated with H1.1 is:

$$\begin{gathered}\text{active maintenance advantage increases} \\ \text{as disturbance demands increase.}\end{gathered} \tag{105}$$

4.12 Test 2: Internal versus external maintenance

Test 2 compares externally maintained Architecture B with internally maintained Architectures C and D.

If both

$$V_B^{\text{path}}(T) \approx V_C^{\text{path}}(T) \approx V_D^{\text{path}}(T) \tag{106}$$

and

$$V_B^{\text{term}}(T) \approx V_C^{\text{term}}(T) \approx V_D^{\text{term}}(T), \tag{107}$$

then viability alone does not distinguish external support from internal maintenance.

The stronger question concerns autonomy under removal of external support.

Define external-support withdrawal at time t_0:

$$m_i^{\text{ext}}(t) = 0 \qquad \text{for } t \geq t_0. \tag{108}$$

An operational prediction associated with H1.2 is:

$$\begin{gathered}\text{internally maintained architectures retain} \\ \text{greater viability after support withdrawal.}\end{gathered} \tag{109}$$

This experiment directly distinguishes

$$\text{viability} \tag{110}$$

from

$$\text{endogenous maintenance.} \tag{111}$$

4.13 Test 3: Internality versus reciprocal maintenance

Test 3 compares internal feed-forward Architecture C with reciprocal Architecture D.

The key hypothesis is not simply that Architecture D outperforms Architecture C on one or both preregistered viability measures.

A strongly connected organisation may or may not outperform a feed-forward architecture in every environment.

A stronger operational prediction associated with H1.3 is:

$$\begin{gathered}\text{targeted disruption of a causally validated} \\ \text{maintenance node in Architecture D produces} \\ \text{predictable propagation through its maintenance graph.}\end{gathered} \tag{112}$$

A corresponding feed-forward architecture should display a different propagation structure.

This test asks whether closure-like topology has causal explanatory content, not whether cycles are aesthetically more "organised".

4.14 Test 4: Single-node ablation

Each candidate maintenance node should be disabled individually.

For each viability endpoint $r \in \{\text{path}, \text{term}\}$, record:

$$\Delta_i V^{(r)}(T) = V^{(r)}_{\text{intact}}(T) - V^{(r)}_{\text{do}(C_i=0)}(T). \tag{113}$$

Also record the induced changes in other component states:

$$\Delta_i q_j(t). \tag{114}$$

If the maintenance graph is explanatory, the observed pattern of downstream effects should correspond to the preregistered causal topology better than simpler alternatives such as raw node degree or component size.

4.15 Test 5: Rescue experiment

After disabling C_i, externally supply the specific maintenance function attributed to that node.

Let

$$F_i^{\text{ext}} \tag{115}$$

denote the matched replacement.

Compare:

$$\begin{gathered} \text{do}(C_i = 0), \\ \text{do}(C_i = 0) + F_i^{\text{ext}}, \\ \text{do}(C_i = 0) + F_{\text{irrelevant}}^{\text{ext}}. \end{gathered} \tag{116}$$

An operational rescue prediction associated with H1.4 is:

$$\begin{gathered} F_i^{\text{ext}} \text{ selectively restores} \\ \text{the predicted downstream function.} \end{gathered} \tag{117}$$

Failure of rescue should trigger reconsideration of the proposed maintenance edge or reveal that C_i contributes through additional mechanisms.

4.16 Test 6: Budget-matched brute-force maintenance

Reciprocal organisation could appear successful merely because it has access to more maintenance activity.

Construct a brute-force architecture receiving an equal or larger total maintenance budget but lacking the hypothesised maintenance topology.

A serious null therefore satisfies approximately

$$\sum_i m_i^{\text{null}}(t) \approx \sum_i m_i^D(t) \tag{118}$$

over matched trajectories.

If the brute-force null matches Architecture D across viability, recovery and perturbation response,

then reciprocal organisation has not demonstrated independent explanatory gain.

4.17 Test 7: Boundary sensitivity

Organisational closure depends upon which components are classified as internal.

Evaluate at least two independently defensible boundaries:

$$\mathcal{B}_1 \subset \mathcal{B}_2. \tag{119}$$

Recalculate:

$$G_{\mathcal{C}}, \qquad \chi_{\text{int}}, \tag{120}$$

under both.

An operational boundary-robustness prediction associated with H1.5 is:

$$\begin{gathered}\text{the principal causal maintenance core survives} \\ \text{reasonable boundary variation.}\end{gathered} \tag{121}$$

If closure appears only under one specially chosen boundary, its interpretation should be restricted.

4.18 Test 8: Timescale sensitivity

Candidate maintenance relations should be evaluated relative to the processes whose continued operation they are proposed to support.

Let τ_C denote the characteristic persistence timescale of the candidate maintenance relation, and let τ_P denote the characteristic timescale of the process whose viability or operation is being maintained.

For each specified relation, define the dimensionless persistence ratio

$$\phi_K = \frac{\tau_C}{\tau_P}. \tag{122}$$

Sampling interval and disturbance timescale should then be varied.

The purpose is not to confirm

$$\phi_K \gg 1. \tag{123}$$

Instead, the experiment tests whether the apparent maintenance relation remains meaningful across scientifically reasonable temporal resolutions.

A result that exists only under one arbitrary sampling interval should be restricted to that

representation.

4.19 Test 9: Bypass search

The strongest challenge to Transition Problem 1 is to search for a system that displays internally generated maintenance without the proposed reciprocal architecture.

Candidate bypass systems may include:

1. highly robust feed-forward repair;
2. centralised self-repair;
3. redundant modular repair without strong connectivity;
4. externally trained but internally executed maintenance policies.

If such an architecture matches or exceeds Architecture D under all decisive tests, then strong organisational closure is not necessary for the artificial self-maintenance capacity being measured.

This would not show that closure is biologically irrelevant.

It would show:

$$\boxed{\text{closure-like reciprocal maintenance} \not\Rightarrow \text{universal necessity for self-maintenance.}} \tag{124}$$

4.20 Primary hypotheses

The block therefore tests the following claims:

Table 2: Primary hypotheses and break conditions for Experimental Block I.

ID	Hypothesis	Break condition
H1.1	Active maintenance provides a selective advantage over passive stability when disturbances require continuing correction.	Passive systems match active systems across preregistered disturbance regimes after degradation rates are properly controlled.
H1.2	Internally maintained systems retain greater operational viability than externally maintained systems after external support is withdrawn.	External and internal systems show equivalent autonomous persistence after matched support withdrawal.
H1.3	Validated reciprocal maintenance topology predicts propagation of targeted component failure better than simpler graph or capacity measures.	Closure topology adds no predictive power beyond component degree, size, energy use or feed-forward dependency.

ID	Hypothesis	Break condition
H1.4	Specific functional replacement rescues deficits caused by removal of a maintenance component.	Matched replacement fails systematically or irrelevant replacement performs equally well.
H1.5	The principal maintenance organisation remains recognisable under reasonable changes of system boundary and temporal resolution.	Closure-like structure appears only under one specially selected boundary or sampling regime.
H1.6	At least some internally maintained systems exhibit operational persistence or recovery that cannot be reduced to passive stability, external support or maintenance budget alone.	Serious null architectures reproduce all decisive outcomes without the proposed internal maintenance structure.

4.21 Pre-registration

The following should be frozen before decisive evaluation:

1. component definitions;
2. viability thresholds;
3. system boundaries;
4. disturbance distributions;
5. maintenance budgets;
6. architecture classes;
7. ablation schedule;
8. rescue conditions;
9. graph-edge threshold ϵ_w;
10. random-seed procedure;
11. primary outcome measures;
12. statistical models;
13. retain, restrict, reduce, suspend and reject criteria.

Exploratory tuning may occur before preregistration, but the final hypotheses must be evaluated on held-out runs.

4.22 Statistical analysis

Primary comparisons should use multiple independent random seeds and report uncertainty rather than relying upon single-run trajectories.

Where survival or time-to-failure is central, the analysis should compare failure-time distributions rather than only mean terminal performance.

Where perturbation response is central, effect sizes should be reported for:

$$\Delta V^{\mathrm{path}}(T), \qquad \Delta V^{\mathrm{term}}(T), \tag{125}$$

together with:

$$\Delta \tau_{\mathrm{rec}}. \tag{126}$$

Multiple-comparison correction should be specified for families of node-level ablation tests.

No post-hoc aggregate organisation score should be constructed.

4.23 Decision framework

The candidate transition

$$\text{persistent organisation} \longrightarrow \text{internally maintained organisation} \tag{127}$$

should receive one of five outcomes.

Retain if internally generated maintenance has reproducible causal effects that survive matched passive, external and budget controls.

Restrict if those effects occur only under specified disturbance regimes, architectures, boundaries or timescales.

Reduce if simpler control, stability or repair models reproduce the same results without an independent organisational construct.

Suspend if causal maintenance cannot be separated experimentally from generic architecture damage or insufficient statistical power.

Reject if the proposed maintenance dependency repeatedly fails or robust bypass architectures demonstrate that the claimed necessary organisation is absent.

4.24 What Block I cannot establish

Importantly, even a strong positive result would not establish:

$$\begin{gathered}\text{life},\\ \text{metabolism in the biological sense},\\ \text{evolutionary individuality},\\ \text{heredity},\\ \text{intelligence},\\ \text{consciousness}.\end{gathered} \tag{128}$$

A successful Block I would establish something narrower:

$$\boxed{\begin{gathered}\text{an artificial system can causally contribute}\\ \text{to maintaining the organisation required}\\ \text{for its continued operation.}\end{gathered}} \tag{129}$$

Whether the same causal architecture explains biological self-maintenance would remain a separate cross-domain hypothesis requiring the validation protocol developed in Paper II.

4.25 Provisional role in the wider architecture

Block I is therefore an experimental test of Transition Problem 1 rather than an attempted recreation of the origin of life.

A positive result would support:

$$\text{reciprocal or internal causal maintenance} \tag{130}$$

as one sufficient artificial route from persistence towards self-maintenance.

A negative result could be equally important. If passive stability, feed-forward repair or simpler control architectures explain all decisive outcomes, organisational closure should be reduced or restricted rather than preserved for theoretical elegance.

The governing interpretation is:

$$\boxed{\begin{gathered}\text{persistence is the starting observation;}\\ \text{self-maintenance is the causal claim;}\\ \text{the experiment must distinguish them.}\end{gathered}} \tag{131}$$

5 Experimental Block II: Heredity, Reproduction and Population Change

The second experimental block tests the transition from an individually maintained system to a population in which reproduction, hereditary transmission, variation and differential reproductive success are sufficiently well defined for selection to operate. The experiment does not begin by assuming that the presence of mutation and changing trait frequencies constitutes evolution by natural selection.

The block instead asks a sequence of logically prior questions:

$$\boxed{\begin{array}{c}\text{What is the reproducing entity?}\\ \downarrow\\ \text{Does reproduction produce identifiable descendants?}\\ \downarrow\\ \text{Is relevant structure transmitted?}\\ \downarrow\\ \text{Does transmitted structure vary?}\\ \downarrow\\ \text{Does that variation causally affect reproductive success?}\\ \downarrow\\ \text{Does differential reproduction alter population composition?}\end{array}} \tag{132}$$

Only after the earlier conditions are satisfied should the final process be described as selection within a heritable artificial population.

5.1 Primary experimental question

Block II asks:

> **Under what experimentally specified conditions does a collection of artificial agents become a heritable reproducing population in which differential reproductive success generates cumulative population change?**

The candidate transition is

$$\text{self-maintaining individual} \longrightarrow \boxed{?} \longrightarrow \text{Darwinian population.} \tag{133}$$

No claim is made that the artificial implementation reproduces the origin of biological reproduction or heredity.

5.2 Artificial ecological environment

The baseline environment is a two-dimensional stochastic world containing:

1. renewable but spatially heterogeneous resources;
2. local and temporally variable hazards;
3. energetic costs for movement and action;
4. environmental regimes changing on controlled timescales;
5. spatial structure limiting interaction and dispersal;
6. partial observability where required by the ecological question.

Environmental state is denoted

$$E_t, \tag{134}$$

and local observations available to individual i are

$$O_t^{(i)} = \mathcal{O}\left(E_t, p_t^{(i)}\right). \tag{135}$$

The ecological environment should be identical across the principal reproductive controls unless environmental variation itself is the manipulated factor.

5.3 Definition of the candidate individual

A population cannot be defined merely by counting whichever objects are easiest for the simulation to enumerate. The candidate individual must be specified before reproduction is analysed.

Let individual i possess state

$$X_t^{(i)} = \left(p_t^{(i)}, e_t^{(i)}, a_t^{(i)}, g_t^{(i)}, \varphi_t^{(i)}\right), \tag{136}$$

where:

- $p_t^{(i)}$ is spatial position;
- $e_t^{(i)}$ is energetic state;
- $a_t^{(i)}$ is age or developmental stage;
- $g_t^{(i)}$ is the transmissible inherited state;
- $\varphi_t^{(i)}$ is the expressed phenotype relevant to the experiment.

Memory and learning variables are deliberately excluded from the core Block II state unless a secondary experiment explicitly introduces them. Their causal roles belong to Block III.

5.4 Individual identity through time

An individual must remain identifiable through ordinary state change.

Define identity label

$$\iota_i \tag{137}$$

that remains associated with one agent from its creation until termination.

Ordinary changes in:

$$p_t^{(i)}, \qquad e_t^{(i)}, \qquad a_t^{(i)}, \qquad \varphi_t^{(i)} \tag{138}$$

do not by themselves create a new individual.

This prevents movement, growth or phenotype change from being mistaken for reproduction.

5.5 What counts as reproduction?

A reproductive event

$$R_i(t) = 1 \tag{139}$$

occurs only when an existing individual produces at least one newly instantiated descendant satisfying preregistered identity criteria.

For parent i and descendant j, reproduction requires:

$$\iota_j \neq \iota_i, \tag{140}$$

together with a recorded parent–offspring relation

$$i \to j. \tag{141}$$

The descendant must be capable, subject to the ordinary experimental conditions, of entering the same reproductive class as its parent.

Consequently,

$$\boxed{\text{growth} \neq \text{repair} \neq \text{state duplication} \neq \text{reproduction.}} \tag{142}$$

5.6 Lineage graph

All successful reproductive events should generate a directed lineage graph

$$G_{\text{lin}} = (V_{\text{ind}}, E_{\text{rep}}), \tag{143}$$

where vertices are individuals and directed edges are verified parent–descendant relations.

The graph permits direct measurement of:

$$\begin{gathered}\text{lineage persistence,} \qquad \text{descendant number,} \\ \text{extinction,} \qquad \text{branching,} \qquad \text{common ancestry.}\end{gathered} \tag{144}$$

Population change should therefore be reconstructed from actual genealogical events rather than inferred only from aggregate reward trajectories.

5.7 Energetic and viability conditions

Agents may reproduce only when independently specified resource and developmental conditions are satisfied.

A simple preregistered criterion is

$$e_t^{(i)} \geq e_{\text{rep}}, \tag{145}$$

together with

$$a_t^{(i)} \geq a_{\text{rep}}. \tag{146}$$

Reproduction incurs energetic cost

$$c_{\text{rep}} > 0. \tag{147}$$

Agents terminate when

$$e_t^{(i)} \leq 0 \tag{148}$$

or another preregistered viability criterion fails.

These rules define the artificial ecology. They are not claimed to reproduce biological metabolism.

5.8 Hereditary transmission

Offspring state must depend reproducibly upon parental transmissible state.

Let parent genotype be

$$G_p \tag{149}$$

and offspring genotype

$$G_o. \tag{150}$$

Transmission is represented by conditional kernel

$$T_{\text{hered}}\left(g_o \mid g_p\right). \tag{151}$$

A simple mutation model is

$$g_o = \mathcal{M}_\mu(g_p), \tag{152}$$

where μ controls mutation or transmission error.

The experiment should not infer heredity merely because parent and offspring resemble one another in a shared environment.

5.9 A statistical heredity diagnostic

One statistical diagnostic of transmission is

$$I\left(G_p; G_o \mid E\right) > 0, \tag{153}$$

where conditioning on relevant environmental variables helps distinguish parent–offspring transmission from resemblance generated by shared external conditions.

This condition is not sufficient by itself.

The simulation architecture must also contain an explicit causal transmission path:

$$G_p \longrightarrow G_o. \tag{154}$$

Thus,

$$\boxed{\text{parent–offspring resemblance} \neq \text{heredity without transmission.}} \tag{155}$$

5.10 Transmission fidelity and variation

Evolution requires neither perfect copying nor unrestricted randomisation.

Define transmission fidelity

$$q = P\left(G_o = G_p\right) \tag{156}$$

for exactly copied states where that definition is appropriate.

More generally, trait-specific fidelity can be reported as

$$q_k = P\left(G_{o,k} = G_{p,k}\right). \tag{157}$$

Mutation rate μ and fidelity q should be varied independently where the representation permits.

The experiment does not presuppose one universal critical fidelity

$$q^*. \tag{158}$$

Instead, it asks whether each tested hereditary architecture exhibits a fidelity–variation region within which inherited structure persists while novel variants continue to arise.

5.11 Phenotype mapping

Selection on inherited variation can be interpreted only if the relation between transmissible state and expressed phenotype is specified.

Let

$$\varphi = F(g, E) \tag{159}$$

define the phenotype generated by inherited state g in environment E.

The mapping may be deterministic or stochastic.

The distinction between inherited state and expressed phenotype should remain explicit:

$$\boxed{\text{genotype} \neq \text{phenotype.}} \tag{160}$$

Selection can act upon phenotypic consequences while heredity operates through the transmissible variables.

Keeping those levels separate prevents acquired or environmentally induced phenotypic states from being misclassified automatically as inherited variation.

5.12 Population representation

At the population level, let

$$p_t(g) \tag{161}$$

denote the frequency distribution of inherited variants among individuals belonging to the preregistered reproducing population at generation or evaluation time t.

Where generations overlap, calendar time and reproductive generation should both be recorded rather than forcing the system into artificial discrete generations.

Population size is

$$N_t = |\mathcal{I}_t|, \tag{162}$$

where $\mathcal{I}_t$ is the set of living candidate individuals at time t.

5.13 Reproductive success

Define the reproductive success of individual i over horizon T as

$$w_i(T) = N_i^{\text{desc}}(T), \tag{163}$$

where $N_i^{\text{desc}}(T)$ is a preregistered measure of viable direct offspring or descendant contribution.

Alternative definitions such as direct offspring number, lineage growth or long-term descendant representation should not be mixed within the same confirmatory analysis.

The relevant measure must be selected before observing which definition gives the strongest association with the focal trait.

5.14 Experimental architectures

Comparatively, Block II should contain at least five matched population conditions.

Architecture E0: Reproduction without heredity. Parents produce descendants, but offspring transmissible states are sampled independently from a fixed reference distribution:

$$G_o \perp G_p. \tag{164}$$

This establishes reproduction while eliminating hereditary transmission.

Architecture E1: Heredity with random reproductive allocation. Offspring inherit parental states through

$$T_{\text{hered}}\left(g_o \mid g_p\right), \tag{165}$$

but reproductive opportunities are allocated independently of focal phenotype or performance.

This permits heredity and mutation without fitness-sensitive reproduction.

Architecture E2: Heredity plus differential reproduction. Inherited variants influence phenotype and thereby alter expected reproductive success under the specified ecology.

This is the principal candidate selection condition.

Architecture E3: Random search without lineage selection. New candidate genotypes are generated according to a matched search or mutation process, but successful variants receive no additional hereditary representation through differential reproduction.

This tests whether apparent adaptation results merely from exploring genotype space.

Architecture E4: Within-lifetime optimisation without inheritance. Agents may improve performance during their own lifetime, but acquired policy changes are reset rather than transmitted to descendants.

This condition is retained only as a control distinguishing population change from non-hereditary within-lifetime adaptation. Detailed learning mechanisms remain the subject of Block III.

5.15 Test 1: Does reproduction exist independently of growth?

The first experiment verifies that the simulation generates genuinely distinct descendants rather than enlarged or copied states of one persistent entity.

The primary criterion is the existence of reproducible parent–offspring events:

$$i \to j \tag{166}$$

with

$$\iota_i \neq \iota_j. \tag{167}$$

The descendant must itself retain reproductive eligibility under the ordinary rules of the ecology.

If this criterion cannot be made non-arbitrary, subsequent selection analysis should be suspended.

5.16 Test 2: Does heredity survive environmental controls?

Comparing E0 with E1–E2.

A genuine hereditary relation should produce:

$$I\left(G_p; G_o \mid E\right) > 0 \tag{168}$$

and predictable changes when the transmission channel is perturbed.

Randomising parental genotype labels should destroy the association.

Disrupting the implemented transmission channel should selectively reduce parent–offspring dependence.

An operational prediction associated with H2.1 is:

$$\begin{gathered}\text{parental transmissible state causally predicts} \\ \text{and influences offspring transmissible state.}\end{gathered} \tag{169}$$

5.17 Test 3: Is variation heritable rather than merely recurrent?

Introduced or spontaneous variant g' should be traced through the lineage graph.

A variant qualifies as heritable variation only if:

$$G_p = g' \tag{170}$$

increases the probability that descendants carry g' or its specified derived state relative to the no-transmission baseline.

Repeated independent appearance of the same state through mutation is not equivalent to

hereditary persistence.

Thus,

$$\boxed{\text{recurrent variation} \neq \text{retained lineage variation.}} \tag{171}$$

5.18 Test 4: Does the focal trait causally affect reproductive success?

Causally, a statistical association between trait z and reproductive success w is insufficient.

Where intervention is meaningful, compare:

$$\begin{aligned} &\mathbb{E}\left[w \mid \mathrm{do}(z = z_1), E\right] \\ &\quad \neq \mathbb{E}\left[w \mid \mathrm{do}(z = z_2), E\right]. \end{aligned} \tag{172}$$

Genetic or phenotypic interventions should preserve unrelated capacities as far as possible.

An operational prediction associated with H2.2 is:

$$\begin{gathered} \text{specified inherited variants produce reproducible} \\ \text{differences in expected reproductive success.} \end{gathered} \tag{173}$$

Failure here prevents a strong causal-selection interpretation even if trait frequencies later change.

5.19 Test 5: Does differential reproduction alter population composition?

Compare E2 with E1.

Let $p_t(g)$ denote the population frequency of inherited variant g at time t, as defined above, and let $\bar{w}_g(t)$ denote the mean reproductive success of individuals carrying that variant under the specified ecology.

Before transmission and mutation are applied, the selection-associated reweighting of variant g can be represented schematically as

$$p_g^{\mathrm{sel}}(t+1) = \frac{p_t(g)\,\bar{w}_g(t)}{\bar{w}(t)}, \tag{174}$$

where mean population reproductive success is

$$\bar{w}(t) = \sum_g p_t(g)\,\bar{w}_g(t). \tag{175}$$

This expression isolates the frequency change associated with differential reproductive success. The complete offspring distribution may additionally depend upon the preregistered heredity and mutation kernels.

The experimental question is whether fitness-sensitive reproduction produces population change

beyond that generated by reproduction, heredity, mutation and finite sampling alone.

A candidate divergence is

$$\Gamma_{\mathrm{sel}}(t) = D_{\mathrm{KL}}\left(p^{\mathrm{obs}}(t+1) \| p^{\mathrm{neutral}}(t+1)\right), \tag{176}$$

where the neutral reference includes the matched non-selective processes.

This remains a model-comparison quantity, not a universal constraint score.

5.20 Test 6: Price-equation decomposition

Population change should also be analysed through the Price equation [7]:

$$\Delta \bar{z} = \frac{1}{\bar{w}} \operatorname{Cov}(w_i, z_i) + \frac{1}{\bar{w}} \mathbb{E}\left[w_i \Delta z_i\right] \tag{177}$$

The first term identifies change associated with differential reproductive success; the second captures change between ancestral and descendant trait values.

This prevents mutation, transformation or transmission effects from being silently attributed to selection.

The covariance term should not by itself be interpreted as proof that z causes reproductive success. That stronger claim belongs to Test 4.

5.21 Test 7: Selection versus random search

Compare E2 with E3 using matched numbers of newly generated variants and comparable computational budgets.

If random exploration of genotype space produces the same rate and durability of adaptation as differential hereditary reproduction, then selection has not demonstrated independent explanatory gain for that task.

An operational comparison associated with H2.3 is:

$$\begin{gathered}\text{fitness-sensitive hereditary reproduction produces} \\ \text{cumulative population adaptation beyond} \\ \text{matched random search.}\end{gathered} \tag{178}$$

The comparison should be performed on held-out environments as well as the selection environment to prevent temporary optimisation from being mistaken for robust adaptation.

5.22 Test 8: Selection versus within-lifetime optimisation

Comparing E2 with E4.

Within-lifetime optimisation may increase individual performance rapidly without producing inherited population change.

Conversely, hereditary selection may change population composition while individual agents possess no learning capacity.

The experiment therefore separates:

$$\boxed{\text{within-lifetime adaptation} \neq \text{cross-generation selection.}} \tag{179}$$

Any architecture combining both processes should be analysed separately after the isolated mechanisms have been established.

5.23 Test 9: Environmental switching

Environmentally, selected populations should be exposed to preregistered regime changes after adaptation has occurred.

Let the environment switch:

$$E_A \longrightarrow E_B. \tag{180}$$

Measure:

$$\begin{gathered} \text{trait-frequency change}, \qquad \text{population viability}, \\ \text{lineage turnover}, \qquad \text{time to renewed adaptation}. \end{gathered} \tag{181}$$

The purpose is not to assume that previously selected populations should always perform better.

A population may instead exhibit:

$$\begin{gathered} \text{robust adaptation}, \qquad \text{specialisation}, \\ \text{extinction}, \qquad \text{reversal of selective direction}. \end{gathered} \tag{182}$$

Such outcomes directly test the rejected idea that selection represents monotonic accumulation of constraint or complexity.

5.24 Test 10: Fidelity–variation sweep

Vary mutation or transmission error across a preregistered range:

$$\mu_1, \mu_2, \ldots, \mu_m. \tag{183}$$

For each regime, measure:

$$\begin{gathered}\text{lineage persistence}, \qquad \text{inherited information}, \\ \text{population diversity}, \qquad \text{adaptation rate}.\end{gathered} \tag{184}$$

The predicted relationship need not be monotonic.

Very low variation may impede exploration, while excessive transmission error may prevent inherited structure from persisting.

The experiment seeks a system-specific fidelity–variation regime rather than a universal value q^*.

5.25 Test 11: Alternative reproducing units

Test 11 challenges the interpretation of selection by changing the candidate unit of reproduction.

Where the simulation permits nested structures, compare whether the relevant hereditary and fitness relations belong most naturally to:

$$\begin{gathered}\text{individual agents}, \qquad \text{sub-agent replicators}, \\ \text{groups}, \qquad \text{persistent compartments}.\end{gathered} \tag{185}$$

If different unit definitions generate contradictory selection claims, the population ontology is underdetermined.

This test operationalises the Paper I warning that a population cannot be defined merely by counting whichever entities are visually salient.

5.26 Bypass search

Block II should actively search for systems showing population-level adaptation without the full proposed sequence.

Candidate bypasses include:

1. population improvement through random search without heredity;
2. inherited improvement without differential reproduction;
3. persistent adaptive traits generated by environmental induction rather than parent–offspring transmission;
4. optimisation by within-lifetime learning without cross-generation change.

If any bypass reproduces the decisive population-level outcome, the claimed necessity of the missing mechanism must be rejected or restricted.

5.27 Primary hypotheses

Block II tests the following claims:

Table 3: Primary hypotheses and break conditions for Experimental Block II.

ID	Hypothesis	Break condition
H2.1	Parent–offspring transmission creates reproducible hereditary dependence that survives environmental controls.	Parent–offspring resemblance disappears after controlling for shared environment or perturbing the proposed transmission pathway.
H2.2	Specified inherited variation causally alters expected reproductive success under at least some preregistered ecological conditions.	Trait–fitness covariance disappears under intervention or is explained by confounding variables.
H2.3	Fitness-sensitive hereditary reproduction generates cumulative population change beyond matched neutral or random reproduction, mutation and finite-sampling effects.	Neutral or random-search controls reproduce the decisive population changes with comparable efficiency and persistence.
H2.4	Population adaptation depends systematically upon the balance between transmission fidelity and generation of variation.	Adaptation shows no reproducible dependence upon fidelity or variation across the tested regime after other factors are controlled.
H2.5	Selection effects remain identifiable under independently defensible definitions of the reproducing unit.	Selection appears or disappears primarily because the unit of reproduction is chosen post hoc.
H2.6	Environmental switching reveals historically structured population response rather than monotonic accumulation of adaptation.	Selected and matched non-selected populations respond identically across the preregistered switching conditions.

5.28 Primary measurements

Block II should report:

1. verified reproduction-event count;
2. lineage depth and branching;
3. parent–offspring transmission fidelity;

4. conditional parent–offspring information;
5. genotype and phenotype frequency distributions;
6. reproductive-success distributions;
7. trait–fitness covariance;
8. Price-equation terms;
9. population size and extinction probability;
10. diversity under specified representation;
11. adaptation and re-adaptation times;
12. divergence from matched neutral population dynamics.

No total evolutionary constraint score should be constructed.

5.29 Pre-registration

The following should be fixed before decisive evaluation:

1. definition of the individual;
2. reproduction criterion;
3. parent–offspring mapping;
4. transmissible variables;
5. mutation operator;
6. phenotype mapping;
7. reproductive-success measure;
8. environmental regimes;
9. population boundaries;
10. comparison architectures;
11. intervention schedule;
12. random-seed procedure;
13. primary statistical tests;
14. break conditions.

Changing the reproducing unit, fitness measure or heredity criterion after observing the population trajectory should generate a new exploratory analysis rather than a confirmatory reinterpretation.

5.30 Decision framework

The transition

$$\text{self-maintaining individual} \longrightarrow \text{heritable reproducing population} \tag{186}$$

and the subsequent operation of selection should receive separate judgements.

Retain the reproduction–heredity transition if independently identified individuals generate descendants with causally validated parent–offspring transmission.

Restrict it if heredity or reproduction depends strongly upon one specific representation, environmental regime or choice of reproducing unit.

Reduce population adaptation if matched random search, environmentally induced change or within-lifetime optimisation explains the same outcome without the proposed hereditary-selection mechanism.

Suspend the claim if individual identity, lineage structure or causal fitness differences cannot be determined adequately.

Reject any strict dependency that is defeated by a robust bypass architecture lacking the supposedly necessary mechanism.

Natural selection itself is retained only where reproduction, heredity, variation and differential reproductive success have each been established independently.

5.31 What Block II cannot establish

Importantly, even a complete positive result would not establish:

$$\begin{gathered}\text{the historical origin of biological reproduction,}\\ \text{universal units of selection,}\\ \text{increasing evolutionary complexity,}\\ \text{progress towards intelligence,}\\ \text{life in every scientifically relevant sense.}\end{gathered} \tag{187}$$

Block II would establish the narrower result that an artificial population can instantiate reproducible parent–offspring transmission, heritable variation and differential reproductive success under controlled ecological conditions.

Whether the same mathematical relations transport to biological populations is a separate cross-domain question.

5.32 Provisional role in the wider architecture

Block II tests Transition Problem 2 and Effective Mechanism 2 from Paper I.

A positive result would support:

$$\boxed{\begin{array}{c}\text{reproduction} + \text{heredity} + \text{variation} + \text{differential reproductive success} \\ \longrightarrow \text{selection-driven artificial population change.}\end{array}} \tag{188}$$

It would not show that self-maintenance inevitably generates reproduction, or that selection necessarily produces learning, agency or intelligence.

The governing interpretation is:

$$\boxed{\begin{array}{c}\text{first identify what reproduces;} \\ \text{then establish what is inherited;} \\ \text{only then ask what selection changes.}\end{array}} \tag{189}$$

6 Experimental Block III: Retention, Feedback and Learning

The third experimental block tests three causal channels that are frequently grouped together under broad labels such as memory, regulation or adaptation. They must nevertheless be distinguished carefully.

Causally consequential retention in the broad sense includes any persistent effect of earlier history on later behaviour. Learning therefore already contains a form of historical retention when experience becomes incorporated into changed policy parameters. For that reason, the factor R used in the present factorial experiment must not be interpreted as "all historical retention".

Instead, the principal design separates three experimentally manipulable channels:

$$\boxed{\begin{array}{c}R = \text{policy-accessible retained state,} \\ F = \text{within-episode closed-loop feedback,} \\ L = \text{between-episode experience-dependent learning.}\end{array}} \tag{190}$$

Here R refers specifically to an internal state that carries task-relevant information across a defined history–decision boundary and remains available to the policy during subsequent control. F refers to the ability of observations arriving after the decisive control interval begins to modify ongoing action. L refers to persistent modification of policy parameters from experience across designated adaptation episodes.

Thus,

$$\boxed{\begin{array}{c} R = 0 \not\Rightarrow \text{absence of every historical influence,} \\ L = 1 \Rightarrow \text{past experience may persist in learned parameters.} \end{array}} \tag{191}$$

The factorial design therefore tests separable causal channels at specified timescales rather than claiming that retention, feedback and learning are ontologically independent processes.

The block continues to reject the assumed sequence

$$\text{memory} \longrightarrow \text{feedback} \longrightarrow \text{learning} \tag{192}$$

as an experimental starting point.

The central question is not whether systems containing all three channels perform best. Greater architectural capacity could make that result trivial. The experiment asks what each channel contributes, under which environmental conditions it matters, and whether any proposed dependency disappears under matched alternatives.

6.1 Primary experimental questions

Block III asks three principal questions.

First:

> **Does policy-accessible retained information about earlier states causally improve behaviour specifically when the current observable state is insufficient?**

Second:

> **Does observation-dependent online action provide an advantage specifically when unpredictable disturbances require correction during the current control interval?**

Third:

> **Does experience-dependent modification of policy parameters across adaptation episodes produce persistent behavioural change that cannot be explained by fixed feedback control, policy-accessible retained state or additional architectural capacity alone?**

These questions correspond to different intervention channels and receive separate verdicts.

6.2 Agent state, timescales and notation

Two timescales are distinguished explicitly.

Let e index adaptation episodes and let t denote time within one episode. The artificial agent receives observation

$$O_{e,t}, \tag{193}$$

selects action

$$A_{e,t}, \tag{194}$$

and may possess policy-accessible retained state

$$M_{e,t}. \tag{195}$$

Policy parameters during episode e are denoted

$$\theta_e. \tag{196}$$

For the principal factorial experiment, θ_e remains fixed throughout each decisive episode. Learning, when enabled, changes parameters only at a preregistered adaptation boundary between episodes.

The environment evolves according to

$$X_{e,t+1} \sim K\left(\cdot \mid X_{e,t}, A_{e,t}, E_{e,t}\right), \tag{197}$$

where $X_{e,t}$ is the underlying environment state and $E_{e,t}$ denotes relevant disturbance or regime variables.

Observation is generated by

$$O_{e,t} \sim G\left(\cdot \mid X_{e,t}\right). \tag{198}$$

Some tasks also contain a history-acquisition interval before the decisive control interval begins. Information encountered during that earlier interval may be carried into the control interval through

$$M_{e,0} \tag{199}$$

when $R = 1$.

This temporal separation is important because it permits retained history, online feedback and slower learning to be manipulated without defining one as the absence of all influence from the others.

6.3 Retention mechanism

A retention-enabled architecture maintains an internal state:

$$M_{e,t} = f_{\theta_e}\left(M_{e,t-1}, O_{e,t}\right). \tag{200}$$

When feedback is also enabled, action may depend upon both the current observation and retained state:

$$A_{e,t} \sim \pi_{\theta_e}\left(\cdot \mid O_{e,t}, M_{e,t}\right). \tag{201}$$

The $R = 0$ control receives no policy-accessible $M_{e,t}$:

$$A_{e,t} \sim \pi_{\theta_e}\left(\cdot \mid O_{e,t}\right) \tag{202}$$

when feedback is enabled.

Capacity matching should provide the $R = 0$ architecture with additional parameters or hidden units where necessary so that performance differences cannot be attributed simply to model size.

Crucially, R concerns the explicit retained-state channel being tested. A learned parameter vector θ_e may itself contain information about earlier experience. Such parameter retention belongs to the learning mechanism L in the factorial design and must not be mistaken for $R = 1$.

Decodability of past information from $M_{e,t}$ is not sufficient to establish functional retention.

The stronger condition is:

$$\boxed{\begin{array}{c}\text{past distinction} \longrightarrow \text{policy-accessible retained state} \\ \longrightarrow \text{causal change in future behaviour.}\end{array}} \tag{203}$$

6.4 Feedback mechanism

A closed-loop controller allows observations arriving during the decisive control interval to affect ongoing action. This use of feedback follows the cybernetic tradition of analysing regulation through ongoing relations between system state, disturbance, observation and action [8].

With $R = 0$:

$$A_{e,t} \sim \pi_{\theta_e}\left(\cdot \mid O_{e,t}\right). \tag{204}$$

With $R = 1$:

$$A_{e,t} \sim \pi_{\theta_e}\left(\cdot \mid O_{e,t}, M_{e,t}\right). \tag{205}$$

The controller is not given the raw sequence $O_{e,\leq t}$ unless that history is encoded through the explicitly manipulated retained-state channel. Otherwise the $R = 0$ condition would receive historical information by another route.

An open-loop condition instead commits to an action sequence or control rule using only information available before the decisive control interval begins:

$$A_{e,0:T} \sim \pi_{\text{open}} \left(\cdot \mid I_{e,0} \right). \tag{206}$$

Where $R = 1$, the initial information set may include retained state:

$$I_{e,0} = \left(I_{e,0}^{\text{ext}}, M_{e,0} \right). \tag{207}$$

No observation arriving after the control interval begins may alter the precommitted action sequence when $F = 0$.

A yoked-replay condition uses an action sequence previously generated by a successful closed-loop system:

$$A_{e,0:T} = A_{e,0:T}^{\text{yoked}}, \tag{208}$$

but prevents newly arriving observations from modifying that sequence.

This preserves action magnitude and temporal structure while breaking the online observation–action relation.

6.5 Learning mechanism

Experience-dependent learning is defined as persistent modification of the behavioural mapping across designated adaptation episodes.

Let the experience collected during episode e be

$$\mathcal{D}_e = \{ O_{e,t}, A_{e,t}, Y_{e,t} \}_{t=0}^{T}, \tag{209}$$

where $Y_{e,t}$ may contain outcome, prediction error, reward, viability signal or another preregistered learning input.

A learning-enabled architecture updates:

$$\theta_{e+1} = \mathcal{L} \left(\theta_e, \mathcal{D}_e \right). \tag{210}$$

A fixed-policy control satisfies:

$$\theta_{e+1} = \theta_e. \tag{211}$$

For the principal factorial comparison, these updates occur between episodes. Within a decisive episode:

$$\theta_e = \text{constant}. \tag{212}$$

This prevents learning from functioning as an unlabelled second within-episode feedback channel.

The distinction is therefore:

$$\boxed{\begin{array}{c}\text{retention carries selected history into policy access;}\\ \text{feedback changes action from newly arriving observations;}\\ \text{learning changes the future policy from accumulated experience.}\end{array}} \tag{213}$$

Learning still constitutes causally consequential historical influence in the broader sense. The factorial design separates where that historical influence is stored and when it becomes available to behaviour.

6.6 Factorial architecture

The principal design treats R, F and L as binary intervention factors under the temporal definitions above.

The full design contains up to

$$2^3 = 8 \tag{214}$$

conditions.

Table 4: Factorial architecture for Experimental Block III.

Condition	R	F	L	**Interpretive role**
R0F0L0	Off	Off	Off	Fixed open-loop reference with no policy-accessible retained state
R1F0L0	On	Off	Off	Tests whether retained pre-control history can improve a precommitted response without new within-episode feedback
R0F1L0	Off	On	Off	Tests current-observation feedback without policy-accessible retained state
R1F1L0	On	On	Off	Tests feedback that can use both current observation and retained state, without learning
R0F0L1	Off	Off	On	Tests between-episode policy adaptation without policy-accessible retained state or within-episode feedback
R1F0L1	On	Off	On	Tests between-episode learning plus retained pre-control history while the decisive episode remains open-loop

Condition	R	F	L	Interpretive role
R0F1L1	Off	On	On	Tests between-episode learning plus current-observation feedback without policy-accessible retained state
R1F1L1	On	On	On	Full factorial condition; tests interactions after the individual channels have been identified

The factors are experimentally separable because they refer to different access channels and timescales. They should not be described as mutually exclusive forms of historical influence.

Not every task will make all eight conditions scientifically meaningful. Undefined combinations should be marked not applicable rather than assigned zero performance.

6.7 Environment families

The environment suite should vary three properties independently where possible:

1. whether relevant historical information is required;
2. whether unpredictable online disturbance is present;
3. whether the behavioural mapping must change through experience.

This produces a task matrix rather than one increasingly difficult benchmark.

Let $\mathcal{H}_{e,t}$ denote the task-relevant history available before the current decision in episode e.

Environment M: Present-state Markov control. The current observation contains all information required for optimal action:

$$P\left(X_{e,t+1} \mid O_{e,t}, \mathcal{H}_{e,t}\right) = P\left(X_{e,t+1} \mid O_{e,t}\right) \tag{215}$$

for the relevant task variables.

Retention should provide little selective advantage here.

Environment H: History-dependent control. The current observation is deliberately ambiguous and optimal action depends upon earlier information:

$$P\left(A^*_{e,t} \mid O_{e,t}, \mathcal{H}_{e,t}\right) \neq P\left(A^*_{e,t} \mid O_{e,t}\right). \tag{216}$$

Examples include delayed cues, hidden regime states and previously observed hazards no longer visible at decision time.

Environment P: Predictable disturbance. Disturbances follow a known or repeatable temporal structure that can be handled by a precomputed action sequence.

Closed-loop feedback should provide limited advantage over appropriately matched open-loop control.

Environment U: Unpredictable disturbance. Disturbance timing or location varies after the episode begins and cannot be known from I_0.

Online feedback should become selectively useful.

Environment S: Stable behavioural mapping. The action–outcome relation remains unchanged across the relevant evaluation period.

A fixed well-trained policy may be sufficient.

Environment C: Changed behavioural mapping. A preregistered causal or task relation changes after calibration:

$$K_A \longrightarrow K_B. \tag{217}$$

Experience-dependent policy modification should become selectively relevant.

6.8 Matched task construction

Decisive task pairs should differ in the need for the candidate mechanism while remaining similar in superficial difficulty.

For retention:

$$E_{\text{history}} \quad \text{vs} \quad E_{\text{Markov}}. \tag{218}$$

For feedback:

$$E_{\text{unpredictable}} \quad \text{vs} \quad E_{\text{predictable}}. \tag{219}$$

For learning:

$$E_{\text{changed}} \quad \text{vs} \quad E_{\text{stable}}. \tag{220}$$

A mechanism earns specific credit only when its advantage is larger in the environment designed to require its hypothesised function.

6.9 Test 1: Retention specificity

Compare retention-enabled and capacity-matched retention-ablated architectures in E_{history} and E_{Markov}.

Define performance difference

$$\Delta P_R(E) = P_{R=1}(E) - P_{R=0}(E). \tag{221}$$

An operational prediction associated with H3.1 is:

$$\Delta P_R\left(E_{\text{history}}\right) > \Delta P_R\left(E_{\text{Markov}}\right). \tag{222}$$

A generic advantage in both task families would suggest additional capacity rather than a specifically historical function.

6.10 Test 2: Retained information versus causal use

The retained state should be manipulated after the relevant history has occurred but before the decisive action.

Candidate interventions include:

1. state erasure;
2. permutation between trials;
3. controlled substitution with another valid historical state;
4. temporal scrambling;
5. insertion of known incorrect retained information.

For episode e, a causal retention effect requires:

$$\begin{aligned} &P\left(A_{e,t+\Delta} \mid \text{do}(M_{e,t} = m_1)\right) \\ &\quad \neq P\left(A_{e,t+\Delta} \mid \text{do}(M_{e,t} = m_2)\right) \end{aligned} \tag{223}$$

for retained states m_1 and m_2 that differ in a preregistered task-relevant historical distinction.

An operational prediction associated with H3.2 is:

$$\begin{gathered} \text{selective manipulation of retained state produces} \\ \text{predictable changes in history-dependent behaviour.} \end{gathered} \tag{224}$$

Decoding historical information from $M_{e,t}$ without this causal effect does not establish causally consequential retention.

6.11 Test 3: Retention profile

Retention should be measured over several preregistered delays.

Let $H_{e,t-\tau}$ denote a task-relevant historical variable, event or distinction occurring at lag τ within episode e. Let $Z_{e,t}$ denote the present-state variables on which the diagnostic is conditioned.

A statistical retention diagnostic is:

$$R(\tau) = I\left(M_{e,t}; H_{e,t-\tau} \mid Z_{e,t}\right). \tag{225}$$

This quantity measures whether the retained state contains information about the nominated historical distinction after conditioning on the specified present-state variables.

A causal counterpart should quantify how intervention on the retained state changes a preregistered later performance or behavioural outcome. Let $P_{e,t+\Delta}$ denote that outcome at future effect horizon Δ. For two experimentally relevant retained states m_1 and m_2, define:

$$\begin{aligned} R_{\text{causal}}(\Delta) = \Big| & \mathbb{E}\left[P_{e,t+\Delta} \mid \text{do}(M_{e,t} = m_1)\right] \\ & - \mathbb{E}\left[P_{e,t+\Delta} \mid \text{do}(M_{e,t} = m_2)\right] \Big|. \end{aligned} \tag{226}$$

The nominated states and outcome must be specified before decisive evaluation. If more than two retained-state interventions are scientifically relevant, the corresponding contrast or divergence measure should be preregistered rather than selected after inspection.

The experiment should not assume:

$$R(\tau) = R_{\text{causal}}(\Delta). \tag{227}$$

Stored historical information can persist after its behavioural relevance has disappeared. Conversely, a small information-theoretic diagnostic under one representation does not by itself prove that historical influence is absent.

6.12 Test 4: Closed-loop feedback specificity

Compare genuine closed-loop control with matched open-loop control and yoked replay.

Define

$$\Delta P_F(E) = P_{\text{closed}}(E) - P_{\text{open}}(E). \tag{228}$$

An operational prediction associated with H3.3 is:

$$\Delta P_F\left(E_{\text{unpredictable}}\right) > \Delta P_F\left(E_{\text{predictable}}\right). \tag{229}$$

A stronger comparison requires:

$$P_{\text{closed}} > P_{\text{yoked}} \tag{230}$$

under disturbances differing from those that generated the original yoked sequence.

This isolates the causal value of online correction.

6.13 Test 5: Feedback-channel perturbation

Manipulate the observation–action channel while preserving actuator capacity.

Candidate manipulations include:

1. observation delay;
2. observation permutation;
3. partial masking;
4. controlled sensor noise;
5. replay of stale observations.

Let feedback delay be

$$\delta. \tag{231}$$

Performance should be measured across several values of

$$\frac{\delta}{\tau_P}, \tag{232}$$

where τ_P is the characteristic process timescale.

An operational prediction associated with H3.4 is:

$$\begin{gathered}\text{feedback effectiveness degrades predictably when} \\ \text{observation delay becomes dynamically consequential.}\end{gathered} \tag{233}$$

Failure of any delay or channel perturbation to affect regulation would weaken the claim that the nominated feedback pathway is causally responsible.

6.14 Test 6: Learning specificity

Compare learning-enabled and frozen-policy architectures under stable and changed behavioural mappings.

Define

$$\Delta P_L(E) = P_{L=1}(E) - P_{L=0}(E). \tag{234}$$

An operational prediction associated with H3.5 is:

$$\Delta P_L\left(E_{\text{changed}}\right) > \Delta P_L\left(E_{\text{stable}}\right). \tag{235}$$

Improvement caused merely by additional trainable parameters is controlled by capacity-matched architectures whose parameters remain fixed during the decisive adaptation period.

6.15 Test 7: Freeze-after-adaptation test

Learning should produce a persistent change in the behavioural mapping.

Allow the learning-enabled architecture to adapt for a preregistered sequence of episodes:

$$e = 1, \ldots, E_A, \tag{236}$$

with between-episode parameter updates

$$\theta_{e+1} = \mathcal{L}\left(\theta_e, \mathcal{D}_e\right). \tag{237}$$

After adaptation episode E_A, freeze the learned parameters:

$$\theta_e = \theta_{E_A+1} \qquad \text{for } e > E_A. \tag{238}$$

Evaluation then proceeds without further learning updates.

If performance remains improved relative to the original frozen policy under the new regime, the adaptation has been incorporated into the behavioural mapping rather than existing only as transient within-episode feedback.

This distinguishes:

$$\boxed{\text{persistent policy modification} \neq \text{temporary corrective action.}} \tag{239}$$

6.16 Test 8: Learning-channel ablation

Causally, disable or randomise the parameter-update pathway while leaving observation, action and retained-state channels intact.

Compare:

$$\text{do}(L = 1) \tag{240}$$

with

$$\text{do}(L = 0). \tag{241}$$

A selective learning deficit should appear after environmental or task structure changes but not

necessarily in the already mastered stable regime.

An operational ablation prediction associated with H3.6 is:

$$\begin{gathered}\text{learning-channel ablation selectively impairs} \\ \text{adaptation to changed mappings.}\end{gathered} \tag{242}$$

Generic degradation in every condition would indicate architectural damage rather than a specific learning function.

6.17 Test 9: Retention–feedback interaction

Interactively, some tasks may require both historical information and online correction.

The factorial interaction can be measured by

$$I_{RF}^{P} = P_{11} - P_{10} - P_{01} + P_{00}, \tag{243}$$

where indices denote retention and feedback status.

A non-zero interaction should be interpreted only relative to the specified performance scale.

The experiment should ask whether retention changes what information feedback can exploit, rather than labelling every super-additive result "synergy".

6.18 Test 10: Feedback–learning interaction

Learning may alter the policy through which later feedback operates, but the two mechanisms should be separated temporally in the principal experiment.

Compare:

$$F0L0, \qquad F1L0, \qquad F0L1, \qquad F1L1, \tag{244}$$

with policy parameters held fixed within each decisive episode.

The F comparison asks whether newly arriving observations alter action online.

The L comparison asks whether experience from earlier adaptation episodes has changed the fixed policy used in the present episode.

The experiment should determine whether learning improves:

1. performance of a later open-loop policy;
2. performance of a later closed-loop policy;
3. adaptation after regime change;
4. some combination of these effects.

If learning provides no additional benefit once a sufficiently capable fixed feedback controller is supplied, the learning claim should be restricted to environments requiring persistent policy modification.

Conversely, if a learned policy performs well after feedback is removed, the result shows that earlier experience has been incorporated into the policy rather than remaining dependent upon immediate corrective feedback.

6.19 Test 11: Retention–learning interaction

Policy-accessible retained state may assist learning even when that state is not required by the final learned policy.

This possibility should be tested by allowing $R = 1$ and $R = 0$ learners to adapt under otherwise matched conditions.

After adaptation, freeze both policies and evaluate them under a common execution condition in which the explicit retained-state channel is removed:

$$R_{\text{eval}} = 0. \tag{245}$$

If the learner trained with retained state adapts more effectively but its post-adaptation policy continues to perform after $M_{e,t}$ is removed, the result supports:

$$\boxed{\begin{array}{l}\text{policy-accessible retention can support learning} \\ \text{without remaining necessary for later execution.}\end{array}} \tag{246}$$

This result would not show that history has ceased to matter altogether. Information acquired during learning may now be stored in

$$\theta_e. \tag{247}$$

The relevant architectural conclusion is therefore that historical influence has changed its mode of storage and causal access.

This distinction prevents the experiment from incorrectly describing a learned reactive policy as history-free merely because it no longer uses an online retained state.

6.20 Test 12: Full factorial comparison

The full $R \times F \times L$ design should test whether the three nominated causal channels remain experimentally distinguishable after their interactions are taken into account.

A suitable factorial model is schematically:

$$\begin{aligned} P = \beta_0 &+ \beta_R R + \beta_F F + \beta_L L \\ &+ \beta_{RF} RF + \beta_{RL} RL + \beta_{FL} FL + \beta_{RFL} RFL + \varepsilon. \end{aligned} \tag{248}$$

The exact statistical family should match the outcome distribution.

The coefficients estimate effects of:

$$\begin{gathered} \text{policy-accessible retained state,} \\ \text{within-episode observation–action feedback,} \\ \text{between-episode policy learning} \end{gathered} \tag{249}$$

under the specified architecture, task family and timescale.

They should not be interpreted as universal strengths of "memory", "feedback" or "learning" in the abstract.

In particular, $R = 0$ does not imply the absence of information retained in learned parameters, and $L = 1$ does not imply that parameter updates occur during the decisive within-episode feedback test.

6.21 Bypass searches

Each mechanism should face a serious bypass architecture.

For retention, possible bypasses include:

1. externally augmented observations containing the relevant history;
2. finite observation windows without recurrent internal state;
3. environment redesign making the task Markovian.

For feedback, possible bypasses include:

1. sufficiently accurate open-loop planning in predictable worlds;
2. precomputed robust policies;
3. disturbance-insensitive strategies.

For learning, possible bypasses include:

1. fixed policies robust across all tested regimes;
2. explicit contextual policy switching without parameter update;

3. larger static policy repertoires selected from present context.

A successful bypass does not show that the mechanism is useless.

It shows that the mechanism is not universally necessary for the target capacity under the tested task class.

6.22 Representation and state-augmentation test

Apparent memory can disappear when omitted state variables are restored.

If a history-dependent task becomes Markovian after augmenting observation with variable Z_t,

$$\widetilde{O}_t = (O_t, Z_t), \tag{250}$$

then performance should be retested using a reactive architecture.

If the retention advantage disappears, the correct conclusion is not that history was irrelevant.

It is:

$$\boxed{\text{the required historical information has been movedinto the present-state representation.}} \tag{251}$$

Accordingly, claims about "memory necessity" must always specify the state representation.

6.23 Primary measurements

Block III should report mechanism-specific quantities rather than one adaptive score.

For retention:

$$\begin{gathered} R(\tau), \qquad R_{\text{causal}}(\Delta), \\ \Delta P_R\left(E_{\text{history}}\right), \qquad \Delta P_R\left(E_{\text{Markov}}\right). \end{gathered} \tag{252}$$

For feedback:

$$\begin{gathered} \Delta P_F, \qquad P_{\text{closed}} - P_{\text{yoked}}, \\ \Gamma_{\text{trans}}, \qquad \Gamma_{\text{path}}, \qquad \text{delay sensitivity}. \end{gathered} \tag{253}$$

For learning:

$$\begin{gathered} \Delta P_L, \qquad \text{adaptation time}, \\ \text{post-freeze performance}, \qquad \text{parameter-update ablation effect}. \end{gathered} \tag{254}$$

Interaction terms should be reported separately and attached to their specified outcomes.

6.24 Typed profile for Block III

The relevant portions of the Paper II profile are:

$$\mathfrak{C}_{\text{Block III}} = [\mathfrak{D}, \mathfrak{F}, \mathfrak{T}]. \tag{255}$$

Here:

$$\mathfrak{D} = \{\Gamma_{\text{trans}}, \Gamma_{\text{path}}\}, \tag{256}$$

$$\mathfrak{F} = \{\Delta P_R, \Delta P_F, \Delta P_L\}, \tag{257}$$

$$\mathfrak{T} = \{R(\tau), R_{\text{causal}}(\Delta), \delta/\tau_P\}. \tag{258}$$

No organisational-closure component is introduced merely because Block I used one.

The profile is modular rather than compulsory.

6.25 Primary hypotheses

Block III tests the following claims:

Table 5: Primary hypotheses and break conditions for Experimental Block III.

ID	Hypothesis	Break condition
H3.1	Policy-accessible retained state provides a selective advantage when task-relevant history is unavailable from the present observable state.	Capacity-matched reactive controls perform equivalently in genuinely history-dependent tasks, or retention advantage is equally large in matched Markov tasks.
H3.2	Manipulation of policy-accessible retained state causally alters later behaviour in a predictable, task-relevant manner.	History can be decoded from internal state, but targeted manipulation of that state produces no relevant behavioural effect.
H3.3	Closed-loop feedback provides a selective advantage under unpredictable disturbance.	Matched open-loop or yoked control performs equivalently across unpredictable-disturbance tests.
H3.4	The causal contribution of feedback depends upon process-relevant observation timing and channel integrity.	Delay, scrambling or masking of the nominated feedback channel produces no predicted selective impairment.

ID	Hypothesis	Break condition
H3.5	Between-episode experience-dependent learning provides a selective advantage when behavioural mappings change.	Frozen capacity-matched policies adapt equally well, or learning advantage is equally large under unchanged mappings.
H3.6	Between-episode learning produces persistent modification of future behaviour rather than only transient within-episode corrective action.	Performance gains disappear immediately when parameter updates are frozen despite retention of the alleged learned state.
H3.7	Policy-accessible retention, within-episode feedback and between-episode learning possess at least partly separable causal roles.	All apparent mechanism-specific effects collapse into one generic capacity difference under matched controls.

6.26 Pre-registration

Before decisive evaluation, Block III should freeze:

1. operational definitions of retention, feedback and learning;
2. factorial architecture conditions;
3. parameter and compute budgets;
4. Markov and history-dependent task pairs;
5. predictable and unpredictable disturbance conditions;
6. stable and changed mapping conditions;
7. intervention schedules;
8. retained-state manipulations;
9. feedback delays and channel perturbations;
10. parameter-freezing procedures;
11. primary outcome measures;
12. interaction models;
13. break conditions;
14. random-seed procedure.

Task variants used to tune the architectures should be separated from the held-out variants used for decisive testing.

6.27 Statistical analysis

Primary effects should be estimated across independent seeds and environment instances.

The principal comparisons should emphasise interaction between mechanism and task requirement:

$$\begin{aligned} &\text{retention} \times \text{history dependence}, \\ &\text{feedback} \times \text{disturbance unpredictability}, \\ &\text{learning} \times \text{mapping change}. \end{aligned} \tag{259}$$

Mechanism specificity is supported more strongly by these interactions than by unconditional main effects.

Where multiple ablations or delay levels are tested, correction for the planned family of comparisons should be preregistered.

6.28 Decision framework

Retention, feedback and learning should receive separate judgements.

Retain retention if historical information is both present and causally used, with selective advantage in tasks that require history.

Retain feedback if observation-dependent online action selectively improves regulation under disturbances that cannot be handled equivalently by matched open-loop or yoked control.

Retain learning if experience-dependent policy modification selectively improves adaptation to changed behavioural mappings and survives freeze-after-adaptation tests.

Restrict any mechanism whose advantage occurs only within narrow architecture classes, task representations or timescales.

Reduce an alleged mechanism if a simpler static, reactive or state-augmented architecture reproduces its decisive effects.

Suspend interpretation when selective intervention cannot be performed without destroying generic capacity.

Reject any strict dependency defeated repeatedly by robust bypass architectures.

6.29 What Block III cannot establish

Importantly, positive results would not establish:

$$\begin{gathered}\text{general intelligence,}\\ \text{explicit world modelling,}\\ \text{counterfactual planning,}\\ \text{reflective self-awareness,}\\ \text{phenomenal consciousness.}\end{gathered} \tag{260}$$

A system can retain history, regulate itself and learn successfully while remaining narrow, reactive at a higher level, or incapable of adapting to structural novelty.

Therefore,

$$\boxed{\text{retention} + \text{feedback} + \text{learning} \not\Rightarrow \text{intelligent agency.}} \tag{261}$$

That transition is the subject of Block IV.

6.30 Provisional role in the wider architecture

Block III tests whether the effective mechanisms identified in Paper I remain causally distinguishable under controlled artificial conditions.

The strongest supported result would be:

$$\boxed{\begin{gathered}\text{retention preserves historically relevant distinctions,}\\ \text{feedback modifies action online,}\\ \text{learning modifies future behavioural mappings,}\\ \text{and these contributions remain experimentally separable.}\end{gathered}} \tag{262}$$

Such a result would validate the distinctions without proving a necessary hierarchical ordering among them.

The governing interpretation is:

$$\boxed{\begin{gathered}\text{do not ask which mechanism is “higher”;}\\ \text{ask which mechanism changes which capability.}\end{gathered}} \tag{263}$$

7 Experimental Block IV: Adaptive Agency and Structural Generalisation

The fourth experimental block addresses the unresolved transition from adaptive regulation to more general agency. Retention, feedback and learning may allow a system to perform successfully within familiar task structures, yet none of those mechanisms alone establishes that the system can identify altered causal relations, evaluate unfamiliar possibilities or reorganise behaviour when previously successful policies cease to apply.

Block IV therefore distinguishes ordinary adaptation from structural generalisation:

$$\boxed{\begin{array}{c}\text{adaptation within familiar structure}\\ \neq \\ \text{adaptation when the structure of the problem changes.}\end{array}} \tag{264}$$

The target capacity is defined as successful behaviour under held-out changes that cannot be solved reliably through interpolation over the training distribution alone.

7.1 Primary experimental question

Block IV asks:

> **Can an artificial agent adapt to previously unseen causal structures, compositions or problem families, and does explicit model-based reasoning make a selective causal contribution to that capacity?**

Mechanistically, the candidate relation is

$$\text{adaptive regulation} \xrightarrow{M} \text{structurally generalising agency}, \tag{265}$$

where M may include latent-state modelling, learned transition structure, counterfactual rollout or another explicitly specified predictive mechanism.

Adversarially, the relation should be restricted or rejected if capacity-matched reactive, model-free or heuristic systems achieve equivalent structural generalisation without the proposed model-based mechanism.

7.2 What counts as structural novelty?

Novelty is divided into increasingly demanding classes.

1. **Interpolation**: new examples drawn from the familiar training distribution.

2. **Parameter variation**: altered values within a familiar causal structure.
3. **Compositional novelty**: unfamiliar combinations of previously encountered components.
4. **Causal novelty**: changed relations among familiar variables or actions.
5. **Problem-family novelty**: tasks requiring a strategy not encountered during training.

Block IV reserves its strongest claims of structural generalisation for categories 3–5.

Consequently,

$$\boxed{\text{held-out data} \not\Rightarrow \text{held-out structure.}} \tag{266}$$

7.3 Task families

Environmentally, the experiment should contain several task families whose decisive variants can be withheld until evaluation.

Suitable families include:

1. navigation with latent hazards whose locations or causal effects change;
2. resource scheduling with delayed depletion or replenishment;
3. multi-step tool-use chains in which intermediate actions alter later possibilities;
4. tasks where actions modify subsequent transition dynamics;
5. compositional tasks built from familiar components arranged in unseen combinations;
6. optional multi-agent tasks involving hidden goals or intentions.

Importantly, no single task family defines intelligence. The purpose of using several families is to test whether the same candidate mechanism contributes across qualitatively different forms of structural novelty.

7.4 Agent and environment model

Formally, let the environment possess latent state

$$S_t \tag{267}$$

and observation

$$O_t \sim G\left(\cdot \mid S_t\right). \tag{268}$$

Actions

$$A_t \tag{269}$$

change the environment through transition kernel

$$S_{t+1} \sim K\left(\cdot \mid S_t, A_t\right). \tag{270}$$

Outcomes relevant to the task are denoted

$$Y_t. \tag{271}$$

A changed causal regime is represented by

$$K_A \longrightarrow K_B, \tag{272}$$

where the decisive regime K_B is withheld during ordinary training.

7.5 Candidate world model

A model-based agent learns an internal predictive model

$$\widehat{K}_\phi\left(S_{t+1}, O_{t+1}, Y_{t+1} \mid S_t, A_t\right). \tag{273}$$

The model may operate directly over observable variables or through learned latent states.

World-model research provides concrete demonstrations that predictive latent models can support behaviour through imagined futures [9]. Paper III does not treat those results as evidence that every successful agent requires a world model. Instead, it attempts to reproduce the relevant causal distinction at a deliberately smaller and more transparent scale.

7.6 Prediction is not sufficient for model-based control

Predictively, accurate internal forecasting alone does not establish that the model contributes to action.

A system may contain representation

$$Z_t \tag{274}$$

from which future observations are highly decodable while the policy ignores that information.

Accordingly,

$$\boxed{\text{predictive representation} \neq \text{causally used world model.}} \tag{275}$$

Model use therefore requires intervention-sensitive evidence linking internal predictions to subsequent action selection.

7.7 Counterfactual rollout

Prospectively, a planning-enabled agent evaluates candidate action sequences

$$A_{t:t+H}^{(j)} \tag{276}$$

over horizon H.

The learned model generates imagined trajectories

$$\widehat{S}_{t+1:t+H}^{(j)} \tag{277}$$

and corresponding predicted outcomes

$$\widehat{Y}_{t+1:t+H}^{(j)}. \tag{278}$$

A planning rule may then select

$$j^* = \arg\max_j \widehat{U}\left(A_{t:t+H}^{(j)}\right), \tag{279}$$

where $\widehat{U}$ is a preregistered evaluation functional.

Counterfactually, the scientific question is whether such imagined alternatives cause better action selection specifically when immediate reactive responses are insufficient.

7.8 Architecture classes

Comparatively, Block IV should include at least five capacity-matched architectures.

Architecture A0: Reactive control. Action depends on current observation:

$$A_t \sim \pi\left(\cdot \mid O_t\right). \tag{280}$$

This architecture provides the simplest baseline.

Architecture A1: History-sensitive model-free control. Action may depend upon retained state:

$$A_t \sim \pi\left(\cdot \mid O_t, M_t\right), \tag{281}$$

but no explicit predictive transition model is queried during action selection.

This controls for retention and recurrent capacity.

Architecture A2: Predictive model without rollout. The agent learns

$$\widehat{K}_\phi, \tag{282}$$

but planning is restricted so that the model is not used to evaluate extended counterfactual action sequences.

This tests whether predictive representation alone is sufficient.

Architecture A3: Model-based rollout control. The learned model is actively queried to evaluate candidate future action sequences.

This is the principal model-based planning condition.

Architecture A4: Capacity-matched model-free control. The agent receives comparable parameters, training experience and inference-time compute, but no explicit learned transition model is used during action selection. This is a serious null against the claim that any A3 advantage is specifically model-based.

7.9 Training regime

Preparatively, all architectures should be trained on a family of environments sharing defined structural regularities.

The decisive evaluation set must contain transformations absent from the training set.

Training may expose the agent to:

$$\mathcal{E}_{\text{train}} = \{E_1, \ldots, E_n\}, \tag{283}$$

while evaluation uses

$$\mathcal{E}_{\text{struct}} \tag{284}$$

containing preregistered novel structural variants.

No architecture may receive additional tuning after performance on $\mathcal{E}_{\text{struct}}$ has been inspected.

7.10 Test 1: Interpolation versus structural generalisation

Initially, evaluate all architectures on both ordinary held-out examples and held-out structural variants.

Define:

$$P_{\text{interp}} \tag{285}$$

for ordinary interpolation performance and

$$P_{\text{struct}} \tag{286}$$

for performance under structural novelty.

A simple structural-generalisation gap is

$$G_{\text{struct}} = P_{\text{interp}} - P_{\text{struct}}. \tag{287}$$

The aim is not necessarily to minimise this gap to zero.

Rather, architectures should be compared on how performance changes when surface novelty becomes structural novelty.

7.11 Test 2: Causal-relation reversal

Causally, construct tasks where a familiar action changes meaning at evaluation.

For example:

$$A \xrightarrow{K_A} Y_+ \tag{288}$$

during training, but

$$A \xrightarrow{K_B} Y_- \tag{289}$$

after a preregistered regime change.

The agent receives sufficient evidence during evaluation to infer that the relation has changed.

An operational prediction associated with H4.1 is:

$$\begin{gathered}\text{model-based agents adapt more rapidly when} \\ \text{previously learned causal relations are altered.}\end{gathered} \tag{290}$$

A mere tendency to relearn through repeated reward feedback is weaker than rapid adaptation using evidence about the changed structure.

7.12 Test 3: Novel composition

Expose the agent to familiar objects, operations or tools combined in arrangements absent from training.

Suppose known components are:

$$C_1, C_2, C_3. \tag{291}$$

Training includes:

$$C_1 \circ C_2 \tag{292}$$

and

$$C_2 \circ C_3, \tag{293}$$

while evaluation requires:

$$C_1 \circ C_3 \tag{294}$$

or a longer unseen composition.

The principal question is whether the agent can exploit learned relations among components rather than memorised whole-task policies.

7.13 Test 4: Action changes future transition structure

Dynamically, create tasks where an early action changes the later transition kernel:

$$A_t \longrightarrow K_{t+1:t+H}. \tag{295}$$

Examples include:

1. opening one route while permanently closing another;
2. consuming a resource that changes future availability;
3. activating a tool that changes later action effects;
4. altering another agent's subsequent policy.

Successful behaviour requires reasoning about how present actions reshape future possibilities rather than only their immediate outcomes.

7.14 Test 5: Delayed consequence planning

Temporally, construct environments where locally attractive actions produce later costs.

For candidate action a,

$$Y_{t+1}(a) > Y_{t+1}(b) \tag{296}$$

may hold while

$$\sum_{\tau=1}^{H} Y_{t+\tau}(a) < \sum_{\tau=1}^{H} Y_{t+\tau}(b). \tag{297}$$

The experiment should vary planning horizon H and test whether performance changes selectively when the relevant consequence lies beyond shorter horizons.

This provides a direct test of temporal planning rather than generic task difficulty.

7.15 Test 6: World-model scrambling

Alter selected internal model predictions at evaluation while preserving policy-network capacity.

Let the normal model be

$$\widehat{K}_\phi, \tag{298}$$

and corrupted model

$$\widetilde{K}_\phi. \tag{299}$$

Corruption should be structured rather than destructive.

Examples include:

1. reversing one learned transition relation;
2. permuting predicted destinations;
3. biasing one outcome probability;
4. changing predicted resource depletion;
5. modifying one latent causal dependency.

The strongest signature of causal use is:

$$\boxed{\begin{array}{c}\text{model corruption}\\ \longrightarrow \text{specific predicted planning error.}\end{array}} \tag{300}$$

If behaviour is unchanged despite substantial model corruption, the model may be epiphenomenal to the tested policy.

7.16 Test 7: Model correction rescue

Restoratively, after inducing a known model error, provide evidence that allows the relevant internal relation to be corrected.

Compare:

$$\widehat{K}_\phi \longrightarrow \widetilde{K}_\phi \longrightarrow \widehat{K}_\phi^{\text{revised}}. \tag{301}$$

If behaviour first changes in the direction predicted by the corruption and then recovers after model correction, the causal model-use interpretation is strengthened.

A passive predictive representation should not necessarily produce this intervention–rescue pattern.

7.17 Test 8: Model accuracy versus planning usefulness

Diagnostically, predictive accuracy and control usefulness should be measured separately.

Let model prediction loss be

$$L_{\text{model}} \tag{302}$$

and planning performance

$$P_{\text{plan}}. \tag{303}$$

The experiment should test whether lower

$$L_{\text{model}} \tag{304}$$

predicts improved

$$P_{\text{plan}} \tag{305}$$

specifically on task-relevant relations.

A highly accurate model of irrelevant environmental detail may contribute little to action.

Therefore,

$$\boxed{\text{world-model accuracy} \neq \text{world-model usefulness.}} \tag{306}$$

7.18 Test 9: Rollout ablation

Ablatively, preserve the learned predictive model while preventing imagined multi-step rollout during action selection.

Compare:

$$A2 \qquad \text{and} \qquad A3. \tag{307}$$

An operational prediction associated with H4.2 is:

$$\begin{gathered}\text{counterfactual rollout produces a selective advantage} \\ \text{when delayed or branching consequences require} \\ \text{evaluation of alternative future trajectories.}\end{gathered} \tag{308}$$

If A2 matches A3 across all decisive planning tasks, explicit rollout is not necessary for the tested capability.

7.19 Test 10: Capacity-matched model-free bypass

Compare A3 against A4 using matched parameter count, training experience and inference-time computational budget as closely as practicable.

A4 should be allowed to use powerful model-free optimisation rather than being designed as a weak control.

An operational comparison associated with H4.3 is:

$$\begin{gathered}\text{explicit model-based control provides additional} \\ \text{structural generalisation beyond a serious} \\ \text{capacity-matched model-free alternative.}\end{gathered} \tag{309}$$

If A4 matches A3 across causal-reversal, compositional and delayed-consequence tests, explicit world-model use should not be treated as necessary for structural generalisation.

7.20 Test 11: Heuristic bypass

Heuristically, construct task-specific strategies that can solve some evaluation environments without general predictive modelling.

Examples include:

1. cached contingency rules;
2. finite-state heuristics;
3. robust reactive policies;
4. explicit search without a learned world model;
5. contextual policy libraries.

If a simpler heuristic reproduces the decisive result, the corresponding claim should be reduced to the simpler mechanism for that task.

This prevents sophisticated architecture labels from receiving explanatory credit where ordinary algorithmic structure is sufficient.

7.21 Test 12: Hidden-state inference

Inferentially, some environments should contain latent regime variable

$$Z_t \tag{310}$$

that is not directly observable.

Evidence available over time permits posterior estimate

$$P\left(Z_t \mid O_{\leq t}\right). \tag{311}$$

The experiment should determine whether successful planning depends upon inferring the hidden regime rather than merely retaining recent observations.

Perturbing the inferred latent state should produce predictable action changes if it is causally used.

7.22 Test 13: Goal-directed flexibility

The same environment should sometimes be paired with different task objectives. Let the objective variable be

$$\mathcal{G}. \tag{312}$$

The agent should select different actions when

$$\mathcal{G} = \mathcal{G}_1 \tag{313}$$

versus

$$\mathcal{G} = \mathcal{G}_2 \tag{314}$$

while environmental dynamics remain fixed.

This test distinguishes flexible use of environmental structure from a policy that merely memorises one preferred trajectory. No claim is made that goal-conditioning alone constitutes agency.

7.23 Test 14: Novel problem-family transfer

The strongest test should withhold an entire problem family rather than merely new instances.

Training might involve:

$$\mathcal{F}_1, \qquad \mathcal{F}_2, \tag{315}$$

while decisive evaluation uses:

$$\mathcal{F}_3. \tag{316}$$

The families should share some reusable causal structure but require different surface actions or compositions.

Success is stronger when the agent exploits transferable relations rather than undergoing extensive new optimisation.

7.24 Adaptation-speed diagnostic

Structural adaptation should be evaluated as a function of new evidence.

Let

$$n_{\text{adapt}} \tag{317}$$

denote the number of post-change interactions required to reach a preregistered performance threshold.

A system that eventually relearns through thousands of trials is scientifically different from one that revises behaviour after a small number of informative observations.

Therefore,

$$n_{\text{adapt}} \tag{318}$$

should be reported alongside final performance.

7.25 Causal model-use signature

Strong evidence for model-based control should require several relations simultaneously:

$$\boxed{\begin{array}{c}\text{predictive model contains task-relevant structure}\\ \text{+model intervention changes planning predictably}\\ \text{+model correction rescues planning}\\ \text{+model use helps selectively under structural novelty.}\end{array}} \tag{319}$$

Any one relation alone is weaker.

For example, accurate prediction without intervention sensitivity establishes representation but not causal use.

7.26 Structural generalisation score should not be universal

Quantitatively, no single universal intelligence or generalisation score is introduced.

Performance should instead be reported separately across:

$$\begin{array}{c}\text{causal reversal,} \qquad \text{novel composition,}\\ \text{delayed consequences,} \qquad \text{hidden state,}\\ \text{problem-family transfer.}\end{array} \tag{320}$$

A system may succeed strongly in one form of novelty and fail another.

Collapsing those results into one scalar would conceal precisely the architecture the experiment is designed to reveal.

7.27 Primary measurements

Block IV should report:

1. ordinary held-out performance;
2. structural-novelty performance;
3. structural-generalisation gap;
4. post-change interactions to adaptation n_{adapt};
5. predictive model loss on task-relevant variables;
6. model-corruption effect size;
7. model-correction rescue effect;
8. rollout-ablation effect;
9. planning-horizon sensitivity;
10. model-free control comparison;
11. transfer across problem families;
12. failure mode by novelty class.

Where trajectory distributions are relevant,

$$\Gamma_{\text{path}} \tag{321}$$

may be reported as a comparative dynamical descriptor, but it is not treated as a measure of intelligence.

7.28 Primary hypotheses

Block IV tests the following claims:

Table 6: Primary hypotheses and break conditions for Experimental Block IV.

ID	Hypothesis	Break condition
H4.1	Model-based architectures adapt selectively faster when familiar causal relations are altered.	Capacity-matched model-free or reactive systems adapt equally quickly across the preregistered causal-reversal tasks.
H4.2	Counterfactual rollout contributes specifically when alternative delayed futures must be compared.	Rollout ablation produces no selective loss, or predictive-model-only systems match rollout systems across decisive tasks.

ID	Hypothesis	Break condition
H4.3	Explicit model-based control provides structural-generalisation benefits beyond serious capacity-matched model-free alternatives.	Model-free architectures match model-based systems across causal, compositional and problem-family novelty.
H4.4	Internal world-model content is causally used during planning.	Targeted model corruption fails to alter behaviour in the predicted direction, or equivalent perturbation of irrelevant states produces the same effect.
H4.5	Correcting a corrupted internal model selectively restores the corresponding planning capability.	Model correction does not rescue behaviour or recovery occurs equally without repairing the nominated representation.
H4.6	Structural generalisation extends beyond ordinary interpolation and parameter variation.	Performance gains disappear when evaluation moves from familiar distributions to genuinely changed causal or compositional structure.
H4.7	Structural-generalisation performance is not reducible to one task-specific heuristic.	Simple preregistered heuristics reproduce the decisive transfer and adaptation results across task families.

7.29 Pre-registration

Before decisive evaluation, Block IV should freeze:

1. definitions of each novelty class;
2. training and held-out environment families;
3. architecture classes;
4. parameter and compute matching procedures;
5. predictive-model targets;
6. planning horizons;
7. model-corruption interventions;
8. model-correction procedures;

9. reactive and model-free nulls;
10. heuristic bypasses;
11. adaptation thresholds;
12. primary metrics;
13. break conditions;
14. random-seed procedure.

Structural test environments should remain inaccessible during ordinary architecture tuning.

7.30 Statistical analysis

The principal comparisons should emphasise interactions between architecture and novelty class.

A schematic model is

$$\begin{aligned} Y_{ijs} = \beta_0 + \beta_1 M_i + \beta_2 N_j + \beta_3 (M_i N_j) \\ + u_s + \epsilon_{ijs}, \end{aligned} \tag{322}$$

where M_i represents architecture class, N_j novelty class and u_s seed or replicate effects.

The key evidential quantity is often the interaction:

$$\beta_3, \tag{323}$$

because a genuine model-based advantage should be strongest where the hypothesised model function is required.

Adaptation time should be analysed separately from final performance rather than folded into one score.

7.31 Decision framework

Model-based control, counterfactual rollout and structural generalisation should receive separate judgements.

Retain model-based control if task-relevant predictive structure is causally used and produces selective benefits under structural novelty.

Retain counterfactual rollout if disabling rollout while preserving the predictive model selectively impairs tasks requiring comparison of alternative delayed futures.

Restrict either mechanism if its advantage occurs only in specific task families, planning horizons or architectural regimes.

Reduce the claimed mechanism if serious model-free, reactive or heuristic alternatives reproduce the decisive behaviour.

Suspend interpretation if internal model interventions cannot be performed without destroying generic computational capacity.

Reject any necessary dependency defeated repeatedly by robust bypass architectures.

7.32 What Block IV cannot establish

Importantly, even strong positive results would not establish:

$$\begin{gathered}\text{general intelligence in an unrestricted sense,}\\ \text{human-like reasoning,}\\ \text{semantic understanding in every philosophical sense,}\\ \text{reflective self-modelled intelligence,}\\ \text{phenomenal consciousness.}\end{gathered} \tag{324}$$

A system may use a world model, plan across counterfactual futures and generalise to structural novelty while lacking any explicit representation of its own competence, uncertainty or limitations.

Therefore,

$$\boxed{\begin{gathered}\text{structurally generalising agency}\\ \not\Rightarrow \text{reflective self-modelling.}\end{gathered}} \tag{325}$$

That distinction is tested in Block V.

7.33 Provisional role in the wider architecture

Block IV tests Transition Problem 3 from Paper I:

$$\text{adaptive regulation} \longrightarrow \boxed{?} \longrightarrow \text{intelligent agency.} \tag{326}$$

A positive result would support one artificial route in which predictive models and counterfactual evaluation contribute to adaptation under structural novelty.

A negative result could be equally significant if simpler architectures achieve the same capabilities without explicit model-based control.

The governing interpretation is:

$$\boxed{\begin{gathered}\text{do not infer intelligence from successful prediction;}\\ \text{test whether representations are causally used}\\ \text{when familiar structure no longer suffices.}\end{gathered}} \tag{327}$$

8 Experimental Block V: Reflective Self-Modelling

The fifth experimental block addresses the proposed transition from intelligent agency to a system that causally represents aspects of its own operation. Successful planning, learning and structural generalisation do not by themselves establish that an agent represents its own competence, uncertainty, internal state or limitations.

The central distinction is:

$$\boxed{\text{successful control} \neq \text{self-prediction} \neq \text{causally used self-model.}} \tag{328}$$

A system may predict whether it is likely to succeed without using that prediction to alter subsequent behaviour. Likewise, a system may generate verbal or symbolic statements about itself without those statements playing any causal role in control.

Block V therefore tests whether self-related information is represented, calibrated and causally used.

8.1 Primary experimental question

Block V asks:

> **Does an artificial agent represent task-relevant facts about its own state, competence, uncertainty or limitations, and does intervention on those representations produce selective changes in subsequent control?**

The candidate transition is

$$\text{intelligent agency} \longrightarrow \boxed{?} \longrightarrow \text{reflective self-modelled intelligence.} \tag{329}$$

The experiment does not assume that reflective self-modelling is necessary for all intelligent behaviour.

8.2 What counts as a self-model?

A candidate self-model must represent at least one property of the agent itself that is relevant to future action.

Candidate variables include:

$$\begin{gathered}\text{estimated probability of success,}\\ \text{uncertainty,}\\ \text{available computational resources,}\\ \text{current sensor reliability,}\\ \text{model confidence,}\\ \text{known capability limits,}\\ \text{recent internal failure state.}\end{gathered} \tag{330}$$

Let Z_t^{self} denote the task-relevant self-related state being estimated, and let H_t^{int} denote the internal information made available to the monitoring process.

A self-model estimate is

$$\widehat{Z}_t^{\text{self}} = h_\psi\left(H_t^{\text{int}}\right). \tag{331}$$

The contents of H_t^{int} must be specified for each experiment. They may include internal activation state, model uncertainty, resource availability, recent error information or other preregistered variables, but should not contain privileged information unavailable to the agent under the tested architecture.

Merely attaching a variable named "self" to the architecture does not satisfy this criterion.

8.3 Passive monitoring versus reflective control

The weakest self-monitoring architecture predicts a self-related quantity but does not feed that quantity back into action selection.

For a confidence-monitoring experiment, define

$$q_t = P\left(\text{current prediction or action is correct} \mid H_t^{\text{int}}\right). \tag{332}$$

A passive monitor outputs q_t, but action remains

$$A_t \sim \pi\left(\cdot \mid O_t, M_t\right). \tag{333}$$

A reflective controller instead allows the monitor output to influence subsequent control:

$$A_t \sim \pi\left(\cdot \mid O_t, M_t, q_t\right). \tag{334}$$

The relevant distinction is therefore:

$$\boxed{\text{self-monitoring output} \neq \text{reflective self-control.}} \tag{335}$$

Accurate prediction by q_t establishes monitoring quality only. Evidence for reflective control additionally requires intervention on the monitor to produce selective changes in later behaviour.

8.4 Architecture classes

Comparatively, Block V should contain at least five matched architecture classes.

Architecture S0: No explicit self-monitor. The agent performs the task using the capacities established in earlier blocks, but no dedicated self-related estimate is produced.

Architecture S1: Passive self-monitor. The agent predicts quantities such as confidence or error probability, but those outputs cannot influence subsequent computation or action.

Architecture S2: Active self-monitor. Self-related estimates can alter behaviour, for example by:

1. abstaining;
2. requesting more information;
3. increasing planning effort;
4. switching strategy;
5. checking an answer;
6. deferring action.

Architecture S3: Shuffled self-monitor control. The architecture receives self-monitoring values drawn from another matched episode, preserving output distribution while breaking episode-specific correspondence.

Architecture S4: Capacity-matched non-self control. The agent receives equivalent extra parameters and compute devoted to task-relevant processing but without an explicit self-related estimate.

This controls for architectural capacity rather than self-modelling.

8.5 Task families

Discriminatively, Block V should include tasks in which self-related information is either relevant or irrelevant.

Suitable task families include:

1. variable-difficulty decisions with an abstention option;
2. tasks where additional computation can be purchased at a cost;
3. environments containing temporary sensor corruption;
4. tasks requiring strategy switching after internal failure;
5. decisions involving known capability boundaries;
6. tasks where confidence should determine whether external assistance is requested.

Matched control tasks should remove the usefulness of self-related information while preserving similar surface difficulty.

8.6 Calibration

Probabilistically, confidence estimate q_t should correspond to empirical accuracy.

For predictions grouped around confidence q,

$$P\left(\text{correct} \mid q_t \approx q\right) \approx q. \tag{336}$$

Calibration can be assessed in part through the Brier score [10]:

$$\text{BS} = \frac{1}{N}\sum_{i=1}^{N}\left(q_i - y_i\right)^2, \tag{337}$$

where

$$y_i \in \{0, 1\} \tag{338}$$

records whether the relevant prediction or action was correct.

Good calibration is useful evidence that the monitor tracks something real, but it does not establish causal self-model use.

8.7 Test 1: Passive versus active monitoring

Initially, compare S1 and S2 on tasks where confidence-sensitive control is useful.

An operational prediction associated with H5.1 is:

$$\begin{gathered}\text{self-monitoring improves behaviour primarily when} \\ \text{its output can alter subsequent computation or action.}\end{gathered} \tag{339}$$

If S1 performs as well as S2 across decisive tasks, then feeding the self-monitor into control has not demonstrated independent benefit.

8.8 Test 2: Selective abstention

Allow the agent to abstain at a known cost.

Let action options include

$$A_t \in \{a_1, \ldots, a_k, a_{\text{abstain}}\}. \tag{340}$$

A useful self-monitor should support:

$$P\left(a_{\text{abstain}} \mid q_t \text{ low}\right) > P\left(a_{\text{abstain}} \mid q_t \text{ high}\right). \tag{341}$$

Selective accuracy should then be measured as a function of coverage.

The purpose is not to reward indiscriminate abstention. Costs should be chosen so that useful calibration, rather than avoiding every difficult decision, is required.

8.9 Test 3: Adaptive computation

Some tasks should allow the agent to allocate additional reasoning or planning effort at a cost.

Let optional additional compute be

$$c_t \geq 0. \tag{342}$$

The self-model should predict when additional computation is worth purchasing.

A strong result would show:

$$\begin{aligned} &\text{low estimated competence} \\ \longrightarrow\ &\text{selective increase in computation} \\ \longrightarrow\ &\text{improved final decision quality.} \end{aligned} \tag{343}$$

A capacity-matched architecture that always spends the same additional compute provides an important control.

8.10 Test 4: Induced internal corruption

Introduce temporary internal impairment that is not directly announced to the self-monitor.

Candidate perturbations include:

1. degraded sensory input;
2. corrupted retained state;
3. partial world-model scrambling;

4. reduced planning horizon;

5. disabled computational resource;

6. temporary noise added to an internal representation.

The system should detect the resulting change in its own reliability through its ordinary monitoring process.

Relevant measures include:

$$\tau_{\text{detect}}, \tag{344}$$

the latency between impairment and reliable detection, and

$$\Delta q, \tag{345}$$

the corresponding change in self-estimated confidence.

8.11 Test 5: Recovery after internal corruption

Restoratively, after detecting impairment, an active self-model should alter control appropriately.

Possible responses include:

$$\begin{gathered}\text{strategy switching,} \qquad \text{additional verification,} \\ \text{reduced action confidence,} \qquad \text{request for assistance,} \\ \text{temporary abstention.}\end{gathered} \tag{346}$$

An operational prediction associated with H5.2 is:

$$\begin{gathered}\text{active self-monitoring improves recovery} \\ \text{after unannounced internal impairment.}\end{gathered} \tag{347}$$

Generic robustness without any corresponding change in self-estimate would not establish reflective monitoring.

8.12 Test 6: Self-model misinformation

Causally, one of the strongest tests is to manipulate the self-model while leaving ordinary task information unchanged.

Let accurate self-estimate be

$$q_t, \tag{348}$$

and experimentally altered estimate

$$\widetilde{q}_t. \tag{349}$$

Two informative manipulations are:

$$\widetilde{q}_t > q_t \tag{350}$$

for induced overconfidence, and

$$\widetilde{q}_t < q_t \tag{351}$$

for induced underconfidence.

If the self-model is causally used, these manipulations should alter behaviours such as abstention, checking, planning effort or assistance seeking in the predicted direction.

Therefore,

$$\boxed{\begin{array}{c}\text{self-model misinformation}\\ \longrightarrow \text{predicted change in reflective control}\end{array}} \tag{352}$$

provides stronger evidence than correlation between confidence and accuracy.

8.13 Test 7: Self-model correction rescue

Restore accurate self-related information after induced misinformation.

The intervention sequence is:

$$q_t \longrightarrow \widetilde{q}_t \longrightarrow q_t^{\text{restored}}. \tag{353}$$

An operational rescue prediction associated with H5.3 is:

$$\begin{array}{c}\text{restoring accurate self-related information}\\ \text{rescues the corresponding control behaviour.}\end{array} \tag{354}$$

This manipulation–rescue pattern provides especially strong evidence that the nominated self-model participates causally in control.

8.14 Test 8: Shuffled-monitor control

Randomly, replace the episode-specific monitor output with values sampled from other episodes while preserving approximately the same marginal distribution.

Let:

$$q_t^{\text{shuffled}} = q_{\sigma(t)}, \tag{355}$$

where σ breaks correspondence between the current internal state and the monitor output.

If S2 outperforms S3 under self-relevant tasks, the result supports the value of state-specific self-monitoring rather than merely an additional numerical input.

8.15 Test 9: Known capability boundary

Construct tasks that cross a preregistered capability boundary.

Suppose the agent performs reliably for difficulty

$$d < d^* \tag{356}$$

but poorly for

$$d > d^*. \tag{357}$$

A useful self-model should track this transition sufficiently well that behaviour changes around the boundary.

Measure whether estimated competence satisfies:

$$q(d) \downarrow \tag{358}$$

as actual performance deteriorates.

More importantly, test whether the agent uses that decline appropriately.

Knowledge of limitation is stronger evidence than merely failing on difficult tasks.

8.16 Test 10: Novel-domain calibration

A self-model may be well calibrated in familiar tasks but fail under distribution shift.

Evaluate calibration and reflective control under:

$$E_{\text{train}} \longrightarrow E_{\text{novel}}. \tag{359}$$

The experiment should distinguish:

$$\begin{gathered} \text{task generalisation,} \\ \text{confidence generalisation,} \\ \text{control generalisation.} \end{gathered} \tag{360}$$

An agent that remains overconfident under novelty may perform well enough on some tasks while possessing a poor transferable self-model.

8.17 Test 11: Error detection without explicit confidence

Construct architectures that detect errors through ordinary environmental feedback without maintaining an explicit confidence variable.

For example, a controller might:

$$\text{detect inconsistency} \longrightarrow \text{retry action} \tag{361}$$

using task-specific rules.

If such systems reproduce the decisive benefits of S2, an explicit probabilistic self-monitor is not necessary for the tested reflective function.

This does not invalidate the broader possibility of self-modelling. It narrows the mechanism required.

8.18 Test 12: External-monitor bypass

Externally, provide some systems with an external confidence or error estimate computed from privileged information.

Let external monitor output be

$$q_t^{\text{ext}}. \tag{362}$$

If externally supplied monitoring reproduces the same behavioural benefits as an endogenous self-monitor, then the experiment has established the functional value of access to reliability information, not yet the necessity of internal self-modelling.

This comparison distinguishes:

$$\boxed{\text{using self-relevant information} \neq \text{generating that information internally.}} \tag{363}$$

8.19 Test 13: Self-model dimensionality

One scalar confidence value may be insufficient.

A richer candidate self-model is:

$$\mathbf{z}_t^{\text{self}} = \left(q_t^{\text{perception}}, q_t^{\text{model}}, q_t^{\text{planning}}, q_t^{\text{action}}\right). \tag{364}$$

The experiment should compare whether domain-specific self-estimates provide better control than a single global confidence score.

However,

$$\boxed{\text{higher-dimensional self-description} \neq \text{better self-model by definition.}} \tag{365}$$

Extra dimensions must earn their place through selective predictive or causal gain.

8.20 Test 14: Self-model consistency across timescales

Self-related estimates may operate over different horizons.

Short-term estimate:

$$q_t^{(1)} \tag{366}$$

may concern the next action.

Longer-term estimate:

$$q_t^{(H)} \tag{367}$$

may concern success over horizon H.

The experiment should test whether self-estimates remain appropriately calibrated when the evaluation horizon changes.

A monitor that predicts immediate correctness well may still fail to represent longer-term capability.

8.21 Test 15: Self-description without causal use

Linguistically or symbolically, where an architecture can generate explicit reports, compare report content with intervention-sensitive self-model use.

Suppose system produces statement

$$R_t^{\text{self}}. \tag{368}$$

The experiment should test:

$$R_t^{\text{self}} \longrightarrow A_{t+\Delta} \tag{369}$$

causally rather than inferring significance from report content alone.

A system may generate accurate self-descriptions because its reporting head reads internal variables while its policy ignores those reports.

Therefore,

$$\boxed{\text{self-report} \neq \text{causally integrated self-model.}} \tag{370}$$

8.22 Test 16: Generic uncertainty versus self-specific representation

Adversarially, some apparent self-modelling may reduce to ordinary uncertainty estimation about the environment.

Compare two forms:

$$q_t^{\text{world}} \tag{371}$$

for uncertainty about external state, and

$$q_t^{\text{self}} \tag{372}$$

for uncertainty about the agent's own competence or internal reliability.

Construct tasks where:

$$q_t^{\text{world}} \tag{373}$$

is held approximately constant while

$$q_t^{\text{self}} \tag{374}$$

changes through internal impairment.

If the system responds appropriately, the result is harder to reduce to generic environmental uncertainty.

8.23 Causal self-model signature

Strong evidence for reflective self-modelling should require several relations:

$$\boxed{\begin{array}{c}\text{self-related variable tracks actual internal capability}\\ \text{+self-model intervention alters control predictably}\\ \text{+correction rescues the altered control}\\ \text{+the effect is selective for self-relevant tasks.}\end{array}} \tag{375}$$

No single measure, including calibration, is sufficient by itself.

8.24 Primary measurements

Block V should report:

1. calibration error;
2. Brier score;
3. selective accuracy versus coverage;
4. abstention efficiency;
5. adaptive-computation allocation;
6. error-detection latency;
7. recovery after internal corruption;
8. self-model misinformation effect size;

9. self-model correction rescue effect;
10. shuffled-monitor performance;
11. capability-boundary calibration;
12. novel-domain calibration;
13. self-report/control dissociation where relevant.

No scalar "self-awareness score" should be constructed.

8.25 Primary hypotheses

Block V tests the following claims:

Table 7: Primary hypotheses and break conditions for Experimental Block V.

ID	Hypothesis	Break condition
H5.1	Self-monitoring improves behaviour primarily when self-related estimates can alter subsequent computation or action.	Passive monitoring performs equivalently to active monitoring across self-relevant tasks.
H5.2	Active self-monitoring improves recovery after unannounced internal impairment.	Recovery is unchanged by monitoring status or occurs without corresponding changes in self-estimated reliability.
H5.3	Manipulating self-related estimates causes predictable changes in reflective control, and restoring accurate estimates rescues behaviour.	Self-model misinformation does not alter control, or behavioural recovery occurs without correction of the nominated self-model.
H5.4	Episode-specific self-monitoring provides more useful control than shuffled or distribution-matched monitoring signals.	Shuffled-monitor controls perform equivalently to state-specific monitoring.
H5.5	The agent tracks at least some internally generated capability changes rather than only environmental uncertainty.	Apparent self-monitoring disappears when external uncertainty is controlled or is reproduced entirely by world-state uncertainty estimates.

ID	Hypothesis	Break condition
H5.6	Reflective self-modelling provides functions not fully reproduced by simpler task-specific error detection or externally supplied confidence.	Bypass architectures reproduce all decisive behavioural effects without endogenous self-representation.

8.26 Pre-registration

Before decisive evaluation, Block V should freeze:

1. operational definition of self-related variables;
2. monitor architecture classes;
3. calibration metrics;
4. task families;
5. abstention and compute costs;
6. induced-corruption procedures;
7. misinformation interventions;
8. correction/rescue procedures;
9. shuffled-monitor construction;
10. external-monitor control;
11. capability-boundary conditions;
12. primary outcome measures;
13. break conditions;
14. random-seed procedure.

Self-model definitions should not be expanded retrospectively whenever a new internal variable correlates with performance.

8.27 Statistical analysis

Primary statistical comparisons should emphasise interactions between monitor type and task requirement.

For example:

$$Y_{ijs} = \beta_0 + \beta_1 S_i + \beta_2 R_j + \beta_3 (S_i R_j) + u_s + \epsilon_{ijs}, \tag{376}$$

where S_i denotes self-monitor condition, R_j denotes whether self-related information is relevant, and u_s represents replicate or seed effects.

Calibration should be analysed separately from behavioural benefit.

An agent can be well calibrated yet gain no control advantage from the monitor, or poorly calibrated while still using the monitor causally.

8.28 Decision framework

Self-prediction and reflective self-model use should receive separate judgements.

Retain self-monitoring if self-related estimates reliably track specified internal capacities or errors.

Retain reflective self-model use only if intervention on those estimates produces selective, predicted changes in subsequent control and appropriate correction rescues behaviour.

Restrict the claim if self-model benefits occur only in narrow task families, architectures or uncertainty regimes.

Reduce the mechanism if ordinary environmental uncertainty estimation, task-specific error detection or externally supplied monitoring reproduces the decisive behaviour.

Suspend interpretation if self-related variables cannot be intervened upon independently of generic architecture damage.

Reject any strict dependency claiming reflective self-modelling is necessary for intelligent agency if robust bypass architectures achieve the target capability without it.

8.29 What Block V cannot establish

Even the strongest possible Block V result would not establish:

$$\begin{gathered} \text{subjective experience,} \\ \text{phenomenal awareness,} \\ \text{sentience,} \\ \text{human-like selfhood,} \\ \text{conscious introspection.} \end{gathered} \tag{377}$$

In particular:

$$\boxed{\begin{array}{c}\text{calibrated self-monitoring} \not\Rightarrow \text{consciousness},\\ \text{causal self-model use} \not\Rightarrow \text{consciousness},\\ \text{accurate self-report} \not\Rightarrow \text{consciousness}.\end{array}} \tag{378}$$

Paper III ends its principal hierarchy here because Paper I identified no validated bridge from reflective self-modelled intelligence to phenomenal experience.

8.30 Provisional role in the wider architecture

Block V tests Transition Problem 4 from Paper I:

$$\text{intelligent agency} \longrightarrow \boxed{?} \longrightarrow \text{reflective self-modelled intelligence}. \tag{379}$$

A positive result would support one artificial route in which explicit self-related representations causally regulate behaviour.

A negative result could be equally significant if intelligent systems achieve flexible correction, uncertainty-sensitive action and robust adaptation without an endogenous self-model.

The governing interpretation is:

$$\boxed{\begin{array}{c}\text{do not infer reflection from self-description;}\\ \text{intervene on the self-model and test}\\ \text{whether control changes accordingly.}\end{array}} \tag{380}$$

9 Auxiliary Test: Conflict Representation and Multi-Constraint Control

This auxiliary experiment tests whether explicit representation of simultaneously competing goals, constraints or action consequences provides a functional advantage over simpler scalarised or model-based alternatives. It does not assume that conflict representation is a necessary stage of intelligence, reflective self-modelling or consciousness.

The experimental question is therefore deliberately narrow:

> **Does explicitly representing conflict among competing objectives improve control when the structure of those trade-offs changes in ways not encountered during training?**

Importantly,

$$\boxed{\begin{array}{c}\text{conflict representation} \neq \text{reflective self-modelled intelligence} \\ \neq \text{phenomenal consciousness.}\end{array}} \tag{381}$$

9.1 Why this is an auxiliary test

Papers I and II provide no basis for treating recursive conflict or paradox negotiation as a universal generative principle.

Accordingly, conflict representation is removed from the main experimental hierarchy.

Its possible value is instead assessed against ordinary alternatives from:

$$\begin{array}{cc}\text{multi-objective control,} & \text{scalar reward optimisation,} \\ \text{model-based planning,} & \text{constraint satisfaction.}\end{array} \tag{382}$$

If those simpler descriptions explain the decisive results, a separate conflict-representation mechanism should be reduced rather than protected.

9.2 Task structure

The test environment should contain genuinely competing requirements.

For example:

$$C_1 : \text{preserve energy}, \tag{383}$$
$$C_2 : \text{meet a deadline}, \tag{384}$$
$$C_3 : \text{avoid hazard}, \tag{385}$$
$$C_4 : \text{maintain partner cooperation}. \tag{386}$$

No single action should satisfy every objective optimally in all states.

Let objective-specific outcomes be

$$r_i(t), \tag{387}$$

for

$$i = 1, \ldots, n. \tag{388}$$

The decisive environments should alter the feasible trade-off structure rather than merely rescale familiar rewards.

9.3 Scalarised baseline

The simplest baseline controller optimises weighted aggregate reward:

$$R_t = \sum_{i=1}^{n} \lambda_i r_i(t), \tag{389}$$

where weights

$$\lambda_i \tag{390}$$

are fixed or context-conditioned according to a preregistered rule.

This baseline must be allowed serious optimisation.

A weak scalar controller would not provide a meaningful test of whether explicit conflict representation adds anything.

9.4 Structured conflict representation

The candidate architecture maintains separate estimates

$$\mathbf{c}_t = (c_{1t}, \ldots, c_{nt}), \tag{391}$$

where each c_{it} describes the predicted status or pressure associated with objective C_i.

Let H_t^{ctrl} denote the task-relevant information state available to the controller at time t. Its contents must be specified in advance and matched across the relevant comparison architectures. Depending upon the task, it may include current observations, permitted retained information and preregistered contextual variables.

Pairwise conflict is represented by

$$Q_{ij,t} = q\left(C_i, C_j \mid H_t^{\text{ctrl}}\right), \tag{392}$$

where q estimates incompatibility, trade-off or mutual obstruction under the controller's current information state.

The matrix

$$\mathbf{Q}_t = [Q_{ij,t}] \tag{393}$$

is not assumed to measure one universal quantity called paradox or conflict.

It is a task-specific representation whose causal utility must be tested.

9.5 Architecture classes

Comparatively, at least four architectures should be used.

Architecture C0: Scalar reward control. The controller receives only aggregate objective

$$R_t. \tag{394}$$

Architecture C1: Multi-objective control without explicit conflict matrix. The controller receives the separate objective vector

$$\mathbf{c}_t \tag{395}$$

but no explicit pairwise conflict representation.

Architecture C2: Structured conflict representation. The controller receives both

$$\mathbf{c}_t \tag{396}$$

and

$$\mathbf{Q}_t. \tag{397}$$

Architecture C3: Capacity-matched model-based control. The controller receives comparable parameters and compute and may model future consequences, but it does not contain an explicit conflict matrix.

This architecture is a serious alternative explanation for any apparent advantage of C2.

9.6 Conflict must be operational rather than linguistic

Operationally, the experiment should define conflict through consequences of joint satisfaction rather than through labels.

For objectives C_i and C_j, one candidate relation is:

$$Q_{ij,t} > 0 \tag{398}$$

when improving expected satisfaction of C_i under the current feasible action set systematically reduces achievable satisfaction of C_j.

Thus,

$$\boxed{\text{two goals named “in conflict” } \neq \text{ demonstrated objective incompatibility.}} \tag{399}$$

9.7 Test 1: Familiar trade-offs

Initially, all architectures should be tested on trade-offs represented during training.

If C2 strongly outperforms C0 here, the result may reflect additional capacity rather than transferable conflict representation.

Consequently, familiar-condition performance is primarily a calibration test, not the decisive experiment.

9.8 Test 2: Novel trade-off structure

Alter which objectives conflict while preserving familiar task elements.

For example, training may establish:

$$C_1 \leftrightarrow C_2, \tag{400}$$

while evaluation introduces:

$$C_1 \leftrightarrow C_3. \tag{401}$$

An operational transfer prediction associated with H6.1 is:

$$\begin{gathered}\text{explicit conflict representation improves transfer} \\ \text{when the structure of objective incompatibility changes.}\end{gathered} \tag{402}$$

The relevant comparison is against both scalarised and multi-objective controls.

9.9 Test 3: Conflict-matrix scrambling

Alter

$$\mathbf{Q}_t \tag{403}$$

while preserving ordinary task information.

Possible manipulations include:

1. permutation of objective-pair entries;
2. sign reversal of selected conflicts;
3. masking one conflict relation;
4. insertion of a false conflict;
5. removal of a genuine trade-off.

If the matrix is causally used, behaviour should change in the direction predicted by the corrupted representation.

Unchanged behaviour would weaken the claim that

$$\mathbf{Q}_t \tag{404}$$

participates directly in control.

9.10 Test 4: Conflict correction rescue

Restoratively, correct a deliberately corrupted conflict representation.

The intervention sequence is:

$$\mathbf{Q}_t \longrightarrow \widetilde{\mathbf{Q}}_t \longrightarrow \mathbf{Q}_t^{\text{restored}}. \tag{405}$$

A strong result would show:

$$\boxed{\begin{array}{c}\text{conflict corruption} \longrightarrow \text{predicted control error} \\ \longrightarrow \text{representation correction} \longrightarrow \text{behavioural rescue.}\end{array}} \tag{406}$$

9.11 Test 5: Scalarisation bypass

Allow the scalar baseline to use context-dependent weights:

$$\lambda_i = \lambda_i \left(H_t^{\text{ctrl}} \right). \tag{407}$$

The scalarised controller must receive the same task-relevant information available to the structured conflict architecture wherever the comparison requires information matching.

If flexible scalarisation matches the structured conflict architecture across all decisive transfer tests, a separate conflict matrix may be unnecessary.

This would support reduction to:

$$\text{context-sensitive multi-objective control} \tag{408}$$

rather than an independent conflict-negotiation mechanism.

9.12 Test 6: Model-based bypass

The C3 model-based bypass should be allowed to infer future trade-offs through ordinary world-model dynamics.

If C3 reproduces the benefits of C2 without representing

$$\mathbf{Q}_t \tag{409}$$

explicitly, then conflict information may already be implicit in predicted consequences.

Accordingly,

$$\boxed{\text{successful trade-off reasoning} \not\Rightarrow \text{explicit conflict representation.}} \tag{410}$$

9.13 Test 7: Higher-order conflict

Pairwise conflict representations may fail when three or more objectives interact.

For example:

$$C_1, \qquad C_2, \qquad C_3 \tag{411}$$

may be pairwise compatible while jointly infeasible.

The experiment should therefore include at least one higher-order case.

If pairwise matrix

$$\mathbf{Q}_t \tag{412}$$

fails systematically, the representation should be restricted rather than generalised beyond its demonstrated scope.

9.14 Test 8: Dynamic conflict

Objective incompatibility should sometimes change during an episode:

$$\mathbf{Q}_t \longrightarrow \mathbf{Q}_{t+1}. \tag{413}$$

The structured controller should update its representation from new evidence rather than retaining a fixed conflict map learned during training.

This distinguishes online trade-off inference from static task labelling.

9.15 Test 9: Conflict representation and self-monitoring

An optional interaction with Block V can test whether an agent recognises when its own conflict representation is unreliable.

Let estimate

$$q_t^Q \tag{414}$$

represent confidence in

$$\mathbf{Q}_t. \tag{415}$$

The experiment may ask whether low

$$q_t^Q \tag{416}$$

appropriately triggers additional observation or planning.

However, this interaction should not be required for the core conflict test.

9.16 Primary measurements

The auxiliary experiment should report:

1. objective-specific outcomes;
2. Pareto efficiency where meaningful;
3. violation rates for hard constraints;
4. transfer under novel trade-offs;
5. conflict-matrix corruption effect;
6. conflict-correction rescue effect;
7. scalarised-control comparison;
8. model-based-control comparison;
9. higher-order conflict failures;
10. adaptation to dynamically changing trade-offs.

No scalar "paradox score" or consciousness metric should be constructed.

9.17 Primary hypotheses

The auxiliary test evaluates:

Table 8: Primary hypotheses and break conditions for the conflict-representation auxiliary test.

ID	Hypothesis	Break condition
H6.1	Explicit conflict representation improves transfer when the structure of objective incompatibility changes.	Capacity-matched scalarised or multi-objective controls transfer equally well.
H6.2	The conflict representation is causally used during control.	Targeted scrambling produces no predicted behavioural change.
H6.3	Correcting a corrupted conflict representation selectively rescues behaviour.	Behaviour recovers without correction or does not recover after accurate restoration.

ID	Hypothesis	Break condition
H6.4	Explicit conflict representation provides information not fully reproduced by ordinary model-based prediction.	Capacity-matched model-based control matches all decisive conflict-transfer effects.
H6.5	The mechanism generalises beyond one fixed pairwise trade-off pattern.	Benefits disappear under higher-order, dynamic or previously unseen trade-offs.

9.18 Decision framework

The conflict-representation hypothesis should be evaluated independently of the five-block hierarchy.

Retain explicit conflict representation if it provides reproducible causal and transfer benefits beyond serious scalarised, multi-objective and model-based controls.

Restrict it if benefits occur only for particular forms of trade-off or task representation.

Reduce it if flexible scalarisation, ordinary constraint satisfaction or model-based prediction reproduces the decisive results.

Suspend interpretation if the conflict representation cannot be intervened upon independently of generic controller capacity.

Reject any claim that an independent conflict mechanism is necessary for the tested functions if robust bypass architectures perform equivalently.

9.19 No inference to consciousness

Phenomenologically, even a strong positive result would establish only that explicit representation of competing objectives can contribute to control under specified artificial conditions.

It would not establish:

$$\begin{gathered}\text{suffering},\\ \text{subjective conflict},\\ \text{phenomenal awareness},\\ \text{conscious deliberation},\\ \text{Paradox–Negotiation Theory as a theory of consciousness.}\end{gathered} \tag{417}$$

Therefore,

$$\boxed{\begin{array}{c}\text{functional conflict management}\\ \not\Rightarrow \text{ phenomenal conflict experience.}\end{array}} \tag{418}$$

9.20 Provisional role in the wider programme

This experiment remains auxiliary.

A positive result could justify retaining explicit conflict representation as one useful control architecture.

A negative result could justify reducing it to established multi-objective, constraint-satisfaction or predictive-control mechanisms.

Neither outcome changes Open Problem 1 concerning phenomenal consciousness.

The governing interpretation is:

$$\boxed{\begin{array}{c}\text{conflict representation must earn a functional role;}\\ \text{it receives no privileged status from PNT}\\ \text{or from the wider Theory of Everything project.}\end{array}} \tag{419}$$

10 Typed Measurement Framework

Paper III does not assume that one scalar or vector measures organisation, constraint, agency or intelligence across all experimental blocks. The experimental blocks investigate different explananda, operate over different state spaces and require different causal tests. Numerical comparability must therefore be demonstrated rather than presumed.

The governing distinction is:

$$\boxed{\begin{array}{c}\text{shared experimental methodology}\\ \neq\\ \text{shared physical or functional quantity.}\end{array}} \tag{420}$$

Where Γ_{trans} appears without an explicit superscript, it denotes the transition-divergence measurement family rather than a fully specified estimand. Confirmatory analyses must state the weighting distribution ρ and report

$$\Gamma^{(\rho)}_{\text{trans}}. \tag{421}$$

Likewise, use of Γ_{path} requires the relevant path distribution, baseline and evaluation horizon to be stated explicitly.

Accordingly, measurements are organised by scientific type and attached to explicit baseline, representation, explanandum, timescale and causal-status metadata.

10.1 The typed profile

Formally, where the Paper II constraint framework is useful, a comparison may be represented schematically by

$$\mathfrak{C} = [\mathfrak{D}, \mathfrak{F}, \mathfrak{T}, \mathfrak{O}] , \tag{422}$$

where

$$\mathfrak{D} = \text{dynamical divergence descriptors}, \tag{423}$$

$$\mathfrak{F} = \text{functional effect descriptors}, \tag{424}$$

$$\mathfrak{T} = \text{temporal and historical descriptors}, \tag{425}$$

$$\mathfrak{O} = \text{organisational and interaction descriptors}. \tag{426}$$

Crucially, these categories are not coordinates of one universal mathematical space.

No operation such as

$$\Gamma_{\text{total}} = \sum_k \Gamma_k \tag{427}$$

is permitted unless a future theory independently establishes why the terms are commensurable and why addition is scientifically meaningful.

10.2 Every measurement requires contextual metadata

The scientifically complete object is not merely a quantity Q.

It is a comparison of the form

$$Q\left[\mathcal{M}_1 \| \mathcal{M}_0; \mathfrak{R}, \Psi, T, \mathcal{K}\right] , \tag{428}$$

where:

1. $\mathcal{M}_1$ is the comparison condition;
2. $\mathcal{M}_0$ is the baseline or counterfactual;
3. $\mathfrak{R}$ is the chosen representation;
4. Ψ is the target explanandum;
5. T is the relevant time horizon;
6. $\mathcal{K}$ records whether the claim is descriptive, predictive, interventional, mechanistic or structural.

Changing any of these elements may change the meaning of the reported number.

Accordingly,

$$\boxed{\text{same numerical value} \not\Rightarrow \text{same scientific quantity.}} \tag{429}$$

10.3 Block I: maintenance measurements

Experimental Block I requires measures capable of distinguishing persistent patterning, operational viability and internally generated maintenance.

Relevant quantities include:

$$\begin{gathered} V^{\mathrm{path}}(T), \qquad V^{\mathrm{term}}(T), \\ \tau_{\mathrm{fail}}, \qquad \tau_{\mathrm{rec}}, \\ \Gamma_{\mathrm{trans}}, \qquad \Gamma_{\mathrm{path}}, \\ G_{\mathcal{C}}, \qquad \chi_{\mathrm{int}}, \qquad \phi_K. \end{gathered} \tag{430}$$

Spatial mutual information and spectral descriptors may additionally describe reaction–diffusion patterning.

However,

$$\boxed{\text{pattern structure} \neq \text{viability} \neq \text{internal maintenance.}} \tag{431}$$

The measures should therefore remain separate even when they covary.

10.4 Block II: evolutionary measurements

Experimental Block II requires lineage and population statistics rather than generic dynamical complexity.

Relevant measurements include:

$$\begin{gathered} \text{reproduction-event count}, \qquad \text{lineage depth}, \\ q, \qquad I(G_p; G_o \mid E), \\ p_t(g), \qquad w_i(T), \\ \mathrm{Cov}(w_i, z_i), \qquad \Gamma_{\mathrm{sel}}(t). \end{gathered} \tag{432}$$

Price-equation terms should additionally distinguish change associated with differential reproductive success from change arising through transmission or transformation.

Selection is therefore inferred only after a structured relation among four separately established conditions has been demonstrated:

$$\boxed{\begin{gathered} \text{reproduction} + \text{heredity} + \text{variation} \\ + \text{differential reproductive success.} \end{gathered}} \tag{433}$$

These conditions should not be inferred from a generic rise in reward, a fitness proxy, changing trait frequencies or increasing apparent complexity.

10.5 Block III: retention, feedback and learning measurements

Experimental Block III requires mechanism-specific measures.

Retention is characterised through:

$$\begin{gathered} R(\tau), \qquad R_{\text{causal}}(\Delta), \\ \Delta P_R\left(E_{\text{history}}\right), \qquad \Delta P_R\left(E_{\text{Markov}}\right). \end{gathered} \tag{434}$$

Feedback is characterised through:

$$\begin{gathered} \Delta P_F, \qquad P_{\text{closed}} - P_{\text{yoked}}, \\ \Gamma_{\text{trans}}, \qquad \Gamma_{\text{path}}, \qquad \delta/\tau_P. \end{gathered} \tag{435}$$

Learning is characterised through:

$$\begin{gathered} \Delta P_L, \qquad \text{adaptation time}, \\ \text{post-freeze performance}, \qquad \text{learning-ablation effect}. \end{gathered} \tag{436}$$

Importantly,

$$\boxed{R(\tau) \neq \Delta P_F \neq \Delta P_L.} \tag{437}$$

These quantities answer different causal questions and should not be added together.

10.6 Block IV: agency and structural-generalisation measurements

Experimental Block IV requires measurements of transfer under structural novelty rather than ordinary held-out accuracy alone.

Relevant quantities include:

$$\begin{gathered} P_{\text{interp}}, \qquad P_{\text{struct}}, \qquad G_{\text{struct}}, \\ n_{\text{adapt}}, \qquad L_{\text{model}}, \\ \text{model-corruption effect}, \qquad \text{model-correction rescue effect}, \\ \text{rollout-ablation effect}, \qquad \text{planning-horizon sensitivity}. \end{gathered} \tag{438}$$

Performance should also be disaggregated across novelty classes:

$$\begin{gathered} \text{compositional novelty}, \quad \text{causal novelty}, \\ \text{delayed consequences}, \quad \text{latent-state inference}, \\ \text{problem-family transfer}. \end{gathered} \tag{439}$$

No total intelligence score is inferred from these measures.

A system can generalise strongly in one dimension while failing another.

10.7 Block V: reflective self-modelling measurements

Experimental Block V requires a distinction between accuracy of self-prediction and causal use of self-related representations.

Descriptive and predictive measures include:

$$\begin{gathered} \text{calibration error}, \quad \text{Brier score}, \\ \text{selective accuracy}, \quad \text{coverage}, \\ \tau_{\text{detect}}. \end{gathered} \tag{440}$$

Intervention-sensitive measures include:

$$\begin{gathered} \text{self-model misinformation effect}, \\ \text{self-model correction rescue effect}, \\ \text{shuffled-monitor performance}, \\ \text{adaptive-computation effect}, \\ \text{recovery after internal corruption}. \end{gathered} \tag{441}$$

Thus,

$$\boxed{\text{self-prediction quality} \neq \text{causal self-model use}.} \tag{442}$$

No scalar self-awareness or consciousness score is constructed.

10.8 Auxiliary conflict measurements

The auxiliary conflict-representation experiment uses its own task-specific measurements.

These include:

$$\begin{gathered}\text{objective-specific outcomes,}\\ \text{Pareto efficiency where defined,}\\ \text{hard-constraint violation rates,}\\ \text{novel trade-off transfer,}\\ \text{conflict-corruption effects,}\\ \text{conflict-correction rescue effects.}\end{gathered} \tag{443}$$

These quantities are not appended to

$$\mathfrak{C} \tag{444}$$

merely because the auxiliary experiment occurs in the same paper.

Conflict representation remains a mechanism-specific hypothesis.

10.9 Cross-block measurement table

The role of the principal measurement families can be summarised as follows:

Table 9: Typed measurement roles across the experimental blocks.

Block	**Primary measurement type**	**Representative quantities**	**What they do not establish**
I	Dynamical, functional, organisational	$V^{\mathrm{path}}(T)$, $V^{\mathrm{term}}(T)$, τ_{fail}, τ_{rec}, $G_{\mathcal{C}}$, χ_{int}	Life, heredity or intelligence
II	Historical and population-level	q, $I(G_p; G_o \mid E)$, $p_t(g)$, w_i, Price terms	Progress, increasing complexity or intelligence
III	Historical, control and adaptation	ΔP_R, ΔP_F, ΔP_L	General agency or explicit world modelling
IV	Generalisation and model-use	G_{struct}, n_{adapt}, model corruption and rescue	Reflective self-modelling or consciousness
V	Metacognitive and intervention-sensitive	Calibration, Brier score, monitor manipulation and rescue	Phenomenal awareness
Auxiliary	Multi-objective control	Trade-off transfer, constraint violations, conflict perturbation	Conscious conflict or PNT

10.10 Repeated quantities do not automatically have repeated meaning

Some quantities may appear in more than one block.

For example,

$$\Gamma_{\text{path}} \tag{445}$$

can compare maintenance trajectories in Block I, feedback trajectories in Block III or planning trajectories in Block IV.

Its mathematical form may be identical while its explanandum differs.

Therefore:

$$\boxed{\begin{array}{c}\text{same formal metric} + \text{different explanandum} \\ \neq \text{one universal measured property.}\end{array}} \tag{446}$$

Cross-block comparison requires the accompanying metadata to be examined explicitly.

10.11 Baseline matching

Every effect measure requires a defensible baseline.

For quantity Q,

$$Q = Q\left(\mathcal{M}_1 \| \mathcal{M}_0\right). \tag{447}$$

Relevant baselines include:

$$\begin{array}{c}\text{passive maintenance control,} \\ \text{non-heritable reproduction,} \\ \text{retention-ablated architecture,} \\ \text{open-loop or yoked control,} \\ \text{capacity-matched model-free agent,} \\ \text{passive self-monitor.}\end{array} \tag{448}$$

Baseline choice should follow the scientific role being tested rather than the baseline that maximises the effect size.

10.12 Representation robustness

Measurements should be tested under scientifically reasonable changes of:

$$\begin{array}{cc}\text{state encoding,} & \text{observation resolution,} \\ \text{system boundary,} & \text{sampling interval,} \\ \multicolumn{2}{c}{\text{evaluation horizon.}}\end{array} \tag{449}$$

A quantity need not remain numerically invariant.

However, if its sign, ranking or substantive interpretation changes arbitrarily under nearby

defensible representations, the corresponding claim should be restricted.

This is particularly important for:

$$R(\tau), \qquad \chi_{\text{int}}, \qquad G_{\text{struct}}, \tag{450}$$

because all depend explicitly upon representation or boundary choices.

10.13 Causal status must be labelled

Paper III distinguishes at least five evidential levels:

$$\begin{gathered} \text{descriptive}, \\ \text{predictive}, \\ \text{interventional}, \\ \text{mechanistic}, \\ \text{structural}. \end{gathered} \tag{451}$$

A descriptive quantity should not be promoted automatically.

For example:

$$I(M_t; H_{t-\tau}) > 0 \tag{452}$$

shows retained information under a specified representation.

It does not by itself establish:

$$M_t \longrightarrow A_{t+\Delta}. \tag{453}$$

Likewise, strong calibration does not establish causal self-model use, and strong trait–fitness covariance does not by itself establish that the trait causes reproductive success.

10.14 Intervention–rescue pairs receive special weight

Several blocks now contain a common causal pattern:

$$\boxed{\begin{gathered} \text{intact function} \\ \longrightarrow \text{targeted disruption} \\ \longrightarrow \text{predicted selective deficit} \\ \longrightarrow \text{functional restoration} \\ \longrightarrow \text{behavioural rescue}. \end{gathered}} \tag{454}$$

Examples include:

$$\begin{gathered}\text{maintenance-node ablation and rescue,}\\ \text{world-model corruption and correction,}\\ \text{self-model misinformation and correction,}\\ \text{conflict-map corruption and correction.}\end{gathered} \tag{455}$$

This pattern is methodologically shared across blocks even though the measured quantities differ.

Successful intervention–rescue structure provides stronger mechanistic evidence than an uncontrolled performance difference.

10.15 No monotonic hierarchy score

The experiment should not expect:

$$Q_{\text{Block I}} < Q_{\text{Block II}} < Q_{\text{Block III}} < Q_{\text{Block IV}} < Q_{\text{Block V}} \tag{456}$$

for any universal quantity Q.

There is presently no justified mathematical object for which such an ordering has meaning.

Accordingly,

$$\boxed{\text{later experimental block} \neq \text{larger value of one universal property.}} \tag{457}$$

A later block poses a different dependency question; it does not occupy a higher point on a numerical scale of organisation.

10.16 Cross-block relations may nevertheless be tested

Rejecting a common scalar does not prevent meaningful cross-block hypotheses.

For example, the programme may test whether:

$$R_{\text{causal}}(\Delta) \tag{458}$$

predicts adaptation speed in some Block III tasks, or whether Block IV structural generalisation correlates with Block V calibration under novelty.

Such hypotheses must be preregistered and interpreted as relations between distinct quantities.

A candidate relation has the form:

$$Q_B = \mathcal{T}_{A\to B}\left(Q_A\right) + \varepsilon, \tag{459}$$

rather than:

$$Q_A = Q_B \tag{460}$$

by assumption.

The transformation

$$\mathcal{T}_{A \to B} \tag{461}$$

must earn empirical support.

10.17 Cross-scale claims require a still higher burden

Successful relationships among artificial blocks do not establish that equivalent relationships hold in biological or physical systems.

Any later mapping must satisfy the Paper II validation sequence:

$$\boxed{\begin{array}{c}\text{within-domain validity} \\ \longrightarrow \text{preregistered mapping} \\ \longrightarrow \text{held-out cross-domain prediction} \\ \longrightarrow \text{replication and generalisation.}\end{array}} \tag{462}$$

Paper III therefore generates candidate relations for later cross-domain testing rather than treating artificial correspondence as universal evidence.

10.18 Measurement hierarchy

Measurement claims should progress through the following levels:

$$\boxed{\begin{array}{c}\text{quantity is well defined} \\ \downarrow \\ \text{quantity is reliably measured} \\ \downarrow \\ \text{quantity predicts the target phenomenon} \\ \downarrow \\ \text{intervention changes the target as predicted} \\ \downarrow \\ \text{mechanism survives matched alternatives.}\end{array}} \tag{463}$$

Failure at one level prevents automatic promotion to the next.

This protects the experiment from interpreting sophisticated metrics as mechanisms merely

because they correlate with successful behaviour.

10.19 Reporting standard

Every principal result should report:

1. the quantity measured;
2. its mathematical definition;
3. baseline and comparison conditions;
4. representation and system boundary;
5. time horizon;
6. causal-status label;
7. effect size and uncertainty;
8. relevant ablation or intervention;
9. robustness to predefined alternatives;
10. the decision status assigned to the claim.

This information should accompany tables and machine-readable result files where practicable.

10.20 Provisional judgement

Paper III requires a common experimental discipline but not common numerical measurements. The correct architecture is:

$$\boxed{\begin{array}{c}\text{common causal standards}\\ \text{+block-specific measurements}\\ \text{+explicit contextual metadata}\\ \text{+tested relations where justified.}\end{array}} \tag{464}$$

Accordingly,

$$\boxed{\begin{array}{c}\text{measure what each mechanism actually changes;}\\ \text{do not manufacture unity by forcing}\\ \text{different phenomena onto one scale.}\end{array}} \tag{465}$$

11 Global Controls Against False Hierarchy and False Emergence

An artificial hierarchy is unusually vulnerable to false positive results because the researcher specifies the architectures, environments, information channels, objectives, training procedures and evaluation criteria. A capability can therefore appear to "emerge" because it was indirectly supplied by the experimental design.

Likewise, a hierarchy can appear cumulative simply because later systems receive more parameters, computation, information or training than earlier ones.

The governing concern is:

$$\boxed{\begin{array}{l}\text{observed progression}\\ \not\Rightarrow \text{ causal hierarchy.}\end{array}} \tag{466}$$

Every block must therefore distinguish genuine mechanism-specific effects from artefacts introduced by experimental construction.

11.1 False hierarchy and false emergence are different problems

Two classes of error should be distinguished.

False emergence occurs when a capability appears to arise from system dynamics but has actually been encoded, privileged or rewarded directly by the experimenter.

False hierarchy occurs when later stages appear more capable because they systematically receive greater resources or because earlier mechanisms have been bundled into later architectures without testing their independent necessity.

Thus:

$$\boxed{\begin{array}{l}\text{false emergence} = \text{capability supplied by design,}\\ \text{false hierarchy} = \text{progression supplied by design.}\end{array}} \tag{467}$$

Both must be controlled.

11.2 Parameter-count control

Adding retention, predictive modelling, self-monitoring or another mechanism usually increases the number of trainable parameters.

Each principal architecture should therefore face a control receiving approximately comparable parameter capacity without the hypothesised structure.

Let:

$$N_\theta^{(A)} \tag{468}$$

and

$$N_\theta^{(B)} \tag{469}$$

denote trainable parameter counts.

Where exact equality is feasible:

$$N_\theta^{(A)} \approx N_\theta^{(B)}. \tag{470}$$

Residual differences should be reported explicitly.

A performance advantage that disappears under capacity matching should be attributed to generic model capacity rather than automatically to the proposed mechanism.

11.3 Compute control

Architectures may differ even when parameter counts are similar.

Relevant resources include:

$$\begin{gathered}\text{training environment steps,}\\ \text{gradient or optimisation updates,}\\ \text{training compute,}\\ \text{inference-time compute,}\\ \text{planning rollouts,}\\ \text{memory-access operations.}\end{gathered} \tag{471}$$

Where practicable, comparisons should match these resources separately.

For example, a model-based planner receiving one hundred simulated rollouts per decision should not be compared only with a model-free controller receiving one forward pass.

If extra computation is itself part of the candidate mechanism, the analysis should report both:

$$\text{raw performance} \tag{472}$$

and

$$\text{performance at matched computational budget.} \tag{473}$$

11.4 Training-data and experience control

Mechanisms should not receive unequal access to relevant training information unless that difference is itself being tested.

Architectures should therefore be matched, where possible, on:

$$\begin{gathered}\text{number of episodes,} \qquad \text{environment interactions,} \\ \text{task-family exposure,} \qquad \text{feedback frequency,} \\ \text{curriculum duration.}\end{gathered} \tag{474}$$

A later architecture should not be credited with structural generalisation if it has simply encountered a wider structural training distribution.

11.5 Information-access control

All capacity-matched architectures should receive equivalent external information unless privileged information is the manipulated factor.

If Architecture A observes:

$$O_t, \tag{475}$$

while Architecture B observes:

$$(O_t, Z_t), \tag{476}$$

then improved performance by B cannot automatically be attributed to internal organisation.

This is particularly important for:

$$\begin{gathered}\text{retention tests,} \\ \text{hidden-state inference,} \\ \text{self-monitoring,} \\ \text{conflict representation.}\end{gathered} \tag{477}$$

Privileged variables must be identified explicitly.

11.6 Hidden-state leakage control

Tasks intended to require memory or latent-state inference must be checked for unintended present-state cues.

Suppose hidden variable is

$$Z_t. \tag{478}$$

If some observable feature

$$O_t^{(k)} \tag{479}$$

predicts Z_t with high accuracy, an apparently history-dependent task may actually be solvable reactively.

Accordingly, each decisive partially observable environment should undergo a leakage audit testing whether:

$$Z_t \tag{480}$$

can be reconstructed from the current observation alone.

A retention advantage is interpretable only if relevant historical information is genuinely absent from the matched present-state observation.

11.7 Target leakage control

Evaluation targets must not be inferable from implementation artefacts unrelated to the intended mechanism.

Potential leakage channels include:

1. fixed task ordering;
2. deterministic file or environment identifiers;
3. episode length;
4. observation padding;
5. action availability;
6. reward timing;
7. seed-dependent visual or numerical artefacts.

A simple control model should be trained on suspected nuisance variables alone.

If nuisance variables predict the decisive target, the environment must be revised before confirmatory testing.

11.8 Reward-shaping control

A capability built directly into the reward function should not be described as emergent.

Let total training reward be:

$$R_t = R_t^{\text{task}} + R_t^{\text{shape}}. \tag{481}$$

All shaping terms

$$R_t^{\text{shape}} \tag{482}$$

must be reported.

Where possible, decisive comparisons should include a sparse or mechanism-neutral objective.

For example, explicitly rewarding calibrated confidence weakens a claim that self-monitoring emerged independently.

Likewise, directly rewarding conflict detection weakens the claim that a distinct conflict representation arose because it was functionally required.

11.9 Hard-coded mechanism control

A mechanism should not receive evidential credit for properties imposed directly by construction.

Examples include:

$$\begin{gathered}\text{predefined maintenance cycles,}\\ \text{explicit parent–offspring labels supplied to the policy,}\\ \text{hand-coded causal graphs,}\\ \text{precomputed confidence thresholds,}\\ \text{fixed conflict matrices.}\end{gathered} \tag{483}$$

Such structures may be useful experimental interventions, but the paper must distinguish:

$$\boxed{\text{engineered mechanism} \neq \text{emergent mechanism.}} \tag{484}$$

The experiment can still test the functional sufficiency of an engineered mechanism.

It cannot claim to have explained its origin.

11.10 Architecture-label control

Architecture names should not substitute for evidence.

A module labelled:

$$\text{memory,} \qquad \text{world model,} \qquad \text{self-monitor,} \qquad \text{conflict representation} \tag{485}$$

earns that functional interpretation only after the relevant intervention tests succeed.

Accordingly,

$$\boxed{\text{implementation label} \neq \text{validated mechanism.}} \tag{486}$$

Where possible, architecture identifiers should be blinded during primary statistical analysis.

11.11 Task-difficulty matching

Matched task pairs should isolate mechanism relevance without introducing large uncontrolled difficulty differences.

For candidate pair:

$$E_A \qquad \text{and} \qquad E_B, \tag{487}$$

one environment should require the proposed mechanism more strongly while remaining comparable in:

$$\begin{gathered} \text{episode length}, \qquad \text{action-space size}, \\ \text{observation complexity}, \qquad \text{baseline success rate}, \\ \text{reward opportunity}. \end{gathered} \tag{488}$$

Exact difficulty equivalence may be impossible.

The remaining mismatch should therefore be quantified rather than ignored.

11.12 Curriculum leakage control

Training curricula can teach the decisive evaluation structure implicitly.

Suppose evaluation contains transformation:

$$T^*. \tag{489}$$

The training curriculum should be audited for transformations sufficiently similar to:

$$T^* \tag{490}$$

that the test reduces to familiar interpolation.

Structural generalisation claims require that the decisive structure genuinely be held out.

Curriculum construction should therefore be frozen before final evaluation.

11.13 Evaluation-set isolation

Decisive environments must not influence architecture tuning.

The separation should be:

$$\boxed{\begin{gathered} \mathcal{D}_{\text{development}} \\ \cap \\ \mathcal{D}_{\text{decisive}} = \varnothing \end{gathered}} \tag{491}$$

at the level of episodes or structural variants specified by the preregistration.

Once decisive results have been inspected, subsequent architectural changes belong to a new experimental version.

11.14 Ablation-integrity control

An ablation can create a false mechanism-specific deficit by damaging generic computational function.

For proposed mechanism M, compare:

$$\mathrm{do}(M = 0) \tag{492}$$

with at least one matched intervention of similar magnitude that does not target the hypothesised function.

Useful controls include:

1. parameter permutation of equal size;
2. matched channel masking;
3. replacement with noise of comparable variance;
4. ablation of an irrelevant module;
5. function-preserving architectural substitution.

A selective deficit supports the mechanism more strongly than generic performance collapse.

11.15 Rescue-specificity control

Rescue experiments can also generate false positives if any additional capacity improves the damaged system.

Therefore, after targeted disruption, compare:

$$\begin{gathered}\text{specific functional rescue,}\\ \text{irrelevant matched replacement,}\\ \text{generic added capacity.}\end{gathered} \tag{493}$$

Mechanistic evidence is strongest when only the functionally appropriate replacement restores the predicted deficit.

11.16 Yoked-control integrity

Yoked replay should preserve the realised action sequence while breaking the online observation–action dependency.

The replay condition must not accidentally receive:

$$\begin{gathered}\text{future observations},\\ \text{disturbance timing},\\ \text{episode-specific privileged information.}\end{gathered} \tag{494}$$

Likewise, the original action sequence should be generated under matched conditions.

A contaminated yoked control would not isolate feedback.

11.17 Selection-null integrity

Neutral and random-search controls must preserve as many non-selective processes as possible.

For the neutral population:

$$\begin{gathered}\text{mutation}, \qquad \text{finite population sampling},\\ \text{reproduction count}, \qquad \text{environmental exposure}\end{gathered} \tag{495}$$

should be matched where feasible.

Otherwise, divergence between selected and neutral populations may reflect different search budgets rather than differential reproductive success.

11.18 World-model null integrity

The model-free control must be allowed to be genuinely competitive.

It should receive comparable:

$$\begin{gathered}\text{training experience}, \qquad \text{parameter capacity},\\ \text{optimisation effort}, \qquad \text{inference-time compute.}\end{gathered} \tag{496}$$

A model-based architecture does not demonstrate explanatory necessity by outperforming a deliberately weak reactive baseline.

11.19 Self-monitor null integrity

Self-monitoring should be compared with:

$$\begin{gathered}\text{passive monitoring,}\\ \text{shuffled monitoring,}\\ \text{external monitoring,}\\ \text{capacity-matched non-self processing.}\end{gathered} \tag{497}$$

This distinguishes the value of:

$$\text{self-related information} \tag{498}$$

from the value of:

$$\text{extra information or compute.} \tag{499}$$

11.20 Bypass controls are mandatory

Every proposed necessary dependency should face a serious architecture designed to realise the target capability by another route.

For:

$$A \longrightarrow B, \tag{500}$$

the experiment should search for:

$$\neg A \wedge B. \tag{501}$$

The bypass must not be artificially crippled.

It should receive sufficient training and computational opportunity to be a credible competitor.

A robust bypass is evidence against necessity even when the preferred mechanism also works.

11.21 No forced monotonic architecture growth

Later blocks should not automatically inherit every mechanism and resource from earlier blocks.

Otherwise:

$$\text{Block } n+1 \tag{502}$$

would almost necessarily have greater expressive capacity than:

$$\text{Block } n. \tag{503}$$

Instead, later experiments should use factorial or matched architectures that remove unnecessary inherited mechanisms where technically possible.

Accordingly,

$$\boxed{\text{later block} \neq \text{strict superset architecture.}} \tag{504}$$

This is a central control against manufacturing an artificial hierarchy.

11.22 Cross-block contamination control

Results from one block should not silently redefine the tasks or metrics of a later block.

For example:

1. Block III retention success should not be built directly into the definition of Block IV intelligence;
2. Block IV world-model success should not be required by definition for Block V reflection;
3. Block V self-monitoring should not be used to define successful conflict representation.

Later capacities must be operationalised independently enough that bypass architectures remain possible.

11.23 Implementation-artefact control

A result may depend upon one software implementation rather than the hypothesised mechanism.

Where feasible, important results should be repeated using at least one variation in:

$$\begin{gathered}\text{network architecture}, \qquad \text{optimiser}, \\ \text{initialisation}, \qquad \text{environment implementation}.\end{gathered} \tag{505}$$

A claim that disappears under minor implementation changes should be restricted accordingly.

11.24 Environment-code validation

Environment logic should be independently tested.

Unit and property tests should verify:

1. transition rules;
2. reward calculations;
3. hidden-state behaviour;
4. reproduction events;

5. intervention implementation;
6. boundary conditions;
7. randomisation procedures.

Where practicable, analytic or hand-computed test cases should be included.

An environment bug that favours one architecture can mimic a theoretical result.

11.25 Random-seed robustness

No principal claim should depend upon one favourable initialisation or environment seed.

For inexpensive experiments, a substantial number of independent seeds should be used.

The original provisional target:

$$N \geq 30 \tag{506}$$

may remain a useful default for low-cost experiments, but it should not be treated as universally sufficient.

More expensive blocks may use fewer replicates only when justified through:

$$\text{power analysis}, \qquad \text{effect-size uncertainty}, \qquad \text{computational feasibility}. \tag{507}$$

Seed distributions and generation procedures should be preregistered.

11.26 Researcher degrees of freedom

Researcher flexibility can create apparent confirmation even when no individual decision is unreasonable.

The following should be frozen before decisive runs:

$$\begin{gathered} \text{environment variants}, \qquad \text{primary metrics}, \\ \text{stopping rules}, \qquad \text{exclusion criteria}, \\ \text{architecture comparisons}, \qquad \text{statistical models}. \end{gathered} \tag{508}$$

Exploratory alternatives remain scientifically useful but must be labelled as exploratory and tested independently before promotion.

11.27 Multiple-hypothesis control

The experiment contains many architectures, perturbations, task families and outcome measures.

Without control, one apparently strong result may arise by chance.

Families of confirmatory comparisons should therefore be identified in advance and analysed with an appropriate multiplicity strategy.

Equally important, null results should not be discarded merely because another metric within the same family produces significance.

Effect size, uncertainty and consistency across replications remain more important than isolated threshold crossing.

11.28 Researcher-blinding where practical

Some analyses can be made less vulnerable to expectation effects.

Where feasible:

1. architecture identities can be coded during primary analysis;
2. decisive test environments can be generated automatically from preregistered seeds;
3. inclusion criteria can be applied before revealing condition labels;
4. primary figures can be generated through fixed scripts.

Full blinding may not be possible in an engineering experiment, but partial blinding can still reduce discretionary interpretation.

11.29 Negative-result preservation

Failed architectures, null findings and unsuccessful replications should be retained when they satisfy preregistered inclusion criteria.

This includes:

$$\begin{gathered}\text{failed mechanism advantages,}\\ \text{successful bypasses,}\\ \text{failed rescue experiments,}\\ \text{representation-sensitive effects.}\end{gathered} \tag{509}$$

Suppressing such outcomes would bias the apparent hierarchy towards successful transitions.

11.30 Adversarial control matrix

The principal false-positive risks can be summarised as follows:

Table 10: Global controls against false hierarchy and false emergence.

False-positive source	**Misleading interpretation**	**Required control**
More parameters	Mechanism appears beneficial	Capacity-matched control
More compute	Planning or modelling appears superior	Matched training and inference budgets
Privileged information	Memory or modelling appears necessary	Information-access and leakage audit
Reward shaping	Capability appears emergent	Mechanism-neutral or sparse objective control
Hard-coded structure	Mechanism appears discovered	Distinguish engineered sufficiency from emergence
Weak null architecture	Preferred theory wins trivially	Serious competitor capable of winning
Destructive ablation	Mechanism appears causally necessary	Matched-damage and rescue controls
Training-test contamination	Generalisation appears stronger than it is	Held-out structural evaluation
Increasing architecture size across blocks	Hierarchy appears cumulative	Factorial and non-superset comparisons
Post-hoc metric choice	Desired effect appears significant	Preregistered primary outcomes
Implementation artefact	Mechanism appears universal	Independent implementation variants
Selective reporting	Hierarchy appears cleaner than evidence	Archive null and failed results

11.31 Global break condition

The experimental architecture should be considered compromised if its principal progression disappears after controlling for generic capacity, information access, compute, reward design and serious bypass architectures.

A global break condition is:

$$\boxed{\begin{array}{c}\text{if the hierarchy survives only because later systems}\\ \text{receive more resources, privileged information}\\ \text{or researcher-specified structure,}\\ \text{the hierarchy has not been demonstrated.}\end{array}} \tag{510}$$

This remains true even if every individual system performs impressively.

11.32 Provisional judgement

Paper III should be harder to confirm than to falsify.

The experiment is credible only if alternative explanations are given genuine opportunities to succeed:

$$\boxed{\begin{array}{c}\text{match capacity} + \text{match information} + \text{match compute}\\ +\text{isolate the mechanism} + \text{search for bypasses} + \text{preserve negative results.}\end{array}} \tag{511}$$

The governing rule is:

$$\boxed{\begin{array}{c}\text{do not ask whether the preferred architecture can work;}\\ \text{ask whether it still earns explanatory credit}\\ \text{after serious alternatives are allowed to work too.}\end{array}} \tag{512}$$

12 Statistical Analysis and Robustness

Paper III is designed to estimate selective causal and functional differences rather than to search for isolated significant performance improvements. The experimental blocks contain different outcome types, dependence structures and intervention designs, so no single statistical model is appropriate for the entire programme.

The governing principle is:

$$\boxed{\begin{array}{c}\text{the statistical model must follow}\\ \text{the experimental question and data-generating structure,}\\ \text{not impose one common analysis across all blocks.}\end{array}} \tag{513}$$

Primary analyses should therefore be preregistered separately for each experimental block while following common standards for estimands, uncertainty, multiplicity, robustness and replication.

12.1 Estimands precede significance tests

Every primary hypothesis should identify the quantity whose difference is scientifically relevant before specifying a test statistic.

For conditions A and B, a generic average treatment contrast is:

$$\Delta_{A,B} = \mathbb{E}[Y \mid A] - \mathbb{E}[Y \mid B]. \tag{514}$$

Where mechanism specificity depends upon environment class, the principal estimand may instead be an interaction:

$$\begin{aligned} \Delta_{\text{int}} = &(\mathbb{E}[Y \mid M = 1, E = 1] - \mathbb{E}[Y \mid M = 0, E = 1]) \\ &- (\mathbb{E}[Y \mid M = 1, E = 0] - \mathbb{E}[Y \mid M = 0, E = 0]) \,. \end{aligned} \tag{515}$$

This quantity asks whether the mechanism provides a larger advantage when its hypothesised function is required.

Accordingly,

$$\boxed{\text{mechanism specificity} > \text{unconditional performance advantage}} \tag{516}$$

as an evidential target.

12.2 Effect sizes and uncertainty are primary

Every principal comparison should report:

1. point estimate;
2. confidence or credible interval;
3. standardised or interpretable raw effect size;
4. replicate count;
5. relevant variance components;
6. prespecified decision threshold where applicable.

Null-hypothesis p-values may be reported where useful but should not be the sole basis for interpretation.

In particular,

$$p < 0.05 \tag{517}$$

does not by itself establish:

$$\begin{gathered}\text{mechanistic importance,}\\ \text{practical relevance,}\\ \text{cross-architecture robustness,}\\ \text{cross-domain validity.}\end{gathered} \tag{518}$$

12.3 Experimental unit must be defined correctly

The unit of statistical replication must match the unit that was independently generated.

Possible units include:

$$\begin{gathered}\text{random seed,} \qquad \text{independent simulation run,}\\ \text{replicate population,} \qquad \text{independently trained agent,}\\ \text{environment instance.}\end{gathered} \tag{519}$$

Repeated observations from one trained agent or one evolving population are not independent replicates merely because many time points or individuals are recorded.

Therefore,

$$\boxed{\text{large trajectory count} \neq \text{large independent sample size.}} \tag{520}$$

12.4 Hierarchical modelling

Many Paper III outcomes naturally contain nested sources of variation.

A generic model is:

$$\begin{aligned}y_{ijks} = \beta_0 + \beta_1 M_i + \beta_2 E_j + \beta_3 (M_i E_j)\\ + u_k + v_s + \epsilon_{ijks},\end{aligned} \tag{521}$$

where:

- M_i is mechanism or architecture condition;
- E_j is environment or task class;
- u_k is an environment-instance effect;
- v_s is an independent seed or replicate effect.

Additional levels should be included only where supported by the experimental design.

The model family should match the outcome distribution rather than assuming Gaussian errors automatically.

12.5 Block I: viability and failure-time analysis

Block I contains outcomes for which ordinary means are insufficient.

For failure time

$$T_{\text{fail}}, \tag{522}$$

the survival function is:

$$S(t) = P\left(T_{\text{fail}} > t\right). \tag{523}$$

The analysis should report:

$$\begin{gathered} \text{survival curves,} \qquad \text{median failure time,} \\ \text{restricted mean survival time where appropriate,} \\ \text{hazard or equivalent model contrasts.} \end{gathered} \tag{524}$$

Censoring rules must be specified before analysis.

Where proportional-hazards assumptions do not hold, alternative survival models or time-varying effects should be used rather than forcing one parametric interpretation.

Recovery time

$$\tau_{\text{rec}} \tag{525}$$

should be analysed separately from survival.

A system can survive disturbance without recovering its previous operational state.

12.6 Block I: intervention and rescue contrasts

Maintenance-node experiments should distinguish three conditions:

$$\begin{gathered} \text{intact,} \\ \text{ablated,} \\ \text{ablated + specific rescue.} \end{gathered} \tag{526}$$

A useful rescue contrast is:

$$\Delta_{\text{rescue}} = \mathbb{E}[Y \mid \text{rescue}] - \mathbb{E}[Y \mid \text{ablated}]. \tag{527}$$

Specificity requires comparison with irrelevant rescue:

$$\begin{aligned} \Delta_{\text{specific}} = \mathbb{E}[Y \mid \text{specific rescue}] \\ - \mathbb{E}[Y \mid \text{irrelevant rescue}]. \end{aligned} \tag{528}$$

Evidence is strongest when both contrasts favour the hypothesised function.

12.7 Block II: populations are the primary replicates

Individuals within one population share ancestry, environmental history and population dynamics. They should therefore not be treated as independent observations for population-level selection claims.

The preferred confirmatory unit is the independently initialised replicate population.

Population-level summaries include:

$$\begin{gathered} p_t(g), \qquad \bar{w}_t, \qquad \mathrm{Cov}(w_i, z_i), \\ \text{lineage depth}, \qquad \text{extinction probability}. \end{gathered} \tag{529}$$

Within-population individual data remain useful for modelling but require hierarchical or lineage-aware treatment.

12.8 Block II: lineage dependence

Related individuals are statistically dependent because they share ancestry.

Where lineage-level outcomes are analysed, models should account for:

$$\begin{gathered} \text{shared ancestry}, \\ \text{generation}, \\ \text{population replicate}, \\ \text{environmental regime}. \end{gathered} \tag{530}$$

Simple regression treating every descendant as an independent sample should be avoided.

Where practical, resampling should occur at the population or lineage level rather than at the individual-agent level.

12.9 Block II: Price-equation quantities

The Price equation should be analysed as a descriptive partition of population change:

$$\Delta \bar{z} = \frac{1}{\bar{w}} \mathrm{Cov}(w_i, z_i) + \frac{1}{\bar{w}} \mathbb{E}\left[w_i \Delta z_i\right]. \tag{531}$$

Uncertainty in both components should be reported across replicate populations.

The selection covariance term should not be promoted into causal evidence without the independent intervention tests specified in Block II.

Thus,

$$\boxed{\text{Price decomposition} \neq \text{causal identification by itself.}} \tag{532}$$

12.10 Block II: extinction is an outcome, not missing data

A population that becomes extinct under a preregistered condition has produced a scientifically meaningful result.

Extinct replicates should not be excluded simply because later trait or performance measurements cannot be collected.

Analyses should distinguish:

$$\begin{gathered}\text{conditional performance among surviving populations} \\ \text{from} \\ \text{overall population-level success including extinction.}\end{gathered} \tag{533}$$

Otherwise survivorship bias could make maladaptive regimes appear successful.

12.11 Block III: factorial analysis

Block III is explicitly designed around:

$$R \times F \times L. \tag{534}$$

A suitable confirmatory model includes:

$$\begin{aligned} Y = {} & \beta_0 + \beta_R R + \beta_F F + \beta_L L \\ & + \beta_{RF} RF + \beta_{RL} RL + \beta_{FL} FL \\ & + \beta_{RFL} RFL + \varepsilon. \end{aligned} \tag{535}$$

Environment requirement should be added as a factor when mechanism specificity is the target.

For example, retention specificity concerns:

$$R \times E_{\text{history}}. \tag{536}$$

Feedback specificity concerns:

$$F \times E_{\text{unpredictable}}. \tag{537}$$

Learning specificity concerns:

$$L \times E_{\text{changed}}. \tag{538}$$

These interactions are more informative than asking whether R, F or L improves mean reward in every environment.

12.12 Block III: interaction terms are scale-dependent

Apparent synergy or antagonism depends upon the measurement scale.

For retention and feedback:

$$I_{RF}^{P} = P_{11} - P_{10} - P_{01} + P_{00}. \tag{539}$$

A non-zero value should be described as an interaction on the specified performance scale.

It should not automatically be interpreted as:

$$\text{emergence}, \tag{540}$$

or:

$$\text{new universal mechanism}. \tag{541}$$

Where conclusions depend strongly upon scale choice, that dependence should be reported.

12.13 Block III: retention-delay analysis

Retention quantities

$$R(\tau) \tag{542}$$

and

$$R_{\text{causal}}(\Delta) \tag{543}$$

should be estimated across preregistered sets of historical lags τ and future effect horizons Δ, respectively.

The resulting profiles should be analysed rather than selecting whichever lag or effect horizon produces the largest association.

Relevant summaries may include:

$$\begin{gathered}\text{peak retention},\\ \text{decay timescale},\\ \text{area under the retention curve},\end{gathered} \tag{544}$$

provided those summaries are defined before decisive evaluation.

12.14 Block IV: structural novelty is a factor

Block IV should model novelty class explicitly rather than treating all held-out tasks as equivalent.

A schematic model is:

$$\begin{aligned} Y_{ijs} = \beta_0 + \beta_1 A_i + \beta_2 N_j + \beta_3 (A_i N_j) \\ + u_s + \epsilon_{ijs}, \end{aligned} \tag{545}$$

where A_i denotes architecture and N_j novelty class.

The decisive interaction asks whether model-based or rollout-enabled architectures gain selectively under:

$$\begin{aligned} \text{causal novelty}, \qquad \text{compositional novelty}, \\ \text{delayed consequences}, \qquad \text{problem-family novelty}. \end{aligned} \tag{546}$$

Ordinary interpolation performance should be reported separately.

12.15 Block IV: adaptation time and final performance are distinct

A system that reaches high performance after extensive relearning is scientifically different from one that adapts after only a small amount of new evidence.

Therefore:

$$n_{\text{adapt}} \tag{547}$$

and:

$$P_{\text{final}} \tag{548}$$

should remain separate outcomes.

Where appropriate, adaptation can be represented as a learning curve:

$$P(n), \tag{549}$$

where n is the number of post-change interactions.

Useful preregistered summaries may include:

$$\begin{aligned} n_{\text{adapt}}, \\ \text{area under the post-change learning curve}, \\ P_{\text{final}}. \end{aligned} \tag{550}$$

12.16 Block IV: model-corruption and rescue effects

Evidence for model use should compare:

$$\begin{aligned} \text{intact model}, \\ \text{corrupted model}, \\ \text{corrected model}. \end{aligned} \tag{551}$$

The corruption effect is:

$$\Delta_{\text{corrupt}} = P_{\text{intact}} - P_{\text{corrupt}}. \tag{552}$$

The rescue effect is:

$$\Delta_{\text{restore}} = P_{\text{restored}} - P_{\text{corrupt}}. \tag{553}$$

Strong evidence requires not merely degradation but degradation in the direction predicted by the altered model content.

12.17 Block V: calibration and control benefit are separate

Calibration quality and behavioural usefulness should not be collapsed into a single result.

Calibration may be assessed through:

$$\text{BS} = \frac{1}{N}\sum_{i=1}^{N}(q_i - y_i)^2 \tag{554}$$

and a preregistered calibration-error statistic.

Behavioural benefit should be measured independently through:

$$\begin{gathered}\text{selective accuracy},\\ \text{abstention efficiency},\\ \text{adaptive-computation allocation},\\ \text{recovery after impairment}.\end{gathered} \tag{555}$$

A well-calibrated passive monitor can therefore be distinguished from a causally useful reflective controller.

12.18 Block V: self-model intervention analysis

Misinformation and restoration experiments should estimate both:

$$\Delta_{\text{misinfo}} \tag{556}$$

and:

$$\Delta_{\text{rescue}}. \tag{557}$$

Where induced overconfidence and underconfidence are both tested, they should be analysed separately because the behavioural consequences need not be symmetric.

The decisive prediction concerns directional consistency:

$$\boxed{\begin{array}{c}\text{overconfidence} \longrightarrow \text{less checking or abstention,}\\ \text{underconfidence} \longrightarrow \text{more checking or abstention,}\end{array}} \tag{558}$$

where those behaviours are available and preregistered.

12.19 Auxiliary conflict analysis

The auxiliary conflict experiment should report each objective separately before considering aggregate summaries.

Where meaningful, Pareto efficiency may be analysed, but the underlying objective vector should remain visible.

Performance under:

$$\begin{array}{c}\text{familiar trade-offs,}\\ \text{novel trade-offs,}\\ \text{dynamic trade-offs,}\\ \text{higher-order conflicts}\end{array} \tag{559}$$

should be treated as separate conditions.

The conflict representation should earn statistical credit only if its advantage exceeds serious scalarised and model-based alternatives.

12.20 Non-inferiority and equivalence tests for bypasses

Failure to reject a difference is not sufficient to claim that a bypass architecture performs equivalently.

Where the scientific hypothesis is:

$$\text{bypass performs comparably,} \tag{560}$$

a preregistered equivalence or non-inferiority margin should be used where possible.

Let margin be:

$$\delta_{\mathrm{NI}}. \tag{561}$$

A bypass may be considered practically non-inferior when the appropriate confidence interval excludes effects worse than:

$$-\delta_{\mathrm{NI}}. \tag{562}$$

The margin must be scientifically justified before decisive testing.

12.21 Absence of evidence is not evidence of absence

A non-significant result can arise for several reasons:

$$\begin{gathered}\text{the mechanism has no meaningful effect,}\\ \text{the effect is small,}\\ \text{the experiment is underpowered,}\\ \text{the intervention is noisy,}\\ \text{the task does not require the mechanism.}\end{gathered} \tag{563}$$

Failure to detect an effect should therefore be classified using uncertainty and power information.

Where the confidence interval remains compatible with scientifically important positive and negative effects, the correct judgement is more likely:

$$\text{Suspend} \tag{564}$$

than:

$$\text{Reject.} \tag{565}$$

12.22 Power analysis

Confirmatory experiments should conduct power or precision analysis before fixing replicate counts.

The relevant target should be the smallest effect size of scientific interest:

$$\Delta_{\min}. \tag{566}$$

Replicate count should be chosen to estimate or detect effects of approximately that scale with acceptable uncertainty.

For stochastic simulations, preliminary variance estimates may be obtained from development runs that are excluded from final confirmation.

The earlier rule:

$$N \geq 30 \tag{567}$$

remains a useful default for inexpensive experiments but is not a substitute for design-specific power analysis.

12.23 Sequential stopping

Stopping rules should be fixed before confirmatory runs.

If sequential analysis is used, decision boundaries must account for repeated inspection of accumulating data.

Otherwise, the experiment should use fixed replicate counts.

Stopping because the desired result has appeared is not permitted.

Likewise, continuing only because a preregistered result is negative introduces asymmetrical researcher freedom.

12.24 Multiplicity

Paper III contains many hypotheses, architectures, task families, ablation targets and outcome measures.

Confirmatory hypotheses should therefore be grouped into prespecified families.

Within each family, an appropriate multiplicity procedure should be defined before evaluation.

Possible strategies include:

$$\begin{gathered}\text{family-wise error control},\\ \text{false-discovery-rate control},\\ \text{hierarchical testing}.\end{gathered} \tag{568}$$

The method should match the scientific role of the hypothesis family.

Exploratory analyses may remain unadjusted if clearly labelled exploratory.

12.25 Primary and secondary outcomes

Each hypothesis should nominate:

1. one primary outcome or a small preregistered set;
2. secondary supportive outcomes;
3. exploratory diagnostics.

A failed primary hypothesis should not be retrospectively rescued because a secondary metric happens to show a favourable effect.

Secondary evidence can motivate a new confirmatory experiment.

12.26 Missing and undefined outcomes

Some outcomes may become undefined.

Examples include:

$$\begin{gathered}\text{recovery time after irreversible failure,}\\ \text{adaptation time when threshold is never reached,}\\ \text{calibration after complete task refusal.}\end{gathered} \tag{569}$$

Rules for such cases should be preregistered.

Undefined outcomes should not simply be removed if their occurrence depends upon experimental condition.

Where possible, failure itself should become part of the analysis.

12.27 Outlier treatment

Exclusion of extreme runs should require an independently justified criterion such as:

$$\begin{gathered}\text{software failure,}\\ \text{corrupted log,}\\ \text{preregistered environment invalidity.}\end{gathered} \tag{570}$$

Extreme but valid biological-like or artificial behaviour is not an outlier merely because it increases variance.

Analyses may report robust summaries alongside conventional estimates without discarding valid runs.

12.28 Sensitivity to representation

Decisive results should be re-estimated under preregistered alternative:

$$\begin{gathered}\text{state encodings,}\\ \text{temporal resolutions,}\\ \text{system boundaries,}\\ \text{evaluation horizons.}\end{gathered} \tag{571}$$

The purpose is not to demand identical numerical estimates.

Instead, robustness requires the substantive conclusion to remain intelligible under nearby scientifically defensible choices.

Where the sign or ranking changes, the claim should be restricted.

12.29 Sensitivity to baseline choice

Key effects should also be examined against more than one defensible baseline where Paper II identified baseline dependence as important.

Suppose:

$$\Delta^{(0)} \tag{572}$$

uses baseline $\mathcal{M}_0$, while:

$$\Delta^{(0')} \tag{573}$$

uses alternative baseline $\mathcal{M}_0'$.

The result need not be invariant, but large qualitative disagreement should be explained.

A claim whose existence depends entirely upon one convenient baseline should be restricted.

12.30 Architecture robustness

Major findings should be tested in more than one implementation where computationally feasible.

For example, retention effects might be compared across two recurrent architectures, or model-based effects across distinct predictive-model implementations.

The scientific target is not identical numerical performance.

It is whether the mechanism-specific relation survives reasonable changes in implementation.

12.31 Environment robustness

Principal hypotheses should not depend upon one handcrafted task instance.

Where possible, decisive conclusions should be tested across:

$$\begin{gathered}\text{multiple environment seeds,}\\ \text{parameter regimes,}\\ \text{independently generated task instances.}\end{gathered} \tag{574}$$

A mechanism that works only in one narrow environment may still be retained, but its status should be:

$$\text{Restrict} \tag{575}$$

rather than universalised.

12.32 Leave-one-environment-family-out analysis

Where several related task families are available, one useful robustness test is to fit or summarise effects without one family and ask whether the overall conclusion survives.

For families:

$$\mathcal{F}_1, \ldots, \mathcal{F}_k, \tag{576}$$

repeat the summary while excluding:

$$\mathcal{F}_j. \tag{577}$$

If one environment family alone drives the entire conclusion, this should be made explicit.

12.33 Seed sensitivity

All principal results should be visualised across independent seeds rather than only through aggregate means.

Where one or two seeds dominate the effect, report:

$$\begin{gathered} \text{median}, \qquad \text{interquartile range}, \\ \text{seed-level effect distribution} \end{gathered} \tag{578}$$

alongside model-based estimates where useful.

Seed heterogeneity may itself reveal unstable mechanisms.

12.34 Replication tiers

Evidence should be classified by replication level.

Tier 1: Internal replication. The effect survives new seeds and held-out environment instances within the same implementation.

Tier 2: Implementation replication. The effect survives a meaningfully different implementation of the same mechanism.

Tier 3: Task-family replication. The effect survives a different environment family requiring the same hypothesised function.

Tier 4: Independent reproduction. The result is reproduced from released code, data or specification by an independent implementation or research group.

Higher tiers strengthen confidence but are not assumed automatically.

12.35 Robustness should not be defined post hoc

Robustness analyses themselves create researcher degrees of freedom if endless variants are tried until a preferred conclusion survives.

The principal robustness dimensions should therefore be preregistered.

Unexpected sensitivity discovered later should still be reported, but it should be labelled exploratory.

12.36 Decision thresholds

Statistical evidence feeds into, but does not replace, the five-state decision framework.

A claim should be considered for **Retain** when:

1. the preregistered primary estimand shows a scientifically meaningful effect;
2. uncertainty excludes important null or contrary effects;
3. the effect is mechanism-specific;
4. serious controls do not reproduce it;
5. key robustness checks survive.

The appropriate judgement is **Restrict** when the effect is robust but conditional upon:

$$\begin{gathered}\text{one architecture class,}\\ \text{one environment regime,}\\ \text{one representation,}\\ \text{one timescale.}\end{gathered} \tag{579}$$

A claim should be considered for **Reduce** when a simpler mechanism matches the decisive effect.

The judgement should remain **Suspend** when uncertainty is too large to distinguish scientifically important alternatives.

A claim should be considered for **Reject** when a preregistered dependency repeatedly fails under adequately powered tests or a robust bypass defeats a claimed necessity relation.

12.37 Statistical significance cannot overrule architectural failure

A statistically strong result cannot rescue a mechanism whose interpretation fails conceptually.

For example:

$$\boxed{\begin{array}{c}\text{significant trait–fitness covariance}\\ \not\Rightarrow \text{causal selection mechanism,}\\ \text{significant confidence–accuracy correlation}\\ \not\Rightarrow \text{causal self-model use.}\end{array}} \tag{580}$$

The statistical and mechanistic requirements must both be satisfied.

12.38 Cross-block synthesis

Paper III may eventually ask whether results from different blocks exhibit reproducible relations.

Such synthesis should occur only after the within-block hypotheses have been evaluated independently.

For quantities:

$$Q_A \tag{581}$$

and:

$$Q_B, \tag{582}$$

a candidate relation may be:

$$Q_B = \mathcal{T}_{A\to B}(Q_A) + \varepsilon. \tag{583}$$

The transformation $\mathcal{T}_{A\to B}$ should be specified before testing whenever the relation is confirmatory.

Exploratory discovery of $\mathcal{T}_{A\to B}$ generates a new hypothesis rather than confirming cross-block unification.

12.39 No pooled hierarchy significance test

Paper III should not construct one omnibus statistic asking whether:

$$\text{the hierarchy is significant.} \tag{584}$$

The hierarchy contains multiple logically separable dependencies.

One arrow may survive while another fails.

Therefore,

$$\boxed{\begin{array}{c}\text{the experimental architecture receives}\\ \text{a vector of judgements, not one } p\text{-value.}\end{array}} \tag{585}$$

For example, a final result might be:

$$\begin{gathered}\text{Block I: Retain,}\\ \text{Block II transition: Restrict,}\\ \text{selection mechanism: Retain,}\\ \text{Block III retention: Retain,}\\ \text{Block III feedback: Retain,}\\ \text{Block III learning: Restrict,}\\ \text{Block IV model necessity: Reject,}\\ \text{Block V self-model use: Suspend.}\end{gathered} \tag{586}$$

Such a mixed outcome is scientifically preferable to forcing the programme into one binary verdict.

12.40 Reporting negative and mixed results

Final reporting should include:

1. successful hypotheses;
2. failed hypotheses;
3. inconclusive hypotheses;
4. successful bypasses;
5. failed rescue experiments;
6. representation-sensitive effects;
7. environment-specific effects;
8. unexpected interactions.

Negative findings should appear in the main results architecture rather than being relegated automatically to supplementary material.

12.41 Provisional statistical workflow

The confirmatory workflow should be:

$$\boxed{\begin{array}{c}\text{define estimand}\\ \downarrow\\ \text{select experimental unit}\\ \downarrow\\ \text{choose model family}\\ \downarrow\\ \text{preregister power and multiplicity}\\ \downarrow\\ \text{run fixed evaluation}\\ \downarrow\\ \text{estimate effect and uncertainty}\\ \downarrow\\ \text{perform preregistered robustness checks}\\ \downarrow\\ \text{compare serious nulls and bypasses}\\ \downarrow\\ \text{assign architectural decision.}\end{array}} \tag{587}$$

12.42 Provisional judgement

The statistical architecture is ultimately subordinate to the scientific question.

Its role is not to manufacture a unified numerical hierarchy from heterogeneous experiments.

Instead, it should determine how strongly the data support each explicitly stated dependency.

The governing rule is:

$$\boxed{\begin{array}{c}\text{estimate the causal or functional contrast}\\ \text{that the hypothesis actually predicts;}\\ \text{quantify uncertainty;}\\ \text{test robustness;}\\ \text{and allow different parts of the hierarchy}\\ \text{to receive different verdicts.}\end{array}} \tag{588}$$

13 Preregistered Decision Framework

The experimental programme requires decisions that are fixed before decisive results are known. Without such rules, an apparent failure could be reinterpreted after the fact as partial success, while a favourable result could be promoted beyond the level actually tested.

Paper III therefore assigns every major mechanism and transition one of five possible outcomes:

$$\boxed{\text{Retain,} \quad \text{Restrict,} \quad \text{Reduce,} \quad \text{Suspend,} \quad \text{Reject.}} \tag{589}$$

These outcomes apply to specified claims, not to whole domains such as "evolution", "intelligence" or "selfhood".

A block may therefore contain several different verdicts.

13.1 Decision states

Retain means that the preregistered claim survives its primary mechanism-specific tests, serious matched controls, relevant robustness checks and any required intervention or rescue criteria within the stated scope.

Restrict means that the claim survives, but only under a narrower range of architectures, environments, representations, timescales or causal conditions than originally proposed.

Reduce means that the observed effect is reproducible but does not require the nominated mechanism because a simpler or already established mechanism explains the decisive evidence equally well.

Suspend means that the available experiment cannot distinguish among scientifically important alternatives with adequate precision or causal specificity.

Reject means that the preregistered dependency repeatedly fails under adequately informative tests, or that a robust counterexample defeats a claim of necessity.

The distinction between these outcomes is important:

$$\boxed{\text{Restrict} \neq \text{Reduce} \neq \text{Suspend} \neq \text{Reject.}} \tag{590}$$

A narrow but real mechanism should not be rejected merely because it is not universal. Likewise, an underpowered experiment should not be interpreted as evidence against the mechanism.

13.2 Claims are judged at the level tested

Evidence does not automatically propagate upwards.

If an experiment establishes:

$$A \xrightarrow{M} B \tag{591}$$

under specified conditions, it does not establish:

$$B \xrightarrow{N} C. \tag{592}$$

Nor does success in an artificial implementation establish that the same mechanism is naturally necessary.

The decision framework therefore attaches each verdict to:

$$\begin{gathered} \text{a defined capacity,} \\ \text{a nominated mechanism,} \\ \text{a specified architecture class,} \\ \text{a stated environment class,} \\ \text{a particular evidential level.} \end{gathered} \tag{593}$$

Claims should not be silently broadened after a favourable result.

13.3 What must be preregistered

Before decisive evaluation begins, each primary claim should specify:

1. the operational definition of the target capacity;
2. the candidate mechanism;
3. the principal comparison architecture;
4. the primary environment or task contrast;
5. the primary estimand;
6. the minimum effect or equivalence margin of scientific interest;
7. the required intervention, ablation or rescue where applicable;
8. the principal robustness checks;
9. the bypass architecture or alternative explanation;
10. the rule assigning Retain, Restrict, Reduce, Suspend or Reject.

Secondary and exploratory tests may refine interpretation but should not replace a failed primary criterion retrospectively.

13.4 A claim can receive more than one component verdict

Some mechanisms contain separable subclaims.

For example, Block V distinguishes:

$$\text{accurate self-monitoring} \tag{594}$$

from:

$$\text{causal use of self-monitoring.} \tag{595}$$

It is therefore possible to obtain:

$$\begin{gathered}\text{self-monitoring: Retain,}\\ \text{reflective self-model use: Reject.}\end{gathered} \tag{596}$$

Similarly, Block IV may produce:

$$\begin{gathered}\text{predictive world model: Retain,}\\ \text{explicit rollout necessity: Reduce.}\end{gathered} \tag{597}$$

The framework should preserve such distinctions rather than forcing one summary verdict upon the entire block.

13.5 Block I decision structure

Block I tests whether persistent organisation can be distinguished from internally generated maintenance.

The principal claims are:

Table 11: Preregistered decision structure for Experimental Block I.

Claim	**Evidence required for retention**	**Principal route to restriction, reduction or rejection**
H1.1	Active maintenance provides a selective advantage over passive stability when disturbances require continuing correction.	Restrict if the effect appears only under narrow disturbance classes; reduce if passive stability reproduces it.
H1.2	Internally maintained systems retain greater operational viability than externally maintained systems after external support is withdrawn.	Reduce if external support or generic robustness explains the same persistence.

Claim	Evidence required for retention	Principal route to restriction, reduction or rejection
H1.3	Validated reciprocal maintenance topology predicts propagation of targeted component failure better than simpler graph or capacity measures.	Reduce if node degree, component size or feed-forward dependency explains the same propagation.
H1.4	Specific functional replacement rescues deficits caused by removal of a maintenance component.	Reject the proposed causal edge if selective ablation and rescue repeatedly fail.
H1.5	The principal maintenance organisation remains recognisable under reasonable changes of system boundary and temporal resolution.	Restrict if the result depends strongly upon one boundary or temporal resolution.
H1.6	At least some internally maintained systems exhibit operational persistence or recovery that cannot be reduced to passive stability, external support or maintenance budget alone.	Reduce or reject if serious null architectures reproduce all decisive outcomes.

A positive Block I result supports artificial self-maintenance only at the functional and causal level tested.

It does not establish life.

13.6 Block II decision structure

Block II contains two logically distinct questions:

$$\begin{gathered}\text{Does a heritable reproducing population exist?}\\ \text{Does differential reproduction generate selection-driven change?}\end{gathered} \tag{598}$$

These questions should receive separate decisions.

Table 12: Preregistered decision structure for Experimental Block II.

Claim	**Evidence required for retention**	**Principal route to restriction, reduction or rejection**
H2.1	Parent–offspring transmission creates reproducible hereditary dependence that survives environmental controls.	Suspend if transmission cannot be distinguished from environmental resemblance; reject if the proposed transmission pathway repeatedly fails.
H2.2	Specified inherited variation causally alters expected reproductive success under at least some preregistered ecological conditions.	Reduce if trait–fitness covariance is explained by confounding or non-heritable variables.
H2.3	Fitness-sensitive hereditary reproduction generates cumulative population change beyond matched neutral or random reproduction, mutation and finite-sampling effects.	Reduce if random search or neutral reproduction produces equivalent cumulative change.
H2.4	Population adaptation depends systematically upon the balance between transmission fidelity and generation of variation.	Restrict if the relation occurs only in a narrow hereditary architecture; reject only if adequately tested regimes show no reproducible relation.
H2.5	Selection effects remain identifiable under independently defensible definitions of the reproducing unit.	Restrict or suspend if the selection claim depends strongly upon post-hoc unit choice.
H2.6	Environmental switching reveals historically structured population response rather than monotonic accumulation of adaptation.	Reduce if selected and non-selected controls show equivalent responses once prior exposure is matched.

Natural selection should be retained as an effective mechanism only where:

$$\boxed{\text{reproduction} + \text{heredity} + \text{variation} + \text{differential reproductive success}} \tag{599}$$

have been independently established.

13.7 Block III decision structure

Block III does not receive one verdict for "adaptation". Retention, feedback and learning are judged separately.

Table 13: Preregistered decision structure for Experimental Block III.

Claim	**Evidence required for retention**	**Principal route to restriction, reduction or rejection**
H3.1	Policy-accessible retained state provides a selective advantage when task-relevant history is unavailable from the present observable state.	Reduce if state augmentation removes the effect; reject necessity if reactive bypasses perform equivalently.
H3.2	Manipulation of policy-accessible retained state causally alters later behaviour in a predictable, task-relevant manner.	Reject the causal-use claim if retained information is decodable but intervention produces no predicted behavioural effect.
H3.3	Closed-loop feedback provides a selective advantage under unpredictable disturbance.	Reduce if open-loop or yoked control reproduces the decisive benefit.
H3.4	The causal contribution of feedback depends upon process-relevant observation timing and channel integrity.	Restrict if the relation is strongly timescale-specific; reject the nominated feedback path if appropriate perturbations have no effect.
H3.5	Between-episode experience-dependent learning provides a selective advantage when behavioural mappings change.	Reduce if fixed capacity-matched policies adapt equivalently.
H3.6	Between-episode learning produces persistent modification of future behaviour rather than only transient within-episode corrective action.	Reject the persistent-learning claim if gains disappear when updates are frozen despite preservation of the alleged learned state.
H3.7	Policy-accessible retention, within-episode feedback and between-episode learning possess at least partly separable causal roles.	Reduce if all effects collapse into one generic capacity difference.

A possible outcome is therefore:

$$\begin{aligned} &\text{retention: Retain,} \\ &\text{feedback: Retain,} \\ &\text{learning: Restrict.} \end{aligned} \tag{600}$$

Such a result is not internally inconsistent.

13.8 Block IV decision structure

Block IV separates predictive modelling, counterfactual rollout and structural generalisation.

Table 14: Preregistered decision structure for Experimental Block IV.

Claim	**Evidence required for retention**	**Principal route to restriction, reduction or rejection**
H4.1	Model-based architectures adapt selectively faster when familiar causal relations are altered.	Reduce if model-free systems adapt equally quickly under matched compute and experience.
H4.2	Counterfactual rollout contributes specifically when alternative delayed futures must be compared.	Reduce if predictive-model-only or model-free systems match rollout-enabled performance.
H4.3	Explicit model-based control provides structural-generalisation benefits beyond serious capacity-matched model-free alternatives.	Reject necessity if capacity-matched model-free bypasses perform equivalently across decisive novelty classes.
H4.4	Internal world-model content is causally used during planning.	Reject causal use if targeted corruption does not produce the predicted planning error.
H4.5	Correcting a corrupted internal model selectively restores the corresponding planning capability.	Reduce or reject the nominated mechanism if behavioural recovery does not depend upon model correction.
H4.6	Structural generalisation extends beyond ordinary interpolation and parameter variation.	Restrict if gains disappear under genuine causal, compositional or problem-family novelty.
H4.7	Structural-generalisation performance is not reducible to one task-specific heuristic.	Reduce if preregistered simple heuristics reproduce the decisive cross-task effects.

A successful world model therefore does not automatically establish the necessity of world-model-based intelligence.

13.9 Block V decision structure

Block V distinguishes self-prediction from causal self-model use.

Table 15: Preregistered decision structure for Experimental Block V.

Claim	**Evidence required for retention**	**Principal route to restriction, reduction or rejection**
H5.1	Self-monitoring improves behaviour primarily when self-related estimates can alter subsequent computation or action.	Reduce if passive monitoring performs equivalently.
H5.2	Active self-monitoring improves recovery after unannounced internal impairment.	Reduce if generic robustness produces the same recovery without self-related tracking.
H5.3	Manipulating self-related estimates causes predictable changes in reflective control, and restoring accurate estimates rescues behaviour.	Reject causal self-model use if misinformation and correction produce no selective effects.
H5.4	Episode-specific self-monitoring provides more useful control than shuffled or distribution-matched monitoring signals.	Reduce if shuffled monitor signals perform equivalently.
H5.5	The agent tracks at least some internally generated capability changes rather than only environmental uncertainty.	Reduce if environmental uncertainty estimates explain all decisive effects.
H5.6	Reflective self-modelling provides functions not fully reproduced by simpler task-specific error detection or externally supplied confidence.	Reject necessity if robust bypasses reproduce all target capabilities.

Even a complete Block V Retain verdict does not bear directly upon phenomenal consciousness.

13.10 Auxiliary conflict-test decision structure

The conflict-representation experiment remains outside the five-block hierarchy.

Its claims should therefore be reported separately.

Table 16: Preregistered decision structure for the auxiliary conflict test.

Claim	**Evidence required for retention**	**Principal route to restriction, reduction or rejection**
H6.1	Explicit conflict representation improves transfer when the structure of objective incompatibility changes.	Reduce if scalarised or ordinary multi-objective control transfers equally well.
H6.2	The conflict representation is causally used during control.	Reject causal use if targeted scrambling produces no predicted behavioural change.
H6.3	Correcting a corrupted conflict representation selectively rescues behaviour.	Reduce or reject if recovery occurs independently of correcting the representation.
H6.4	Explicit conflict representation provides information not fully reproduced by ordinary model-based prediction.	Reduce if capacity-matched model-based control reproduces all decisive effects.
H6.5	The mechanism generalises beyond one fixed pairwise trade-off pattern.	Restrict if benefits disappear under novel, dynamic or higher-order conflicts.

No conflict-test result modifies the consciousness verdict of Paper I.

13.11 Mechanism-level decision rules

A mechanism should receive **Retain** only when all of the following are satisfied:

1. the primary preregistered effect is present at scientifically meaningful magnitude;
2. uncertainty is sufficiently narrow to exclude important contrary interpretations;
3. the effect is selective for conditions where the mechanism should matter;
4. targeted intervention behaves as predicted where intervention is technically possible;
5. matched alternatives do not reproduce the decisive effect;
6. preregistered robustness checks do not overturn the interpretation.

Failure of one requirement does not always imply rejection.

For example, lack of implementation replication may justify:

$$\text{Restrict} \tag{601}$$

rather than:

$$\text{Reject.} \tag{602}$$

13.12 Necessary-dependency claims face a higher threshold

Necessity claims require more than showing that mechanism M helps produce capacity B.

Evidence that:

$$M \longrightarrow B \tag{603}$$

under one architecture establishes at most a route of synthetic sufficiency.

A necessary-dependency claim additionally requires serious attempts to produce:

$$B \tag{604}$$

without:

$$M. \tag{605}$$

The relevant adversarial target is:

$$\boxed{\neg M \wedge B.} \tag{606}$$

A robust bypass defeats universal necessity even when M remains useful.

13.13 Sufficiency claims remain architecture-bound

Artificial sufficiency should be stated at the narrowest justified level.

A result of the form:

$$A + M \longrightarrow B \tag{607}$$

supports:

$$\begin{gathered} M \text{ can contribute to } B \\ \text{within architecture class } A \\ \text{under the tested conditions.} \end{gathered} \tag{608}$$

It does not justify:

$$M \text{ universally generates } B. \tag{609}$$

Any later results report should preserve this architecture-bound wording in its Results and Conclusion.

13.14 Gate logic is replaced by dependency logic

The previous experimental design used phase gates that could be read as:

$$\text{Phase I passes} \longrightarrow \text{Phase II becomes legitimate} \longrightarrow \text{Phase III follows.} \tag{610}$$

That logic is no longer appropriate.

The revised programme asks instead:

$$\boxed{\begin{array}{c}\text{Is the later question logically dependent} \\ \text{upon the earlier result?}\end{array}} \tag{611}$$

If yes, the earlier failure constrains how the later experiment may be interpreted.

If no, the later experiment may proceed independently.

13.15 Three progression cases

Three cases should be distinguished.

Case A: strict experimental dependence.

Suppose a later experiment requires the actual output of an earlier mechanism:

$$A \to B. \tag{612}$$

If A cannot be instantiated or measured adequately, the dependent test of B may need to be:

$$\text{Suspend.} \tag{613}$$

Case B: theoretical dependence but independent construction.

A later capacity can be engineered directly even though the proposed natural dependency from an earlier capacity has failed.

In that case, the later experiment may proceed, but it tests:

$$\text{synthetic sufficiency} \tag{614}$$

rather than confirming the failed transition.

Case C: logical independence.

The later question does not require the earlier mechanism.

The experiment should proceed normally.

This distinction prevents failure in one part of the architecture from artificially terminating unrelated questions.

13.16 Examples of non-blocking failures

Several important failures should not stop later work.

If reciprocal organisational closure is rejected as necessary for self-maintenance, Block II may still test reproduction and heredity in artificial agents constructed by another route.

If explicit retention is unnecessary for some tasks, Block IV may still test world-model use.

If model-based planning is reduced to a powerful model-free mechanism, Block V may still ask whether the resulting agent uses self-related information.

If reflective self-modelling is rejected as necessary for flexible agency, the auxiliary conflict experiment may still be performed.

Thus:

$$\boxed{\text{failure of one arrow} \neq \text{automatic failure of every later node.}} \tag{615}$$

13.17 Examples of interpretation-changing failures

Some failures do alter later interpretation.

If Block II cannot establish heredity, then observed population change must not later be described as selection on inherited variation.

If Block III fails to demonstrate experience-dependent policy modification, later performance should not be described as arising from learning unless learning is independently re-established in the later architecture.

If Block IV fails to show causal world-model use, Block V should not describe the tested agent as a model-based reasoner merely because a predictive module is present.

The programme may continue, but its terminology must follow the surviving evidence.

13.18 Decision inheritance is downward only

A failed or restricted lower-level claim can constrain how dependent higher claims are worded.

Evidence does not travel in the opposite direction.

For example, successful Block IV performance cannot retrospectively establish that Block III retention or learning was necessary.

Likewise, successful Block V self-monitoring cannot prove that model-based planning caused the development of self-modelling.

Therefore:

$$\boxed{\begin{array}{c}\text{later success cannot rescue}\\ \text{an earlier failed dependency.}\end{array}} \tag{616}$$

13.19 Mixed verdicts are expected

The research architecture should expect mixed results.

A plausible final decision set could be:

$$\begin{array}{c}\text{internal maintenance: Retain,}\\ \text{reciprocal closure necessity: Reject,}\\ \text{heredity: Retain,}\\ \text{selection: Retain,}\\ \text{retention: Restrict,}\\ \text{feedback: Retain,}\\ \text{learning: Retain,}\\ \text{explicit world-model necessity: Reduce,}\\ \text{structural generalisation: Retain,}\\ \text{reflective self-model use: Suspend.}\end{array} \tag{617}$$

Such a result would refine the dependency map rather than count as a failure of the Theory of Everything research programme.

13.20 Decision table for the whole experiment

The five outcome states can be summarised across the programme as follows:

Table 17: General interpretation of preregistered decision outcomes.

Decision	**Meaning**	**Required response in the research architecture**
Retain	The claim survives its decisive tests within the stated scope.	Keep the claim active, preserving its scope and evidential level.
Restrict	The claim survives only under specified conditions.	Narrow the claim explicitly; do not describe it as universal.
Reduce	A simpler mechanism explains the decisive effect.	Replace the higher-level explanatory claim with the simpler mechanism unless independent evidence remains.

Decision	Meaning	Required response in the research architecture
Suspend	The experiment cannot yet discriminate among important alternatives.	Keep the claim quarantined and specify the missing decisive test.
Reject	The proposed dependency repeatedly fails or a robust bypass defeats necessity.	Remove the rejected dependency from the active architecture and revise any later claims that relied upon it.

13.21 Quarantine rule

Claims that are suspended, untested or supported only by exploratory evidence remain outside the active architecture.

They may be retained in a quarantine ledger with:

$$\begin{gathered}\text{current evidential status},\\ \text{reason for quarantine},\\ \text{decisive evidence still required},\\ \text{conditions for reconsideration}.\end{gathered} \tag{618}$$

The quarantine ledger is not a list of discarded ideas.

It is a mechanism for preventing unresolved claims from entering later arguments as though they had already been established.

13.22 Decision revisions require new evidence

Verdicts may change as evidence changes.

A previously rejected claim can be reconsidered if:

$$\begin{gathered}\text{a materially different mechanism is proposed},\\ \text{a failed assumption is corrected},\\ \text{new evidence defeats the original counterexample}.\end{gathered} \tag{619}$$

Likewise, a retained claim can later be restricted or rejected by stronger experiments.

Revision is therefore permitted, but not through reinterpretation alone.

A decision changes because the evidence changes.

13.23 No privileged outcome

The preregistration must not define theoretical confirmation as the preferred result.

For every principal claim:

$$\text{Retain}, \quad \text{Restrict}, \quad \text{Reduce}, \quad \text{Suspend}, \quad \text{Reject} \tag{620}$$

must be treated as legitimate scientific outcomes.

A programme in which only Retain counts as success would create pressure to reinterpret ambiguous evidence in favour of the hierarchy.

Paper III instead defines success as obtaining a justified verdict.

13.24 Final decision record

After all confirmatory analyses are complete, the project should publish a machine-readable and human-readable decision record.

For each hypothesis, the record should contain:

1. hypothesis identifier;
2. preregistered wording;
3. primary estimand;
4. primary result;
5. uncertainty interval;
6. intervention result where applicable;
7. bypass result;
8. robustness summary;
9. final decision;
10. scope of that decision;
11. any unresolved qualification.

This record should preserve failed and mixed results alongside retained ones.

13.25 Provisional judgement

The purpose of the decision framework is not to force the experimental programme through five stages.

It is to determine which proposed relations survive.

The final architecture may therefore be:

$$\text{a chain}, \tag{621}$$

a branching structure,

$$\text{a network}, \tag{622}$$

or a collection of partly independent mechanisms.

The experiment should allow the evidence to decide.

The governing rule is:

$$\boxed{\begin{array}{c}\text{do not preserve a transition because later sections need it;}\\ \text{do not terminate a later question merely because}\\ \text{an earlier transition failed;}\\ \text{judge each dependency at the level actually tested.}\end{array}} \tag{623}$$

14 Minimal Implementation Plan

A useful experimental programme must be implementable without requiring the entire five-block dependency architecture to be implemented at once. Paper III should therefore begin with the smallest experiments capable of producing decisive evidence against individual dependency claims, then extend the infrastructure only when a later test requires additional machinery.

The implementation sequence is consequently practical rather than ontological:

$$\boxed{\begin{array}{c}\text{implementation order}\\ \neq\\ \text{evidence for natural dependency order.}\end{array}} \tag{624}$$

Building Block I before Block II is an efficient engineering decision. It does not imply that self-maintenance necessarily generates reproduction.

14.1 Reference implementation

The reference implementation should use standard Python together with a mainstream numerical or tensor library. The precise framework is less important than transparency, deterministic configuration where possible and the ability to reproduce every reported result from stored specifications.

The repository should contain:

1. environment definitions;
2. architecture definitions;
3. configuration files for every confirmatory condition;
4. seed-generation and seed-registration files;
5. fixed held-out evaluation specifications;
6. raw trajectory logging;
7. intervention and ablation definitions;
8. rescue-condition definitions;
9. statistical-analysis scripts;
10. scripts that regenerate every figure and table;
11. unit and property tests for environment dynamics;
12. a machine-readable decision record.

The implementation should make it possible to distinguish exactly which results came from exploratory development and which came from preregistered confirmatory runs.

14.2 Repository structure

A simple repository layout could be:

```
paper3/
    configs/
        development/
        confirmatory/
        held_out/
    environments/
        block1/
        block2/
        block3/
        block4/
        block5/
        auxiliary_conflict/
    agents/
        controls/
        retention/
        feedback/
        learning/
        world_models/
        self_models/
```

```
    conflict_models/
interventions/
    ablations/
    corruptions/
    rescues/
    bypasses/
experiments/
    block1/
    block2/
    block3/
    block4/
    block5/
    auxiliary_conflict/
analysis/
    block1/
    block2/
    block3/
    block4/
    block5/
    cross_block/
tests/
results/
    raw/
    processed/
    figures/
    tables/
    decisions/
```

This is illustrative rather than mandatory. The important requirement is that experimental definitions, interventions, raw results and derived analyses remain separable and traceable.

14.3 Experiment manifests

Every confirmatory experiment should be defined by a machine-readable manifest.

At minimum, the manifest should specify:

$$\begin{aligned} &\text{hypothesis identifier}, && \text{block identifier},\\ &\text{environment version}, && \text{architecture version},\\ &\text{training budget}, && \text{evaluation conditions},\\ &\text{seed procedure}, && \text{primary estimand},\\ &\text{interventions}, && \text{break conditions}. \end{aligned} \tag{625}$$

Once a confirmatory manifest has been frozen, changing any field should produce a new experimental version rather than silently altering the old one.

14.4 Shared simulation interface

Where practical, all agent-based blocks should use a common environment interface:

$$\begin{aligned} O_t &= \mathrm{observe}(E_t), \\ A_t &= \mathrm{act}(O_t), \\ E_{t+1}, Y_t &= \mathrm{step}(E_t, A_t). \end{aligned} \tag{626}$$

The interface should also expose controlled experimental operations:

$$\begin{aligned} &\text{intervene}, \qquad \text{ablate}, \\ &\text{corrupt}, \qquad \text{restore}\,. \end{aligned} \tag{627}$$

Using a common interface reduces unnecessary engineering differences without forcing the substantive state spaces of the blocks to be identical.

14.5 Shared logging standard

Every run should preserve enough information to reconstruct the experiment.

Where applicable, logs should include:

$$\begin{gathered} \text{environment state}, \qquad \text{observation}, \qquad \text{action}, \\ \text{outcome or reward}, \qquad \text{viability state}, \\ \text{accessible internal state}, \qquad \text{parameter-update event}, \\ \text{intervention label}, \qquad \text{random seed}, \\ \text{architecture and environment version}. \end{gathered} \tag{628}$$

Population experiments should additionally preserve:

$$\begin{aligned} &\text{individual identity}, \qquad \text{parent–offspring relations}, \\ &\text{genotype}, \qquad \text{phenotype}, \qquad \text{reproductive events}. \end{aligned} \tag{629}$$

Summary statistics should be regenerated from the raw records rather than stored as the only surviving evidence.

14.6 Stage A: shared experimental infrastructure

The first implementation stage should contain no substantive hierarchy claim.

Its purpose is to establish:

1. deterministic or controlled stochastic environments;

2. configuration loading;
3. trajectory logging;
4. intervention hooks;
5. seed management;
6. statistical-analysis interfaces;
7. automated figure and table generation;
8. reproducibility tests.

A small dummy environment should verify that the entire pipeline can run from configuration to final decision output before any major experiment begins.

The Stage A criterion is therefore:

$$\boxed{\begin{array}{c}\text{one command or reproducible workflow}\\ \longrightarrow \text{raw data} \longrightarrow \text{analysis} \longrightarrow \text{figure/table} \longrightarrow \text{decision record.}\end{array}} \tag{630}$$

14.7 Stage B: minimal Block I implementation

The first scientifically substantive implementation should be Block I because it can be made small, transparent and inexpensive.

It should contain two linked experiments.

B1: driven pattern baseline. Implement the Gray–Scott reaction–diffusion system and reproduce preregistered pattern-forming and non-pattern-forming regimes.

The minimum manipulations are:

1. sustained driving;
2. removal of driving;
3. local perturbation;
4. boundary modification.

This establishes the weaker fact that driven dynamics can sustain persistent organisation.

B2: causal maintenance network. Implement a small network of approximately three to five functional components under:

$$\begin{gathered}\text{passive persistence,}\\ \text{external maintenance,}\\ \text{internal feed-forward maintenance,}\\ \text{reciprocal internal maintenance.}\end{gathered} \tag{631}$$

The first decisive experiment should then perform:

$$\boxed{\begin{gathered}\text{targeted node ablation}\\ \downarrow\\ \text{predicted selective deficit}\\ \downarrow\\ \text{specific functional rescue.}\end{gathered}} \tag{632}$$

This is the smallest implementation capable of producing direct evidence about Transition Problem 1.

14.8 Block I should produce the first complete preregistered study

Before extending the programme, Stage B should demonstrate that Paper III's full methodology works in practice.

The first complete study should include:

1. preregistered hypotheses;
2. capacity- or resource-matched controls;
3. held-out disturbances;
4. intervention;
5. rescue;
6. robustness analysis;
7. Retain/Restrict/Reduce/Suspend/Reject judgement.

The project should not wait until Blocks IV or V to discover whether its preregistration and logging architecture is workable.

14.9 Stage C: minimal Block II implementation

Once the infrastructure has been validated, a minimal evolutionary ecology can be added independently of whether reciprocal closure survived Block I.

The smallest useful Block II environment requires:

$$\begin{gathered}\text{spatially distributed resources},\\ \text{energetic costs},\\ \text{agent death},\\ \text{reproductive events},\\ \text{parent–offspring identity},\\ \text{transmissible state},\\ \text{mutation}.\end{gathered} \tag{633}$$

The first implementation should avoid learning so that population-level selection is not confounded immediately with within-lifetime optimisation.

The minimal comparison is:

$$\boxed{\begin{gathered}\text{heredity + fitness-sensitive reproduction}\\ \text{versus}\\ \text{heredity + random reproduction}.\end{gathered}} \tag{634}$$

A third condition should remove heredity while retaining reproduction.

These conditions allow reproduction, heredity and differential reproduction to be separated before more complicated evolutionary behaviour is introduced.

14.10 Block II lineage infrastructure

The implementation should generate an explicit lineage graph:

$$G_{\text{lin}} = (V_{\text{ind}}, E_{\text{rep}}). \tag{635}$$

Every reproductive event should write:

$$\begin{gathered}\text{parent ID}, \qquad \text{offspring ID},\\ \text{parent transmissible state}, \qquad \text{offspring transmissible state},\\ \text{environmental state}, \qquad \text{time}.\end{gathered} \tag{636}$$

This should be implemented before any claim involving lineage persistence or selection is analysed.

14.11 Stage D: minimal Block III factorial experiment

Block III should be implemented as a factorial experiment rather than as three successively larger agents.

The first version can use a compact navigation or control environment with three independently configurable properties:

$$\begin{aligned} H &= \text{history requirement}, \\ U &= \text{unpredictable disturbance}, \\ C &= \text{changed behavioural mapping}. \end{aligned} \tag{637}$$

The three intervention channels vary factorially:

$$R \times F \times L, \tag{638}$$

where R denotes policy-accessible retained state, F denotes within-episode observation–action feedback and L denotes between-episode policy learning. Policy parameters remain fixed within each decisive episode.

This yields a compact test of whether:

$$\begin{aligned} R &\text{ helps specifically when history matters}, \\ F &\text{ helps specifically under unpredictable disturbance}, \\ L &\text{ helps specifically after mapping change}. \end{aligned} \tag{639}$$

Not every possible environment combination needs to be implemented in the first version.

The priority is to construct well-matched discriminating task pairs.

14.12 Minimal retention test

The simplest retention experiment should use two environments that are identical except for whether information from an earlier cue is needed later.

Compare:

$$E_{\text{history}} \tag{640}$$

with:

$$E_{\text{Markov}}. \tag{641}$$

A recurrent or retained-state controller should face a capacity-matched present-state controller.

The decisive intervention should erase, permute or substitute the retained state immediately before the history-dependent decision.

14.13 Minimal feedback test

The simplest feedback experiment should compare:

$$\begin{gathered}\text{closed-loop control},\\ \text{open-loop control},\\ \text{yoked replay}.\end{gathered} \tag{642}$$

Two disturbance regimes are sufficient initially:

$$\begin{gathered}\text{predictable disturbance},\\ \text{unpredictable disturbance}.\end{gathered} \tag{643}$$

The experiment succeeds as a discriminator only if the closed-loop advantage is selective for the condition in which online correction is actually useful.

14.14 Minimal learning test

A compact learning experiment should switch a known action–outcome relation after the agent has reached stable performance:

$$K_A \longrightarrow K_B. \tag{644}$$

Compare a learning-enabled controller with a capacity-matched frozen controller.

After adaptation, freeze the learning-enabled controller and test whether the behavioural change persists.

This provides the minimal distinction between:

$$\text{online feedback} \tag{645}$$

and:

$$\text{persistent policy modification}. \tag{646}$$

14.15 Stage E: minimal Block IV implementation

Block IV should not begin with a large general-purpose world model.

A small discrete or low-dimensional environment is preferable because its causal structure can be inspected directly.

The first task should include:

1. a latent or predictive transition structure;

2. at least two possible future action sequences;

3. one held-out causal change;

4. delayed consequences;

5. a model-free control.

Implement:

$$\begin{gathered} A1 : \text{history-sensitive model-free control}, \\ A2 : \text{predictive model without rollout}, \\ A3 : \text{model-based rollout}, \\ A4 : \text{capacity-matched model-free alternative}. \end{gathered} \tag{647}$$

A purely reactive baseline may also be retained for orientation but should not be the only competitor.

14.16 Minimal world-model intervention

The first causal model-use test should alter one interpretable transition in the learned model.

For example:

$$s_i \xrightarrow{a} s_j \tag{648}$$

can be replaced internally by:

$$s_i \xrightarrow{a} s_k. \tag{649}$$

The experiment should predict in advance how this false internal relation should change action selection.

Correcting the transition afterwards should test rescue.

A small interpretable model is preferable to a large opaque system for this first implementation.

14.17 Stage F: minimal Block V implementation

Block V can initially use the successful agent architecture from Block IV or another sufficiently capable agent. It need not depend upon Block IV having established that world models are necessary.

Add a monitor predicting:

$$q_t = P\left(\text{success} \mid H_t^{\text{int}}\right). \tag{650}$$

The first task should provide:

$$\begin{gathered}\text{variable difficulty,}\\ \text{an abstention or checking option,}\\ \text{known cost for using that option.}\end{gathered} \tag{651}$$

Compare:

$$\begin{gathered}S0 : \text{no monitor,}\\ S1 : \text{passive monitor,}\\ S2 : \text{active monitor,}\\ S3 : \text{shuffled monitor.}\end{gathered} \tag{652}$$

The decisive causal test should then manipulate q_t directly.

14.18 Minimal self-model misinformation and rescue test

Choose episodes in which task information is held fixed while the monitor signal is altered.

Test:

$$q_t \longrightarrow \widetilde{q}_t. \tag{653}$$

The preregistered prediction should specify how induced overconfidence or underconfidence changes:

$$\begin{gathered}\text{checking,} \qquad \text{abstention,}\\ \text{additional computation,} \qquad \text{assistance seeking.}\end{gathered} \tag{654}$$

Then restore the accurate estimate:

$$\widetilde{q}_t \longrightarrow q_t^{\text{restored}}. \tag{655}$$

This should be the first decisive Block V experiment.

14.19 Stage G: auxiliary conflict experiment

The conflict-representation experiment should be implemented only after the core five-block infrastructure is stable.

Its first version can use three objectives:

$$C_1, \qquad C_2, \qquad C_3, \tag{656}$$

with a trade-off structure that changes between training and evaluation.

Compare:

$$\begin{gathered}\text{scalar reward control,}\\ \text{multi-objective control,}\\ \text{explicit conflict representation,}\\ \text{capacity-matched model-based control.}\end{gathered} \tag{657}$$

There is no requirement to run Stage G if earlier theoretical or empirical work shows that the proposed conflict mechanism adds nothing beyond the simpler controls.

14.20 Implementation stages are not phase gates

The stages above describe an economical development sequence.

They do not imply:

$$B \longrightarrow C \longrightarrow D \longrightarrow E \longrightarrow F \tag{658}$$

as a causal hierarchy.

For example, Stage E may proceed even if a Block III necessity claim has been rejected, provided the Block IV architecture can be constructed and its question remains well defined.

Likewise, Stage F may test self-related control even if explicit world-model necessity fails in Stage E.

Therefore:

$$\boxed{\begin{gathered}\text{implementation sequence optimises effort;}\\ \text{the dependency experiments determine theory.}\end{gathered}} \tag{659}$$

14.21 Minimal common experiment runner

The original phase-based execution loop should be replaced by a manifest-driven block runner.

A high-level implementation is:

```
for experiment in confirmatory_manifest:

    specification = load_specification(experiment)

    for architecture in specification.architectures:
        for environment in specification.environments:
            for seed in specification.seeds:

                agent = initialise_agent(
                    architecture=architecture,
                    seed=seed
                )
```

```
                env = initialise_environment(
                    environment=environment,
                    seed=seed
                )

                train(
                    agent=agent,
                    env=env,
                    budget=specification.training_budget
                )

                freeze_training_state(agent)

                baseline = evaluate(
                    agent=agent,
                    env=env,
                    test_suite=specification.held_out_tests
                )

                interventions = run_interventions(
                    agent=agent,
                    env=env,
                    interventions=specification.interventions
                )

                rescues = run_rescues(
                    agent=agent,
                    env=env,
                    rescues=specification.rescues
                )

                save_raw_run(
                    specification,
                    baseline,
                    interventions,
                    rescues
                )

run_preregistered_statistics()
run_preregistered_robustness_checks()
compare_bypass_architectures()
assign_preregistered_decisions()
generate_tables_and_figures_from_raw_data()
```

This architecture treats the experiment specification rather than the hierarchical stage as the unit of execution.

14.22 Development and confirmatory modes

The software should distinguish explicitly between:

$$\texttt{development} \tag{660}$$

and:

$$\texttt{confirmatory} \tag{661}$$

modes.

Development mode may permit:

$$\begin{gathered}\text{architecture tuning},\\ \text{environment debugging},\\ \text{metric exploration},\\ \text{visual inspection}.\end{gathered} \tag{662}$$

Confirmatory mode should load a frozen manifest and reject unregistered changes to:

$$\begin{gathered}\text{architecture}, \quad \text{environment}, \quad \text{seed set},\\ \text{evaluation suite}, \quad \text{primary metric}.\end{gathered} \tag{663}$$

This separation reduces accidental contamination between exploration and confirmation.

14.23 Automatic provenance recording

Each run should automatically record:

$$\begin{gathered}\text{experiment ID}, \quad \text{software version},\\ \text{configuration hash}, \quad \text{architecture hash},\\ \text{environment hash}, \quad \text{seed},\\ \text{date and runtime metadata}.\end{gathered} \tag{664}$$

Where practical, the source-code commit identifier should also be stored.

The purpose is to make every reported result traceable to the exact implementation that produced it.

14.24 Intervention registry

Every ablation, corruption or rescue should have a unique identifier.

A registry entry should state:

1. target mechanism;
2. exact implementation change;
3. hypothesised selective effect;
4. matched-damage control;
5. rescue condition;
6. affected hypotheses.

This prevents intervention terminology from drifting across experiments.

For example,

$$\texttt{WORLD_MODEL_CORRUPT_01} \tag{665}$$

should always refer to the same preregistered transformation within one experimental version.

14.25 Bypass registry

Counterexample architectures should receive the same implementation status as preferred mechanisms.

A bypass registry should record:

$$\begin{gathered}\text{target necessity claim},\\ \text{mechanism deliberately absent},\\ \text{alternative mechanism supplied},\\ \text{capacity-matching procedure},\\ \text{non-inferiority or equivalence criterion.}\end{gathered} \tag{666}$$

Bypasses should not be added only after preferred mechanisms fail.

At least one serious bypass should be specified before decisive testing of every important necessity claim.

14.26 Environment validation

Before agents are compared, each environment should pass automated validation.

Tests should verify:

1. transition rules;
2. observation generation;
3. reward or viability calculations;

4. hidden-state properties;
5. disturbance schedules;
6. reproduction logic where applicable;
7. intervention behaviour;
8. randomisation procedures.

Task pairs intended to isolate one mechanism should also be checked for unexpected information leakage.

An environment that fails validation should not proceed to confirmatory runs.

14.27 Reference baselines should remain simple

Simple reference models are useful for detecting experimental artefacts.

Examples include:

$$\begin{gathered}\text{random policy},\\ \text{fixed heuristic},\\ \text{oracle where scientifically informative},\\ \text{constant predictor}.\end{gathered} \tag{667}$$

These are diagnostic baselines, not substitutes for serious capacity-matched controls.

A preferred architecture should not earn mechanistic credit merely by beating a random agent.

14.28 Computational scaling

The first implementation should remain deliberately small.

Only after the causal distinction survives should the experiment scale:

$$\begin{gathered}\text{larger populations},\\ \text{larger state spaces},\\ \text{deeper networks},\\ \text{longer planning horizons},\\ \text{richer task families}.\end{gathered} \tag{668}$$

Scaling before the mechanism is validated risks spending substantial computational resources on an effect that simpler controls would already explain.

The preferred order is therefore:

$$\boxed{\begin{array}{c}\text{small interpretable system}\\ \longrightarrow \text{causal discrimination}\\ \longrightarrow \text{replication}\\ \longrightarrow \text{scaling.}\end{array}} \tag{669}$$

14.29 Minimum viable research programme

If resources permit only a small first study, Paper III does not require all five blocks to be implemented.

The minimum viable programme is:

1. implement the shared infrastructure;
2. reproduce the driven Gray–Scott baseline;
3. implement the four Block I maintenance architectures;
4. preregister one maintenance-edge hypothesis;
5. perform targeted ablation;
6. perform specific and irrelevant rescue;
7. run the matched nulls;
8. publish the raw trajectories and decision record.

That experiment alone could produce a meaningful result about whether internally generated maintenance provides causal explanatory value beyond passive persistence and external support.

14.30 Criteria for extending the implementation

A later implementation stage should be added when at least one of the following applies:

1. the next block tests an unresolved high-priority dependency;
2. an earlier result generates a specific new discriminating prediction;
3. a bypass requires a new architecture;
4. replication requires a different implementation;
5. a robustness test requires a broader environment family.

Expansion should not occur merely because the next block appears later in the paper.

14.31 What should not be implemented prematurely

Several tempting additions should be deferred until a clear experiment requires them:

$$\begin{gathered}\text{large language models,}\\ \text{high-dimensional embodied simulators,}\\ \text{large multi-agent societies,}\\ \text{complex neural self-report systems,}\\ \text{quantum or quantum-inspired components.}\end{gathered} \tag{670}$$

None is excluded in principle.

The point is that architectural complexity should enter only when it increases discriminating power.

A complicated system that makes causal interpretation harder is not an experimental advance merely because it appears more realistic.

14.32 No consciousness implementation target

The implementation roadmap stops at functional self-modelling.

There should be no software milestone labelled:

$$\texttt{consciousness} \tag{671}$$

or equivalent.

No behavioural module, self-report channel, recursive controller or conflict mechanism has been established as a valid proxy for phenomenal experience.

Consequently:

$$\boxed{\begin{gathered}\text{Paper III may implement increasingly rich}\\ \text{functional architectures;}\\ \text{it does not implement a consciousness test.}\end{gathered}} \tag{672}$$

14.33 Implementation success criteria

The implementation should be judged successful if it can:

1. instantiate the intended mechanism;
2. instantiate a serious matched alternative;
3. perform selective interventions;

4. restore disrupted functions where required;
5. preserve complete experimental provenance;
6. regenerate all analyses from raw data;
7. assign the preregistered decision without manual reinterpretation.

Whether the theoretical hypothesis is retained is a separate matter.

A well-implemented experiment that rejects the hypothesis is a successful implementation.

14.34 Provisional judgement

The practical strategy for Paper III is therefore not to build an artificial universe containing every proposed level of organisation.

It is to construct the smallest system capable of discriminating between competing explanations.

The implementation roadmap is:

$$\boxed{\begin{array}{c}\text{shared infrastructure}\\ \downarrow\\ \text{small interpretable experiment}\\ \downarrow\\ \text{matched alternatives}\\ \downarrow\\ \text{intervention and rescue}\\ \downarrow\\ \text{preregistered decision}\\ \downarrow\\ \text{replicate before scaling.}\end{array}} \tag{673}$$

The governing rule is:

$$\boxed{\begin{array}{c}\text{implement only as much complexity as is needed}\\ \text{to place the hypothesis at genuine risk.}\end{array}} \tag{674}$$

15 What the Experiment Cannot Establish

Paper III is intentionally ambitious in experimental design and conservative in interpretation. Even if every principal experiment succeeds, the results would establish only a set of controlled functional, causal and dependency relations within specified artificial systems.

The experiment does not test everything represented in the wider Theory of Everything research programme, nor can artificial construction by itself settle questions about natural history, ontology, life, intelligence or phenomenal consciousness.

The governing limitation is:

$$\boxed{\begin{array}{c}\text{successful artificial construction} \\ \neq \text{ demonstration of universal natural necessity.}\end{array}} \tag{675}$$

15.1 Synthetic sufficiency is not natural necessity

Suppose artificial architecture A, supplemented by mechanism M, reliably produces capacity B:

$$A + M \longrightarrow B. \tag{676}$$

The strongest immediate conclusion is that M can contribute to B within the tested architecture and environment.

This does not establish:

$$M \text{ is universally necessary for } B, \tag{677}$$

nor:

$$M \text{ is the route by which natural systems produce } B. \tag{678}$$

A necessity claim requires serious attempts to construct:

$$\neg M \wedge B. \tag{679}$$

Even failure to find such a bypass in Paper III would establish necessity only within the tested architecture classes, not across every physically possible system.

15.2 Artificial reconstruction does not recover historical origin

A successful experiment can demonstrate that a transition is possible under specified artificial conditions.

It cannot, by itself, show that the same sequence occurred historically in the origin of:

$$\begin{gathered}\text{life},\\ \text{heredity},\\ \text{Darwinian evolution},\\ \text{learning},\\ \text{intelligence},\\ \text{reflective cognition}.\end{gathered} \tag{680}$$

Historical origin claims require independent evidence from the relevant natural sciences.

Accordingly,

$$\boxed{\text{synthetic reconstruction} \neq \text{historical reconstruction}.} \tag{681}$$

15.3 Block ordering does not establish an evolutionary ladder

The five-block ordering is an experimental organisation of questions.

It does not establish that natural systems must develop through:

$$\text{maintenance} \to \text{heredity} \to \text{learning} \to \text{agency} \to \text{self-modelling} \tag{682}$$

in that order.

Later blocks may be implementable independently of earlier ones.

Bypass architectures may reveal:

$$A \not\to B \tag{683}$$

even when both capacities exist.

The final dependency structure may therefore be a chain, a branching graph, a network of partly independent mechanisms or no coherent hierarchy at all.

15.4 The experiment cannot establish a universal measure of organisation

Paper II rejected the original proposal that heterogeneous constraint quantities should be interpreted as independent coordinates of one universal cross-scale vector.

Paper III therefore does not establish:

$$\Gamma_{\text{total}}, \tag{684}$$

a universal scalar of organisation, complexity, intelligence or constraint.

The typed profile

$$\mathfrak{C} = [\mathfrak{D}, \mathfrak{F}, \mathfrak{T}, \mathfrak{O}] \tag{685}$$

is a comparative organisational device, not a discovered fundamental state variable.

Consequently,

$$\boxed{\text{successful measurement across blocks} \neq \text{discovery of one universal quantity.}} \tag{686}$$

15.5 Artificial self-maintenance does not establish life

Block I may demonstrate that an artificial system contributes causally to maintaining the organisation required for its continued operation.

Such a result would not determine whether the system is alive.

Life may involve additional properties such as:

$$\begin{gathered}\text{metabolism},\\ \text{autonomous material organisation},\\ \text{reproduction},\\ \text{open-ended heredity},\\ \text{evolutionary individuality}.\end{gathered} \tag{687}$$

The experiment can isolate formal properties associated with living systems without resolving the general definition of life.

Thus,

$$\boxed{\text{self-maintenance} \not\Rightarrow \text{life}.} \tag{688}$$

15.6 Artificial selection does not establish the origin or universality of Darwinism

Block II can show that a specified artificial population instantiates:

$$\begin{gathered}\text{reproduction},\\ \text{heredity},\\ \text{variation},\\ \text{differential reproductive success}.\end{gathered} \tag{689}$$

It may also show that these relations generate cumulative population change.

That result would support natural-selection-like dynamics in the artificial system.

It would not establish:

$$\begin{gathered}\text{how biological heredity first arose,}\\ \text{one universal unit of selection,}\\ \text{inevitable increase in complexity,}\\ \text{inevitable progression towards intelligence.}\end{gathered} \tag{690}$$

Selection remains an effective population-level mechanism once its enabling conditions are present.

Paper III does not convert it into a fundamental generative law of reality.

15.7 Retention, feedback and learning do not establish intelligence

Block III may demonstrate causally distinct roles for:

$$\begin{gathered}\text{retention,}\\ \text{closed-loop feedback,}\\ \text{experience-dependent learning.}\end{gathered} \tag{691}$$

Even all three together do not establish general intelligence.

A system may successfully remember, regulate and learn while remaining highly task-specific.

Therefore,

$$\boxed{\text{retention} + \text{feedback} + \text{learning} \not\Rightarrow \text{general intelligent agency.}} \tag{692}$$

The transition to structural generalisation requires separate evidence.

15.8 World-model use does not establish unrestricted understanding

Block IV may demonstrate that an internal predictive model is causally used in planning and supports adaptation under structural novelty.

That would be an important result.

It would not establish that the system:

$$\begin{gathered}\text{understands in every philosophical sense,}\\ \text{possesses human-like concepts,}\\ \text{has unrestricted general intelligence,}\\ \text{represents reality in the same way as humans.}\end{gathered} \tag{693}$$

A world model is an experimentally testable functional construct.

Questions about semantic understanding require additional criteria beyond successful prediction and control.

15.9 Structural generalisation does not establish AGI

Strong performance under:

$$\begin{gathered}\text{causal novelty,}\\ \text{compositional novelty,}\\ \text{problem-family transfer}\end{gathered} \tag{694}$$

would support a broader form of agency than ordinary interpolation.

It would still not establish artificial general intelligence in an unrestricted sense.

The tested task families remain finite and experimentally defined.

Accordingly,

$$\boxed{\text{structural generalisation} \neq \text{unbounded general intelligence.}} \tag{695}$$

15.10 Self-monitoring does not establish reflective selfhood

Block V may show that self-related estimates such as confidence, competence or internal reliability are:

$$\begin{gathered}\text{accurately represented,}\\ \text{causally used,}\\ \text{selectively useful in control.}\end{gathered} \tag{696}$$

Even this does not establish human-like selfhood.

The experiment tests a narrower functional property:

$$\text{causal use of representations about the system itself.} \tag{697}$$

It does not establish:

$$\begin{gathered}\text{autobiographical identity,}\\ \text{social selfhood,}\\ \text{narrative selfhood,}\\ \text{subjective introspection.}\end{gathered} \tag{698}$$

15.11 Self-report does not establish consciousness

Where an artificial system can generate reports about its own state, those reports may be experimentally informative.

They are not sufficient evidence of phenomenal consciousness.

A self-report may be generated because:

$$\begin{gathered}\text{a reporting module reads an internal variable,}\\ \text{a policy has learned a linguistic mapping,}\\ \text{the architecture predicts expected evaluator responses.}\end{gathered} \tag{699}$$

Therefore,

$$\boxed{\text{self-report} \not\Rightarrow \text{phenomenal awareness.}} \tag{700}$$

15.12 No experiment in Paper III establishes phenomenal consciousness

The strongest limitation concerns consciousness.

Paper III contains no validated operational bridge from

$$\text{functional or reflective architecture} \tag{701}$$

to:

$$\text{subjective experience.} \tag{702}$$

Accordingly:

$$\boxed{\begin{gathered}\text{memory} \not\Rightarrow \text{consciousness},\\ \text{learning} \not\Rightarrow \text{consciousness},\\ \text{world modelling} \not\Rightarrow \text{consciousness},\\ \text{self-modelling} \not\Rightarrow \text{consciousness},\\ \text{conflict representation} \not\Rightarrow \text{consciousness.}\end{gathered}} \tag{703}$$

Phenomenal consciousness remains Open Problem 1 in Paper I.

Paper III does not attempt to solve that problem by defining a behavioural proxy.

15.13 Conflict representation does not validate PNT

The auxiliary conflict experiment may show that explicit representation of competing objectives improves control under some artificial conditions.

Such a finding would not validate Paradox–Negotiation Theory as a theory of consciousness.

At most it would establish:

$$\text{a functional role for explicit conflict representation.} \tag{704}$$

If ordinary multi-objective optimisation, model-based control or constraint satisfaction explains

the same result, the conflict mechanism should be reduced accordingly.

Thus,

$$\boxed{\text{functional conflict management} \neq \text{evidence for phenomenal paradox negotiation.}} \tag{705}$$

15.14 The experiment cannot establish that artificial and biological mechanisms are identical

Formal similarity across artificial and biological systems can motivate comparison.

It does not establish mechanistic identity.

For example, an artificial retention mechanism and biological memory may share a relation such as:

$$\text{past state} \longrightarrow \text{present retained distinction} \longrightarrow \text{future behavioural consequence.} \tag{706}$$

The underlying substrates, causal pathways and developmental histories may still differ profoundly.

Any claim of cross-domain equivalence must therefore pass the validation protocol developed in Paper II.

15.15 Cross-scale resemblance does not establish universality

Suppose similar formal relationships appear in:

$$\begin{gathered}\text{physical systems,}\\ \text{biological systems,}\\ \text{artificial systems.}\end{gathered} \tag{707}$$

That resemblance is a hypothesis-generating observation.

Universality requires stronger evidence, such as:

$$\begin{gathered}\text{invariants,}\\ \text{monotonicities,}\\ \text{controlled transformation laws,}\\ \text{causally preserved mappings.}\end{gathered} \tag{708}$$

Paper III alone cannot establish such universality.

15.16 The experiment cannot establish fundamental ontology

Nothing in Paper III determines that:

$$\begin{gathered}\text{constraint},\\ \text{information},\\ \text{computation},\\ \text{organisation},\\ \text{agency}\end{gathered} \tag{709}$$

is a fundamental substance or primitive constituent of reality.

The experiment tests relations among operationally defined artificial variables.

Ontological claims require independent argument and empirical support.

In particular, successful artificial modelling would not establish:

$$\text{the universe is literally a computer.} \tag{710}$$

15.17 Paper III cannot establish a fundamental informational field

The experiments do not test the existence of a fundamental informational scalar, informational field or information-based gravitational degree of freedom.

No result in Paper III should therefore be interpreted as evidence for:

$$I(x), \tag{711}$$

UIFA, UIFA+, or any related physical-field hypothesis.

Those proposals require their own theoretical and empirical tests.

The present experiment concerns artificial dependency architecture, not quantum gravity.

15.18 Success cannot establish inevitability

Even if the same sequence of capacities appears repeatedly across many artificial runs, this would not establish that the sequence is inevitable.

Repeated occurrence may reflect:

$$\begin{gathered}\text{shared architecture},\\ \text{shared objective functions},\\ \text{shared environment design},\\ \text{shared optimisation procedure}.\end{gathered} \tag{712}$$

Claims of inevitability require exploration of alternative architectures and counterfactual conditions far beyond one experimental family.

15.19 Failure cannot establish impossibility

The interpretive restriction applies equally to negative results.

If mechanism M fails to produce capacity B in the tested systems, the result does not establish:

$$B \text{ is impossible.} \tag{713}$$

It establishes only that:

$$\begin{gathered}\text{the tested implementation of } M \\ \text{did not produce the preregistered evidence for } B \\ \text{under the specified conditions.}\end{gathered} \tag{714}$$

A stronger impossibility claim would require independent mathematical or physical argument.

15.20 Absence of one mechanism does not establish absence of the capacity

A failed necessity claim should be distinguished from failure of the target capacity itself.

If explicit world-model rollout is rejected as necessary for structural generalisation, the conclusion is:

$$\text{rollout is not necessary under the tested conditions.} \tag{715}$$

It is not:

$$\text{structural generalisation does not exist.} \tag{716}$$

Likewise, rejection of explicit self-modelling as necessary for flexible control would not imply that intelligent agents cannot regulate uncertainty.

15.21 The experiment cannot validate the whole Theory of Everything

Paper III tests a limited part of the wider research architecture.

Even complete success would not validate:

$$\begin{gathered}\text{the whole Minimal Principles architecture,} \\ \text{the whole Mathematics of Constraint programme,} \\ \text{a physical Theory of Everything,} \\ \text{a unified account of life, intelligence and consciousness.}\end{gathered} \tag{717}$$

Its contribution is narrower and more useful:

$$\boxed{\begin{array}{c}\text{it can eliminate unsupported dependency claims,}\\ \text{identify mechanisms that survive controlled challenge,}\\ \text{and reveal where the proposed hierarchy breaks.}\end{array}} \tag{718}$$

15.22 Summary boundary table

The principal interpretive limits are summarised below.

Table 18: Claims that Paper III does not establish automatically.

Possible experimental result	**Conclusion that does not follow automatically**
Persistent driven organisation	Self-maintenance or life
Internal causal maintenance	Biological life
Heritable differential reproduction	Universal evolutionary progress
Selection-driven population adaptation	Inevitable increase in complexity or intelligence
Retention, feedback and learning	General intelligence
World-model-based structural generalisation	Unrestricted understanding or AGI
Causal self-model use	Phenomenal self-awareness
Accurate self-report	Consciousness
Functional conflict representation	PNT as a consciousness theory
Formal artificial–biological similarity	Mechanistic or ontological identity
Repeated dependency pattern across artificial systems	Universal or inevitable natural hierarchy
Successful cross-block measurement	Universal scalar of organisation
Complete success of Paper III	Validation of the whole Theory of Everything programme

15.23 Provisional judgement

The epistemic value of Paper III comes partly from what it refuses to infer.

Artificial experiments can reveal:

$$\begin{gathered}\text{possible mechanisms,}\\ \text{causal dependencies,}\\ \text{robust bypasses,}\\ \text{failed necessity claims,}\\ \text{conditional sufficiency relations.}\end{gathered} \tag{719}$$

They cannot, without additional evidence, convert those findings into claims about universal natural law, historical origin, fundamental ontology or phenomenal experience.

The governing rule is:

$$\boxed{\begin{gathered}\text{claim only what the intervention establishes;}\\ \text{do not infer natural necessity from artificial sufficiency;}\\ \text{do not infer consciousness from functional complexity;}\\ \text{and do not treat one successful artificial dependency structure}\\ \text{as the architecture of reality.}\end{gathered}} \tag{720}$$

16 Relationship to the Minimal Principles and Constraint Architectures

Paper III is the experimental component of a three-paper research architecture. Its role can be understood only by distinguishing what the preceding papers propose from what the present experiments are capable of testing.

The three papers perform different scientific tasks:

$$\boxed{\begin{gathered}\text{Paper I: identify candidate conditions, mechanisms}\\ \text{and unresolved transitions;}\\ \text{Paper II: test whether constraint concepts admit}\\ \text{mathematically defensible comparison;}\\ \text{Paper III: expose selected dependency claims}\\ \text{to controlled artificial intervention.}\end{gathered}} \tag{721}$$

None of the three papers is intended to confirm the other two by definition. Their relationship is therefore one of constrained interaction rather than mutual presupposition.

16.1 Paper I provides the candidate dependency architecture

The Minimal Principles Paper asks which claims deserve a place in a research architecture extending from physical dynamics through organisation, evolution and intelligence towards the unresolved problem of consciousness.

Its adversarial review produced several different evidential categories rather than one continuous hierarchy:

$$\begin{aligned} \text{FC1}: &\ \text{dynamical state evolution and physical admissibility},\\ \text{EC1}: &\ \text{sustained non-equilibrium driving},\\ \text{EM1}: &\ \text{causally consequential retention},\\ \text{EM2}: &\ \text{natural selection in heritable populations},\\ \text{EM3}: &\ \text{closed-loop regulation, learning and model-based control.} \end{aligned} \tag{722}$$

It also retained four unresolved transition problems:

$$\begin{aligned} \text{TP1}: &\ \text{dissipative organisation} \rightarrow \text{self-maintenance},\\ \text{TP2}: &\ \text{self-maintenance} \rightarrow \text{Darwinian population},\\ \text{TP3}: &\ \text{adaptive regulation} \rightarrow \text{intelligent agency},\\ \text{TP4}: &\ \text{intelligent agency} \rightarrow \text{reflective self-modelled intelligence}. \end{aligned} \tag{723}$$

Phenomenal consciousness was not converted into another transition mechanism. It remained:

$$\text{OP1}: \text{The Physical Basis of Phenomenal Consciousness}. \tag{724}$$

Paper III inherits these distinctions as questions to test, not propositions to demonstrate.

16.2 Paper III does not experimentally reproduce all of Paper I

Not every component of the Minimal Principles architecture has a corresponding artificial experiment.

Framework Condition 1 concerns physical admissibility at a level more general than the artificial systems constructed here. Paper III presupposes ordinary lawful computation and simulation; it does not experimentally test whether reality itself must obey a particular fundamental state-evolution principle.

Likewise, sustained non-equilibrium driving appears in the Gray–Scott baseline and maintenance experiments, but Paper III does not attempt to establish EC1 as a universal condition for every possible form of organisation.

The relationship is therefore selective:

$$\boxed{\begin{array}{c}\text{Paper I architecture}\\ \supset \\ \text{subset accessible to controlled artificial testing.}\end{array}} \tag{725}$$

Experimental tractability should not be mistaken for theoretical priority.

16.3 Block I tests part of Transition Problem 1

Experimental Block I corresponds most directly to:

$$\text{TP1} : \text{dissipative organisation} \rightarrow \text{self-maintaining organisation.} \tag{726}$$

The Gray–Scott baseline first demonstrates why persistent pattern formation is insufficient. Externally driven dynamics can sustain organised structure without establishing that the structure causally maintains itself.

The subsequent maintenance architectures then distinguish:

$$\begin{array}{c}\text{passive persistence,}\\ \text{external maintenance,}\\ \text{internal feed-forward maintenance,}\\ \text{reciprocal internal maintenance.}\end{array} \tag{727}$$

A successful result would therefore narrow TP1 by demonstrating at least one artificial route from persistence towards internally generated maintenance.

It would not solve the historical origin of biological self-maintenance.

16.4 Block II separates Transition Problem 2 from Effective Mechanism 2

Paper I distinguishes the emergence of a Darwinian population from the later operation of natural selection within such a population.

Paper III preserves that distinction.

Block II first asks:

$$\boxed{\text{What reproduces?}} \tag{728}$$

It then tests:

$$\begin{gathered}\text{individual identity,}\\ \text{reproduction,}\\ \text{parent–offspring transmission,}\\ \text{heritable variation.}\end{gathered} \tag{729}$$

Only after those relations are established does the experiment test:

$$\text{EM2 : natural selection in a heritable population.} \tag{730}$$

This ordering prevents selection from being treated as the generative principle that creates the very heredity and reproduction required for selection to operate.

16.5 Block III connects Effective Mechanism 1 with selected components of Effective Mechanism 3

Paper I treats causally consequential retention as Effective Mechanism 1 and places closed-loop regulation, learning and model-based control within Effective Mechanism 3.

Block III therefore does not decompose EM3 into retention, feedback and learning. Instead, it brings one experimentally tractable implementation of EM1 into direct comparison with selected components of EM3.

The principal factorial design is:

$$R \times F \times L, \tag{731}$$

where:

$$\begin{aligned} R &= \text{policy-accessible retained state,}\\ F &= \text{within-episode closed-loop feedback,}\\ L &= \text{between-episode experience-dependent learning.}\end{aligned} \tag{732}$$

This notation is experimental rather than ontological. Learning itself retains the effects of earlier experience in changed policy parameters, so R is not intended to represent every possible form of historical influence.

The experiment instead asks whether three differently located causal channels make distinguishable contributions:

$$\boxed{\begin{gathered}\text{retained state changes which past distinctions}\\ \text{remain directly available to the policy;}\\ \text{feedback changes action from newly arriving information;}\\ \text{learning changes the policy used in future episodes.}\end{gathered}} \tag{733}$$

If these distinctions collapse under matched intervention, the corresponding architecture in

Paper I should be revised.

If they remain experimentally separable, the result supports functional distinctions among EM1 and the tested components of EM3 without establishing a necessary hierarchy among them.

16.6 Block IV addresses Transition Problem 3

Transition Problem 3 asks what, if anything, separates successful adaptive regulation from more general intelligent agency.

Paper III operationalises one candidate distinction through:

$$\text{structural generalisation.} \tag{734}$$

The decisive tests move beyond ordinary held-out examples towards:

$$\begin{gathered}\text{novel composition,}\\ \text{altered causal relations,}\\ \text{delayed consequences,}\\ \text{hidden-state inference,}\\ \text{new problem families.}\end{gathered} \tag{735}$$

World models and counterfactual rollout are treated as candidate mechanisms, not definitional requirements for intelligence.

Consequently, Paper III can discover either:

$$\text{world-model use contributes to structural generalisation,} \tag{736}$$

or:

$$\boxed{\text{structural generalisation can bypass explicit world-model use.}} \tag{737}$$

Either result modifies TP3 constructively.

16.7 Block V addresses Transition Problem 4

Transition Problem 4 concerns the move from intelligent agency to reflective self-modelled intelligence.

Block V gives that transition an intervention-sensitive interpretation.

The target is not merely:

$$\text{self-description,} \tag{738}$$

nor:

$$\text{confidence prediction.} \tag{739}$$

The stronger candidate relation is:

$$\boxed{\begin{array}{c}\text{representation of a property of the system itself} \\ \longrightarrow \text{causal modification of subsequent control.}\end{array}} \tag{740}$$

Self-model misinformation and correction provide the key intervention–rescue test.

A positive result would support one functional route to reflective self-modelling. A successful bypass would show that some apparently reflective capacities can be realised without the proposed endogenous representation.

16.8 Open Problem 1 has no Paper III block

The absence of a consciousness block is theoretically significant.

Paper I concluded that the present architecture provides no validated bridge from

$$\text{reflective self-modelled intelligence} \tag{741}$$

to

$$\text{phenomenal consciousness.} \tag{742}$$

Paper III therefore stops its principal experimental sequence at Block V.

There is no:

$$\text{Block VI: Consciousness.} \tag{743}$$

Such a block would create the appearance that phenomenal experience is simply the next experimentally defined capacity in the hierarchy, when no justified operational transition has been established.

16.9 The auxiliary conflict test does not restore a consciousness bridge

The conflict-representation experiment is intentionally outside Blocks I–V.

Its purpose is to test whether explicit representation of competing objectives provides functional value beyond:

$$\begin{array}{c}\text{scalar reward optimisation,} \\ \text{ordinary multi-objective control,} \\ \text{model-based prediction.}\end{array} \tag{744}$$

This preserves a testable part of the wider Paradox–Negotiation research programme without granting it privileged status.

Even a strong positive result would establish only:

$$\text{functional value of explicit conflict representation.} \tag{745}$$

It would not move PNT into Open Problem 1 as the demonstrated origin of phenomenal consciousness.

16.10 Paper II changes what Paper III is allowed to measure

The Mathematics of Constraint Paper began with a candidate cross-scale vector:

$$\mathbf{\Gamma} = (\Gamma_{\text{state}}, \Gamma_{\text{trans}}, \Gamma_{\text{path}}, \Pi_C, \phi_K, \chi_{\text{int}}, R(\tau)) \,. \tag{746}$$

Adversarial review showed that the vector interpretation does not survive.

The proposed quantities differ in mathematical role, dependence upon baseline, representation, timescale and explanandum. Some are nested; others describe different kinds of functional or historical relation.

Paper III therefore does not attempt to validate:

$$\mathbf{\Gamma} \tag{747}$$

as a universal state variable.

Instead, it inherits the weaker typed profile:

$$\mathfrak{C} = [\mathfrak{D}, \mathfrak{F}, \mathfrak{T}, \mathfrak{O}]. \tag{748}$$

16.11 The typed profile constrains experimental interpretation

Paper II classifies relevant comparisons broadly into:

$$\begin{aligned} \mathfrak{D} &: \text{ dynamical divergence,} \\ \mathfrak{F} &: \text{ functional effects,} \\ \mathfrak{T} &: \text{ temporal and historical descriptors,} \\ \mathfrak{O} &: \text{ organisational and interaction structure.} \end{aligned} \tag{749}$$

Paper III uses these categories only where they match the experimental explanandum.

For example, Block I can legitimately use:

$$G_{\mathcal{C}} \tag{750}$$

and:

$$\chi_{\text{int}}, \tag{751}$$

because maintenance topology and internal contribution are part of the target question.

Block V has no reason to invent corresponding organisational-closure quantities merely to preserve numerical symmetry.

Thus:

$$\boxed{\text{shared framework} \neq \text{mandatory shared metric.}} \tag{752}$$

16.12 Paper III provides intervention tests for Paper II quantities

The relationship also operates in the other direction.

Paper II identifies quantities whose interpretation requires causal validation. Paper III supplies controlled environments in which some of those requirements can be tested directly.

For example:

$$R(\tau) \tag{753}$$

may indicate that information about earlier history remains recoverable.

Block III then asks whether intervention on the retained state changes later behaviour:

$$R_{\text{causal}}(\Delta). \tag{754}$$

Likewise:

$$\chi_{\text{int}} \tag{755}$$

may describe the relative contribution of internal maintenance.

Block I then tests whether nominated internal maintenance relations survive:

$$\begin{gathered}\text{ablation},\\ \text{matched damage},\\ \text{specific rescue.}\end{gathered} \tag{756}$$

This turns candidate descriptors into intervention-sensitive research questions.

16.13 Paper III does not validate constraint as a universal principle

Even widespread experimental usefulness of constraint language would not show that constraint is fundamental.

Suppose Paper III finds meaningful restriction or divergence relations in all five blocks.

That would establish:

$$\text{constraint descriptions are useful across these experiments.} \tag{757}$$

It would not establish:

$$\text{constraint is the universal generative principle of reality.} \tag{758}$$

The stronger Stage 1 programme identified in Paper I and Paper II still requires evidence for:

$$\begin{gathered}\text{cross-scale invariants,}\\ \text{monotonic relations,}\\ \text{controlled transformation laws,}\\ \text{causally preserved mappings.}\end{gathered} \tag{759}$$

Paper III can generate candidate relations for those tests but cannot assume their universality.

16.14 Artificial results can generate cross-scale hypotheses

A robust artificial dependency may nevertheless become scientifically useful outside Paper III.

Suppose Block III finds a reproducible relation:

$$Q_B = \mathcal{T}_{A\to B}(Q_A) + \varepsilon. \tag{760}$$

A biological analogue may then motivate a preregistered cross-domain hypothesis.

The required progression is:

$$\boxed{\begin{gathered}\text{validate artificial relation}\\ \downarrow\\ \text{define independent natural-domain quantities}\\ \downarrow\\ \text{specify mapping before decisive comparison}\\ \downarrow\\ \text{test held-out cross-domain prediction.}\end{gathered}} \tag{761}$$

Similarity noticed after the fact is not sufficient.

16.15 The three papers must not form a circular confirmation system

A serious risk in a multi-paper programme is evidential circularity.

Paper I could propose a mechanism because it appears compatible with Paper II; Paper II could choose measurements because they fit Paper I; Paper III could then engineer the proposed mechanism and claim experimental confirmation of the original architecture.

That sequence would be scientifically weak.

The programme therefore requires:

$$\boxed{\begin{array}{c}\text{Paper I supplies hypotheses, not answers;}\\ \text{Paper II supplies admissibility tests, not validation by notation;}\\ \text{Paper III supplies interventions, not engineered confirmation.}\end{array}} \tag{762}$$

A hypothesis must remain capable of failing at every stage.

16.16 Evidence can flow back into Paper I

Although the evidential structure must not be circular, it is iterative.

Results from Paper III should modify Paper I when they directly bear upon its dependency claims.

For example, a robust artificial bypass of:

$$A \longrightarrow B \tag{763}$$

would provide evidence against treating A as universally necessary for B.

Depending upon scope, Paper I should then:

$$\text{Restrict,} \qquad \text{Reduce,} \qquad \text{or Reject} \tag{764}$$

the corresponding dependency claim.

The architecture is therefore revisable.

It is not a fixed hierarchy awaiting experimental illustration.

16.17 Evidence can also flow back into Paper II

Paper III may reveal that a candidate measurement is:

$$\begin{gathered}\text{unstable under representation change,}\\ \text{redundant with another descriptor,}\\ \text{insensitive to causal intervention,}\\ \text{useful only within a narrow domain.}\end{gathered} \tag{765}$$

Such results should feed back into the Mathematics of Constraint programme.

A descriptor retained provisionally in Paper II can therefore later be:

$$\begin{gathered}\text{restricted,}\\ \text{redefined,}\\ \text{replaced,}\\ \text{removed.}\end{gathered} \tag{766}$$

Mathematical notation does not protect a quantity from empirical failure.

16.18 The research loop is iterative rather than circular

The legitimate interaction among the papers can be represented as:

$$\boxed{\begin{gathered}\text{Paper I: propose dependency}\\ \downarrow\\ \text{Paper II: clarify measurable mathematical structure}\\ \downarrow\\ \text{Paper III: intervene experimentally}\\ \downarrow\\ \text{revise Paper I and/or Paper II}\\ \downarrow\\ \text{generate a sharper new hypothesis.}\end{gathered}} \tag{767}$$

The crucial difference from circular confirmation is that every return step must be driven by new evidence or a demonstrated logical correction.

Nothing is preserved merely to maintain coherence among the papers.

16.19 Paper III can distinguish possible from necessary transitions

One of the most useful contributions of artificial experimentation is the ability to separate two questions:

$$\begin{gathered}\text{Can this mechanism produce the capacity?} \\ \text{Must this mechanism produce the capacity?}\end{gathered} \tag{768}$$

Successful construction addresses the first.

Bypass experiments address the second.

For:

$$A \xrightarrow{M} B, \tag{769}$$

Paper III may eventually support:

$$\begin{gathered}M \text{ is one sufficient artificial route to } B, \\ M \text{ is not necessary because another route exists.}\end{gathered} \tag{770}$$

That is a scientifically informative result even though the original hierarchy has been weakened.

16.20 The papers operate at different levels of claim

The division of labour can be summarised as follows:

Table 19: Division of labour across the three-paper research architecture.

Paper	**Primary question**	**Main output**	**What it does not establish**
Paper I	Which conditions, mechanisms and transitions deserve active consideration?	Dependency map, evidential categories, transition problems and falsification programme	That the proposed dependencies are experimentally true
Paper II	Can constraint-related quantities be defined and compared without collapsing distinct mathematical roles?	Typed comparative profile, failure tests and cross-scale validation protocol	A universal constraint scalar or ontology
Paper III	Which selected dependencies survive controlled artificial intervention?	Ablation, rescue, bypass and robustness verdicts	Universal natural necessity, historical origin or consciousness

Together, the papers define a research programme rather than a completed Theory of Everything.

16.21 Relationship to the wider Theory of Everything programme

The three-paper architecture occupies an intermediate level within the wider programme.

It does not attempt directly to unify:

$$\begin{gathered}\text{quantum mechanics,}\\ \text{gravity,}\\ \text{particle physics,}\\ \text{chemistry,}\\ \text{biology,}\\ \text{cognition,}\\ \text{consciousness}\end{gathered} \tag{771}$$

through one equation.

Instead, it asks a prior methodological question:

$$\boxed{\begin{gathered}\text{which proposed bridges between levels}\\ \text{survive precise definition, mathematical scrutiny}\\ \text{and adversarial experiment?}\end{gathered}} \tag{772}$$

Only relations surviving those tests deserve consideration in a later unified architecture.

16.22 A failed Paper III transition can be a successful ToE result

The wider programme should not measure progress by how much of the proposed hierarchy survives.

Suppose Paper III demonstrates that:

$$\begin{gathered}\text{self-maintenance does not require reciprocal closure,}\\ \text{structural generalisation does not require explicit rollout,}\\ \text{reflective control does not require an endogenous self-model.}\end{gathered} \tag{773}$$

These would be negative results for several candidate dependencies.

They would nevertheless improve the Theory of Everything programme by removing unnecessary structure.

In this sense:

$$\boxed{\text{explanatory subtraction} = \text{scientific progress}} \tag{774}$$

when the removed element was not required by the evidence.

16.23 Positive results must remain conditional

Conversely, even strong positive results should retain their experimental scope.

A result such as:

$$\text{world-model intervention changes structural generalisation} \tag{775}$$

supports a causal role for that representation in the tested architecture.

It should enter Paper I initially as:

$$\text{supported artificial mechanism,} \tag{776}$$

not immediately as:

$$\text{universal principle of intelligence.} \tag{777}$$

Further architectures, task families and natural systems must determine how far the result transports.

16.24 Relationship to the quarantine ledger

Paper I's quarantine principle also applies to results generated here.

An experimental claim remains quarantined when:

$$\begin{gathered}\text{only exploratory evidence exists,}\\ \text{the decisive intervention has not been performed,}\\ \text{the effect is underpowered,}\\ \text{serious alternatives remain untested,}\\ \text{cross-domain extrapolation exceeds the evidence.}\end{gathered} \tag{778}$$

Paper III therefore supplies evidence that can move claims:

$$\begin{gathered}\text{into the active architecture,}\\ \text{into a restricted form,}\\ \text{back into quarantine,}\\ \text{or out of the architecture entirely.}\end{gathered} \tag{779}$$

16.25 Provisional judgement

The relationship among the three papers is deliberately asymmetric.

Paper I asks what might be required.

Paper II asks whether the proposed mathematical language can bear the weight placed upon it.

Paper III asks whether selected mechanisms and dependencies survive controlled attempts to break them.

The intended synthesis is therefore:

$$\boxed{\begin{array}{c}\text{conceptual architecture}\\ +\text{mathematical discipline}\\ +\text{adversarial experiment}\\ \longrightarrow \text{revisable dependency map.}\end{array}} \tag{780}$$

The three papers should converge only where independent lines of evidence permit convergence.

The governing rule is:

$$\boxed{\begin{array}{c}\text{Paper I may propose;}\\ \text{Paper II may formalise;}\\ \text{Paper III may test;}\\ \text{but only surviving evidence may unify.}\end{array}} \tag{781}$$

17 Open-Science Requirements

The adversarial value of Paper III depends upon other researchers being able to reconstruct not only the reported results but the chain of decisions that produced them. Releasing a final code repository is therefore insufficient. The experiment should preserve hypotheses, configurations, interventions, failed runs, bypass architectures, raw trajectories, analysis scripts and decision records in forms that permit independent inspection and reproduction.

These requirements are consistent with the broader principle that scientific data and metadata should remain findable, accessible, interoperable and reusable where practicable [11].

The central requirement is:

$$\boxed{\begin{array}{c}\text{claim} \longrightarrow \text{preregistered specification} \longrightarrow \text{implementation}\\ \longrightarrow \text{raw result} \longrightarrow \text{analysis} \longrightarrow \text{decision}\end{array}} \tag{782}$$

with every transition in that chain recoverable.

Open science is therefore not treated as an administrative appendix to the experiment. It is one of the controls against post-hoc theoretical reconstruction.

17.1 The reproducibility target

A complete release should allow an independent researcher to answer at least five questions.

1. What exactly was hypothesised before the decisive experiment?
2. Which code, environment and architecture instantiated the test?
3. Which raw observations produced the reported statistics?
4. Which analysis transformed those observations into the published result?
5. Why did that result receive its final Retain/Restrict/Reduce/Suspend/Reject judgement?

If one of these questions cannot be answered from the archived materials, the evidential chain is incomplete.

17.2 Preregistration archive

Every confirmatory hypothesis should have a preregistration record created before decisive evaluation.

The record should contain:

$$\begin{gathered}\text{hypothesis wording,} \qquad \text{operational definitions,} \\ \text{architectures,} \qquad \text{environment classes,} \\ \text{primary estimand,} \qquad \text{sample or replicate plan,} \\ \text{interventions,} \qquad \text{bypass tests,} \\ \text{robustness analyses,} \qquad \text{decision rules.}\end{gathered} \tag{783}$$

The preregistration should be timestamped and preserved in an immutable or versioned repository.

Later clarification is permitted, but it must not overwrite the original record.

17.3 Amendment ledger

Changes made after preregistration should be recorded explicitly.

For amendment A_j, store:

$$A_j = (t_j, r_j, c_j, e_j)\,, \tag{784}$$

where:

- t_j is the time of the amendment;
- r_j is the reason for the change;

- c_j records the changed specification;
- e_j records whether decisive data had already been inspected.

A correction made before decisive results are visible differs epistemically from a change introduced after the preferred result has failed.

Both may be legitimate, but they should remain distinguishable.

17.4 Machine-readable experiment manifests

Each confirmatory experiment should have a machine-readable manifest linked to its preregistration.

The manifest should include:

$$\begin{gathered}\text{experiment ID}, \quad \text{hypothesis IDs},\\ \text{architecture versions}, \quad \text{environment versions},\\ \text{training budget}, \quad \text{seed list or seed procedure},\\ \text{held-out evaluation suite}, \quad \text{intervention IDs},\\ \text{rescue IDs}, \quad \text{bypass IDs},\\ \text{primary analysis script}, \quad \text{decision rule}.\end{gathered} \tag{785}$$

Human-readable prose explains the science.

The manifest records exactly what the computer executed.

Both are required.

17.5 Versioned source code

All code used for confirmatory experiments should be version controlled.

Each published result should identify:

$$\begin{gathered}\text{source-code commit},\\ \text{repository release},\\ \text{dependency specification},\\ \text{runtime environment}.\end{gathered} \tag{786}$$

Where practical, the release should include a reproducible software environment through a lock file, container specification or equivalent mechanism.

The purpose is not merely convenience. Software dependencies can change numerical behaviour and therefore belong to the experimental provenance.

17.6 Configuration hashes

Configuration files should be treated as experimental objects.

For configuration C, record cryptographic digest:

$$h_C = H(C). \tag{787}$$

Equivalent hashes should be recorded for:

$$\begin{gathered}\text{architecture definitions,}\\ \text{environment definitions,}\\ \text{held-out test suites,}\\ \text{intervention specifications.}\end{gathered} \tag{788}$$

This allows the published result to be tied to an exact experimental specification even when the repository later evolves.

17.7 Raw trajectories are primary records

The primary empirical record should be the raw experimental trajectory, not the final summary table.

For agent-based experiments, preserve where applicable:

$$\begin{gathered}O_t, \qquad A_t, \qquad X_t, \qquad Y_t,\\ M_t, \qquad \theta_t, \qquad \text{intervention status,}\\ \text{environment state,} \qquad \text{architecture state accessible to analysis.}\end{gathered} \tag{789}$$

Block I should additionally preserve component states and maintenance relations.

Block II requires population and lineage records.

Blocks IV and V require the internal states needed to reproduce model and self-model intervention analyses.

If storage limits make complete preservation impossible, the retention policy should be fixed before confirmatory runs rather than after inspecting which trajectories are interesting.

17.8 Lineage records for Block II

Population experiments require especially careful archival structure because aggregate trait frequencies cannot reconstruct ancestry.

The lineage record should include:

$$\begin{gathered}\text{individual identifier},\\ \text{parent identifier or identifiers},\\ \text{birth and termination times},\\ \text{transmissible parental state},\\ \text{offspring transmissible state},\\ \text{mutation event},\\ \text{reproductive success},\\ \text{environmental regime}.\end{gathered} \tag{790}$$

These records should be sufficient to reconstruct:

$$G_{\mathrm{lin}} \tag{791}$$

independently from the published summary statistics.

17.9 Model checkpoints

Where trained artificial agents are used, the exact checkpoints employed in decisive evaluation should be preserved.

For checkpoint W_j, archive:

$$\begin{gathered}\text{architecture ID}, \qquad \text{training configuration},\\ \text{training seed}, \qquad \text{training step},\\ \text{checkpoint hash}.\end{gathered} \tag{792}$$

This is particularly important for Blocks IV and V because internal interventions may depend upon the precise trained representation.

Retraining a nominally identical architecture is not necessarily equivalent to preserving the tested model.

17.10 Intervention registry

Every confirmatory ablation, corruption, perturbation and rescue should appear in a public intervention registry.

For intervention I_j, record:

$$\begin{gathered}\text{target mechanism},\\ \text{implementation operation},\\ \text{predicted effect},\\ \text{matched-damage control},\\ \text{rescue operation},\\ \text{associated hypothesis}.\end{gathered} \tag{793}$$

The registry should preserve interventions that fail as well as those that produce the predicted result.

Otherwise the published intervention set would itself become subject to selection bias.

17.11 Bypass registry

Bypass architectures deserve the same archival status as preferred architectures.

For every necessity claim:

$$M \longrightarrow B, \tag{794}$$

the corresponding bypass record should state:

$$\begin{gathered}\text{which mechanism } M \text{ is absent},\\ \text{which alternative architecture is used},\\ \text{how capacity is matched},\\ \text{how compute is matched},\\ \text{which equivalence or non-inferiority criterion applies}.\end{gathered} \tag{795}$$

A successful bypass should be preserved prominently rather than treated as a failed version of the preferred experiment.

17.12 Null architectures must be archived

Control systems are part of the evidential result.

The release should therefore include:

$$\begin{gathered}\text{passive controls},\\ \text{external-maintenance controls},\\ \text{neutral evolutionary controls},\\ \text{open-loop and yoked controllers},\\ \text{capacity-matched model-free systems},\\ \text{passive and shuffled self-monitors},\\ \text{scalarised conflict controls}.\end{gathered} \tag{796}$$

Publishing only the successful focal architecture would make it impossible to assess whether its explanatory advantage was genuine.

17.13 Development runs should remain distinguishable

Exploratory development data should not be mixed silently with confirmatory data.

The archive should distinguish:

$$\begin{gathered}\texttt{development},\\ \texttt{pilot},\\ \texttt{confirmatory},\\ \texttt{replication}.\end{gathered} \tag{797}$$

Development runs may be valuable for understanding architecture behaviour and estimating variance.

They should not be presented later as though they had been held out from design decisions.

17.14 Held-out environments must remain identifiable

Structural-generalisation claims depend critically upon what was withheld during development.

The release should therefore identify:

$$\begin{gathered}\mathcal{E}_{\text{train}},\\ \mathcal{E}_{\text{development}},\\ \mathcal{E}_{\text{decisive}}.\end{gathered} \tag{798}$$

The transformation rules used to generate decisive environments should also be archived.

Where disclosure before completion could compromise a future evaluation, the specification may be embargoed temporarily, but the timestamped version should still exist before testing.

17.15 Environment validation records

Environment unit tests and validation reports should accompany the experiment.

These should document checks of:

$$\begin{gathered}\text{transition dynamics},\\ \text{observation generation},\\ \text{hidden-state leakage},\\ \text{reward or viability calculations},\\ \text{reproduction logic},\\ \text{intervention implementation},\\ \text{randomisation behaviour}.\end{gathered} \tag{799}$$

A result produced by a later-discovered environment defect should remain in the historical archive but be marked invalid rather than silently removed.

17.16 Analysis scripts should regenerate published outputs

Every principal figure and table should be reproducible from archived raw or declared intermediate data.

The preferred relation is:

$$\boxed{\begin{gathered}\text{raw data} + \text{fixed analysis code}\\ \longrightarrow \text{published figure or table}.\end{gathered}} \tag{800}$$

Manual editing of numerical results after analysis should be avoided.

Where graphical annotation requires manual adjustment, the underlying plotted values should remain machine-readable.

17.17 Derived data must retain provenance

Processed datasets should identify the raw records from which they were derived.

For derived object D_j, record:

$$D_j = f_j\left(R_1, \ldots, R_n\right), \tag{801}$$

where f_j is the archived transformation and R_i are raw inputs.

This allows later researchers to inspect whether filtering, aggregation or normalisation altered the interpretation.

17.18 Data dictionaries

Every released dataset should include a data dictionary defining:

$$\begin{gathered}\text{field name}, \qquad \text{data type},\\ \text{unit}, \qquad \text{allowed range},\\ \text{missing-value meaning}, \qquad \text{experimental interpretation}.\end{gathered} \tag{802}$$

Internal variables should not be given theoretically loaded labels unless their function has been established.

For example, a tensor stored under implementation name:

$$\texttt{self_state} \tag{803}$$

should not automatically be described in the archive as:

$$\text{self-awareness}. \tag{804}$$

17.19 Seed records

Randomness must remain reproducible.

For every confirmatory run, preserve:

$$\begin{gathered}\text{training seed},\\ \text{environment seed},\\ \text{evaluation seed},\\ \text{mutation seed where relevant}.\end{gathered} \tag{805}$$

If seeds are generated algorithmically, preserve the generator and master seed.

Seed sets should not be filtered according to performance after the fact.

17.20 Computational provenance

Where numerical reproducibility may depend upon hardware or low-level libraries, the archive should record sufficient computational metadata to diagnose discrepancies.

Relevant metadata can include:

$$\begin{gathered}\text{operating system},\\ \text{language and library versions},\\ \text{accelerator type where used},\\ \text{numerical precision},\\ \text{determinism settings}.\end{gathered} \tag{806}$$

Exact bitwise reproduction may not always be possible.

The requirement is that deviations can be traced and scientifically assessed.

17.21 Negative results are first-class outputs

The archive should preserve results that weaken the research architecture.

These include:

$$\begin{gathered}\text{failed primary hypotheses},\\ \text{successful null models},\\ \text{successful bypasses},\\ \text{failed intervention effects},\\ \text{failed rescue experiments},\\ \text{replication failures},\\ \text{representation-sensitive findings}.\end{gathered} \tag{807}$$

A negative result that satisfies the preregistered protocol should receive the same archival status as a positive result.

Otherwise the public record would systematically exaggerate the coherence of the artificial hierarchy.

17.22 Invalid runs and negative runs must be distinguished

A scientifically negative result is not the same as an invalid experiment.

An invalid run may result from:

$$\begin{gathered}\text{software failure},\\ \text{corrupted data},\\ \text{failed environment validation},\\ \text{violation of a preregistered execution condition}.\end{gathered} \tag{808}$$

A valid run in which the hypothesis fails is evidence.

The archive should therefore preserve separate labels:

$$\boxed{\texttt{VALID_NEGATIVE} \neq \texttt{INVALID_RUN}.} \tag{809}$$

This distinction prevents inconvenient evidence from being removed under the language of data cleaning.

17.23 Machine-readable decision record

The final decision framework should be published both in prose and in a machine-readable ledger.

For hypothesis H_j, the record should contain:

$$\begin{gathered}\text{hypothesis ID},\\ \text{preregistered wording},\\ \text{primary estimand},\\ \text{observed effect},\\ \text{uncertainty interval},\\ \text{intervention result},\\ \text{rescue result},\\ \text{bypass result},\\ \text{robustness summary},\\ \text{final verdict},\\ \text{scope restriction}.\end{gathered} \tag{810}$$

The record should also indicate which version of Paper I or Paper II, if any, is affected by the result.

17.24 The quarantine ledger should be public

Claims placed in quarantine should remain visible.

For each quarantined claim, preserve:

$$\begin{gathered}\text{claim},\\ \text{current evidential status},\\ \text{reason for quarantine},\\ \text{missing decisive test},\\ \text{conditions for reconsideration}.\end{gathered} \tag{811}$$

This prevents unresolved ideas from disappearing when unfavourable while also allowing them

to be reconsidered if genuinely new evidence arises.

17.25 Cross-paper revision history

Because Papers I–III form a revisable programme, changes propagated from one paper to another should be recorded.

For example:

$$\begin{gathered}\text{Paper III result} \\ \longrightarrow \text{Paper I dependency revised}\end{gathered} \tag{812}$$

should generate a traceable revision note explaining:

1. which experimental result triggered the revision;
2. which earlier claim changed;
3. whether it was restricted, reduced, suspended or rejected;
4. which version first incorporated the change.

This creates an auditable history of theoretical evolution.

17.26 Figures and tables should carry source provenance

Every published figure and principal table should be traceable to:

$$\begin{gathered}\text{experiment manifest,} \\ \text{raw or processed dataset,} \\ \text{analysis script,} \\ \text{software version.}\end{gathered} \tag{813}$$

Where practical, figure-generation scripts should write this provenance automatically into accompanying metadata.

17.27 Open formats

Long-term preservation is stronger when core records are stored in widely documented, non-proprietary or easily convertible formats.

Where appropriate, the archive should favour:

$$\begin{gathered}\text{plain text,} \qquad \text{CSV,} \qquad \text{JSON,} \\ \text{standard image formats,} \qquad \text{documented binary arrays.}\end{gathered} \tag{814}$$

Specialised formats may also be retained when necessary, but they should not be the only representation of indispensable metadata.

17.28 Persistent archival releases

Major experimental releases should receive stable version identifiers and, where possible, persistent citations.

A practical publication architecture is:

$$\begin{gathered}\text{active development repository}\\ +\text{versioned archival release}\\ +\text{persistent bibliographic identifier.}\end{gathered} \tag{815}$$

The active repository can continue to evolve.

The archived release associated with a published claim should remain fixed.

17.29 Checksums and archive integrity

Large data releases should include checksums for important artefacts.

For file F_j:

$$h_j = H(F_j). \tag{816}$$

This allows later users to verify that downloaded or mirrored data remain identical to the published archive.

Archive-integrity information is especially useful when model checkpoints or large raw trajectory files are stored separately from the manuscript.

17.30 Independent reproduction package

The strongest release should include a minimal reproduction package requiring no knowledge of the original development workflow.

The package should provide:

1. a frozen confirmatory manifest;
2. required code and environment specification;
3. one small reference dataset or deterministic test;
4. instructions for reproducing one principal analysis;

5. expected checksum or numerical tolerance;

6. instructions for regenerating one published figure or table.

This package tests whether the public materials are actually usable rather than merely technically available.

17.31 Reproduction and replication should be distinguished

Two forms of confirmation should remain separate.

Computational reproduction asks whether the same code, data and specification regenerate the reported result.

Scientific replication asks whether the same claim survives new runs, new implementations or new task families.

Thus:

$$\boxed{\text{reproducible calculation} \neq \text{replicated mechanism.}} \tag{817}$$

Paper III requires both where feasible, but they answer different questions.

17.32 Replication tiers

The replication hierarchy introduced in the statistical framework should also be reflected in the archive.

Table 20: Open-science replication tiers for Paper III.

Tier	Replication level	Required archived material
1	New seeds and environment instances	Frozen configuration, seed procedure, raw data and analysis scripts
2	Alternative implementation	Mechanism specification independent of one code architecture, plus original reference implementation
3	Alternative task family	Operational hypothesis and decision rule transportable beyond the original environment
4	Independent reproduction	Complete public specification sufficient for an external researcher to reconstruct the experiment

A claim should report the highest replication tier actually achieved rather than implying full replication from internal reruns alone.

17.33 Resource limitations should be recorded

Open science also requires transparency about experiments that were not run.

If a bypass, replication or robustness test is omitted because of:

$$\begin{gathered}\text{compute limits,}\\ \text{storage limits,}\\ \text{implementation difficulty,}\\ \text{time constraints,}\end{gathered} \tag{818}$$

the omission should be stated explicitly.

An unperformed adversarial test is not evidence that the preferred mechanism would have survived it.

17.34 No selective release of internal states

Internal activations or representations should not be released only for runs that support the desired interpretation.

Where internal-state analysis is part of a preregistered hypothesis, the release policy should apply symmetrically across:

$$\begin{gathered}\text{successful runs,}\\ \text{failed runs,}\\ \text{control architectures,}\\ \text{bypass architectures.}\end{gathered} \tag{819}$$

Otherwise mechanistic interpretation could be biased by selective inspectability.

17.35 Interpretive annotations should remain separate from raw data

The archive may contain researcher annotations, but they should not be mixed with primary observations.

For example:

$$\text{“possible planning failure”} \tag{820}$$

is an interpretation.

The underlying:

$$\text{state,} \quad \text{action,} \quad \text{model prediction,} \quad \text{outcome} \tag{821}$$

are the primary records.

Keeping these layers separate allows later researchers to challenge the original interpretation.

17.36 Long-term preservation

The programme should plan for preservation beyond the lifetime of one software repository.

At minimum, final releases should preserve:

$$\begin{gathered}\text{manuscript},\\ \text{source code},\\ \text{configuration files},\\ \text{preregistrations},\\ \text{decision records},\\ \text{core raw datasets},\\ \text{analysis scripts},\\ \text{documentation}.\end{gathered} \tag{822}$$

Very large artefacts may require tiered storage, but metadata describing them should remain permanently available.

The archival aim is that a future researcher can understand not only what the project concluded but how those conclusions were generated.

17.37 No deletion of superseded theory

When the dependency architecture changes, earlier versions should remain archived.

A rejected hypothesis should not disappear from the historical record.

Instead:

$$\begin{gathered}\text{original claim}\\ \longrightarrow \text{experimental challenge}\\ \longrightarrow \text{revised claim}\end{gathered} \tag{823}$$

should remain traceable.

This is especially important for a programme whose explicit purpose is to remove unsupported assumptions.

The history of subtraction is part of the scientific result.

17.38 No consciousness-labelled dataset

Paper III should not publish Block V or conflict-test data under labels that imply phenomenal consciousness has been measured.

Terms such as:

$$\texttt{conscious}, \quad \texttt{sentient}, \quad \texttt{phenomenal_state} \tag{824}$$

should not be used as ground-truth labels unless a future independent scientific basis for such labels exists.

The datasets should retain the actual operational variables:

$$\begin{gathered} \text{confidence}, \\ \text{self-model estimate}, \\ \text{abstention}, \\ \text{checking behaviour}, \\ \text{conflict representation}. \end{gathered} \tag{825}$$

This prevents terminology from smuggling an unresolved theoretical conclusion into the data itself.

17.39 Minimum release for each principal result

Every principal Paper III result should therefore be accompanied by at least:

1. preregistration or frozen confirmatory specification;
2. experiment manifest;
3. exact code version;
4. environment and architecture configuration;
5. seed information;
6. raw or sufficiently complete primary data;
7. intervention and bypass records;
8. analysis code;
9. figure and table generation scripts;
10. final machine-readable decision record.

A result lacking one of these elements should state why it is unavailable.

17.40 Open-science audit table

The release can be checked against the following audit structure:

Table 21: Open-science audit requirements for Paper III.

Artefact	**Purpose**	**Failure prevented**
Preregistration	Preserve hypotheses and decisions before decisive results	Post-hoc hypothesis revision
Experiment manifest	Specify exactly what was run	Configuration ambiguity
Versioned code	Preserve implementation	Software drift
Raw trajectories	Preserve primary observations	Irreversible summary bias
Lineage records	Reconstruct evolutionary history	Aggregate-only selection claims
Model checkpoints	Preserve tested trained systems	Irreproducible internal-state interventions
Intervention registry	Record ablation and rescue definitions	Selective mechanistic reporting
Bypass registry	Preserve serious alternatives	Preferred-architecture bias
Negative-result archive	Preserve failed hypotheses and replications	Publication and confirmation bias
Decision record	Connect evidence to theoretical verdict	Retrospective reinterpretation
Checksums and persistent release	Preserve archive integrity	Silent file or version drift

17.41 Provisional judgement

Open science in Paper III should preserve more than successful outcomes.

It should preserve the complete history by which a proposed dependency was allowed to succeed or fail.

The required architecture is:

$$\boxed{\begin{array}{c}\text{preregister}\\ \downarrow\\ \text{version}\\ \downarrow\\ \text{execute}\\ \downarrow\\ \text{preserve raw evidence}\\ \downarrow\\ \text{analyse reproducibly}\\ \downarrow\\ \text{archive positive, negative and mixed results}\\ \downarrow\\ \text{record the theoretical consequence.}\end{array}} \tag{826}$$

The governing rule is:

$$\boxed{\begin{array}{c}\text{if the evidence is strong enough to change}\\ \text{the Theory of Everything architecture,}\\ \text{the full evidential path should be strong enough}\\ \text{for another researcher to inspect and reproduce.}\end{array}} \tag{827}$$

18 Possible Outcome Space

Paper III is not designed around one preferred empirical story. The experiments may support parts of the proposed dependency architecture, weaken others, identify alternative routes, collapse apparently distinct mechanisms into simpler explanations or fail to reveal any coherent hierarchy at all.

The possible outcome space is therefore broader than:

$$\text{hypothesis confirmed} \tag{828}$$

or:

$$\text{hypothesis rejected.} \tag{829}$$

The central aim is to discover which dependency structure is best supported after matched controls, interventions, rescues, bypass searches and robustness tests have been completed.

The governing principle is:

$$\boxed{\begin{array}{c}\text{the experiment does not ask whether}\\ \text{the proposed hierarchy can be made to appear;}\\ \text{it asks what dependency architecture remains}\\ \text{after the hierarchy has been given serious opportunities to fail.}\end{array}} \tag{830}$$

18.1 Outcome 1: substantial support for the proposed architecture

One possible result is that many of the candidate dependencies survive.

For example:

$$\begin{array}{c}\text{internal maintenance}\\ \downarrow\\ \text{heritable reproducing population}\\ \downarrow\\ \text{retention, feedback and learning}\\ \downarrow\\ \text{structurally generalising agency}\\ \downarrow\\ \text{reflective self-modelled intelligence.}\end{array} \tag{831}$$

Such a result would be important because the transitions would have survived serious null architectures, selective interventions and bypass searches.

Even then, the conclusion would remain conditional.

Paper III would support:

$$\text{one experimentally validated artificial dependency architecture,} \tag{832}$$

not:

$$\text{the unique universal architecture of life and intelligence.} \tag{833}$$

18.2 Outcome 2: a partial hierarchy

A more likely outcome may be that some transitions survive while others do not.

For example:

$$\begin{gathered}\text{self-maintenance: Retain,}\\ \text{heredity: Retain,}\\ \text{selection: Retain,}\\ \text{retention: Retain,}\\ \text{feedback: Retain,}\\ \text{learning: Restrict,}\\ \text{world-model necessity: Reject,}\\ \text{reflective self-model use: Suspend.}\end{gathered} \tag{834}$$

This would not represent a failed Theory of Everything programme.

It would show that the original architecture contained both robust and unnecessary structure.

The revised dependency map should then preserve only the surviving relations.

18.3 Outcome 3: branching dependencies

The experiments may reveal that one lower-level capacity supports several higher-level routes without determining a single sequence.

For example:

$$\boxed{\begin{gathered}\text{retention} \longrightarrow \text{learning} \longrightarrow \text{adaptive agency},\\ \text{retention} \longrightarrow \text{model-free control} \longrightarrow \text{adaptive agency}.\end{gathered}} \tag{835}$$

This would indicate that retention can contribute to more than one route towards adaptive agency, rather than functioning as one step in a mandatory linear progression.

A branching result would therefore weaken any claim that the system must develop through a single ordered ladder. The appropriate response would be to replace the linear transition map with a branching dependency structure whose separate routes are tested independently.

18.4 Outcome 4: convergent routes to the same capacity

Different mechanisms may independently generate similar higher-level capabilities.

Suppose both:

$$M_1 \longrightarrow B \tag{836}$$

and:

$$M_2 \longrightarrow B \tag{837}$$

survive serious tests.

The correct conclusion would be:

$$\boxed{\begin{array}{c}\text{capacity } B \text{ admits multiple sufficient routes;}\\ \text{neither route is automatically necessary.}\end{array}} \tag{838}$$

This possibility is especially relevant to:

$$\begin{array}{c}\text{self-maintenance,}\\ \text{structural generalisation,}\\ \text{reflective control.}\end{array} \tag{839}$$

A Theory of Everything architecture that permits multiple realisation may be more accurate than one built around a unique mechanism.

18.5 Outcome 5: successful bypass of a proposed necessity

A particularly informative outcome would be a robust system satisfying:

$$\neg M \wedge B, \tag{840}$$

where M had been proposed as necessary for capacity B.

Examples include:

$$\begin{array}{c}\text{self-maintenance without reciprocal closure,}\\ \text{learning without persistent recurrent memory during execution,}\\ \text{structural generalisation without explicit world-model rollout,}\\ \text{reflective control without an endogenous confidence model.}\end{array} \tag{841}$$

Such a bypass would be a major result.

It would not show that the bypassed mechanism is useless.

It would show:

$$\boxed{M \text{ is not universally necessary for } B} \tag{842}$$

within the scope reached by the experiment.

18.6 Outcome 6: reduction to a simpler mechanism

Some higher-level explanatory constructs may turn out to add no independent causal value.

For example:

$$\begin{gathered}\text{reciprocal closure} \rightarrow \text{ordinary causal dependency},\\ \text{explicit conflict representation} \rightarrow \text{multi-objective control},\\ \text{self-modelling} \rightarrow \text{task-specific uncertainty estimation}.\end{gathered} \tag{843}$$

Where a simpler mechanism reproduces the decisive effects, the appropriate verdict is:

$$\text{Reduce}. \tag{844}$$

Reduction is not a lesser scientific result.

It identifies where the research architecture contains explanatory machinery that the evidence does not require.

18.7 Outcome 7: architecture-dependent mechanisms

A mechanism may work reliably but only within a narrow implementation class.

For example, world-model rollout may provide a clear advantage in one planning architecture but no benefit in another.

Likewise, reciprocal maintenance may matter only when:

$$\begin{gathered}\text{component failure is local},\\ \text{repair resources are constrained},\\ \text{disturbance timescales fall within a specific range}.\end{gathered} \tag{845}$$

The correct response is:

$$\text{Restrict}, \tag{846}$$

not to discard the mechanism or universalise it.

Such results help identify the actual domain of validity of each proposed dependency.

18.8 Outcome 8: environment-dependent mechanisms

Some effects may be driven more strongly by environment structure than by the intrinsic hierarchy of the agent.

For example:

$$\begin{gathered}\text{retention matters only when present observation is insufficient},\\ \text{feedback matters only under unpredictable disturbance},\\ \text{learning matters only when mappings change},\\ \text{planning matters only when consequences are delayed or branching}.\end{gathered} \tag{847}$$

This would support an ecological interpretation:

$$\boxed{\text{mechanism value depends upon problem structure.}} \tag{848}$$

The hierarchy would then be better understood as a set of conditional capabilities than as an intrinsic universal ranking.

18.9 Outcome 9: measurement-sensitive results

Paper II predicts that some candidate measures may depend strongly upon:

$$\begin{gathered}\text{baseline},\\ \text{representation},\\ \text{timescale},\\ \text{system boundary}.\end{gathered} \tag{849}$$

Paper III may confirm that sensitivity experimentally.

A result that changes sign, ranking or interpretation under nearby defensible measurement choices should not be described as robust.

The appropriate verdict may be:

$$\text{Restrict} \tag{850}$$

or:

$$\text{Suspend}. \tag{851}$$

Such an outcome would feed directly back into Paper II.

18.10 Outcome 10: descriptor without mechanism

A quantity may predict behaviour without identifying its cause.

For example:

$$R(\tau) > 0 \tag{852}$$

may predict future performance while:

$$R_{\text{causal}}(\Delta) \approx 0. \tag{853}$$

Likewise, confidence may correlate strongly with accuracy while manipulation of the confidence representation produces no control change.

The correct distinction is:

$$\boxed{\text{predictive descriptor} \neq \text{causal mechanism.}} \tag{854}$$

Paper III should preserve useful descriptors even when their mechanistic interpretation fails.

18.11 Outcome 11: mechanism without broad generalisation

The reverse outcome is also possible.

A mechanism may have a clear causal effect while remaining narrow.

For example:

$$\text{world-model corruption} \longrightarrow \text{specific planning deficit} \tag{855}$$

may be reproducible even though the system fails on novel problem families.

This would support:

$$\text{causal model use} \tag{856}$$

while rejecting:

$$\text{broad structural generalisation.} \tag{857}$$

Mechanism validation and capability breadth should therefore remain separate.

18.12 Outcome 12: no clean distinction between candidate mechanisms

Some mechanisms may prove experimentally inseparable.

For example, retention and learning may be so deeply coupled in a particular architecture that selective intervention cannot remove one without damaging the other.

The correct outcome would be:

$$\text{Suspend,} \tag{858}$$

unless a new experimental design resolves the confounding.

Paper III should not force a mechanistic distinction simply because the conceptual framework contains one.

18.13 Outcome 13: collapse of the proposed hierarchy

A more radical result would occur if:

$$\begin{gathered}\text{later capacities can be implemented independently,}\\ \text{bypass architectures repeatedly succeed,}\\ \text{earlier mechanisms provide no necessary contribution.}\end{gathered} \tag{859}$$

The original hierarchy could then collapse into:

$$\boxed{\text{a collection of largely independent functional mechanisms.}} \tag{860}$$

This would be a major theoretical result.

It would show that the appearance of a hierarchy came from conceptual organisation rather than underlying causal dependence.

18.14 Outcome 14: emergence of a different hierarchy

The experiments may reveal a dependency architecture not anticipated by the original programme.

For example:

$$\begin{gathered}\text{retention} \longrightarrow \text{learning},\\ \text{learning} \longrightarrow \text{self-monitoring},\\ \text{self-monitoring} \longrightarrow \text{structural generalisation}\end{gathered} \tag{861}$$

might receive stronger support than the expected ordering.

Such a result should not be forced back into the original transition map.

Instead, Paper I should be revised around the new dependency structure.

18.15 Outcome 15: no coherent artificial hierarchy

The most adversarial possibility is that no stable dependency hierarchy survives.

Results might show:

$$\begin{gathered}\text{strong architecture dependence,}\\ \text{multiple bypasses,}\\ \text{weak cross-block relations,}\\ \text{large representation sensitivity,}\\ \text{no reproducible monotonic progression.}\end{gathered} \tag{862}$$

The conclusion would then be:

$$\boxed{\begin{array}{c}\text{the tested capacities do not form one}\\ \text{experimentally supported artificial hierarchy.}\end{array}} \tag{863}$$

That would be a highly informative result.

It would prevent the wider Theory of Everything programme from importing a hierarchy that the evidence does not support.

18.16 Outcome 16: unexpectedly strong cross-block relations

The opposite result is also possible.

Distinct block-level quantities may exhibit reproducible relations that were not assumed in advance.

For example:

$$Q_B = \mathcal{T}_{A\to B}(Q_A) + \varepsilon \tag{864}$$

may hold across architectures and task families.

Such a relation would be scientifically interesting, particularly if:

$$\begin{array}{c}\text{it survives held-out prediction,}\\ \text{it is robust to representation,}\\ \text{it has intervention-sensitive interpretation.}\end{array} \tag{865}$$

The result should first be treated as a new candidate relation.

Exploratory discovery does not become confirmation until independently tested.

18.17 Outcome 17: candidate cross-scale invariant

A particularly important long-term outcome would be evidence that one relation identified in artificial systems transports to a natural domain.

Suppose:

$$Q_B = \mathcal{T}_{A\to B}(Q_A). \tag{866}$$

is validated artificially and later predicts a biological relation under a preregistered mapping.

This could provide the first serious evidence for a deeper cross-scale regularity.

Such a result would be relevant to the Stage 1 research programme identified in Papers I and II.

It would still require replication before being described as universal.

18.18 Outcome 18: failure of the cross-scale programme

Paper III may also contribute to the opposite conclusion.

If candidate relations repeatedly fail to transport across:

$$\begin{gathered}\text{architectures},\\ \text{representations},\\ \text{task families},\\ \text{natural domains},\end{gathered} \tag{867}$$

then the search for a universal cross-scale constraint architecture may need to be reduced or abandoned.

This would be a legitimate outcome of the wider programme.

A Theory of Everything should not retain cross-scale unity merely because unity was the original objective.

18.19 Outcome 19: conflict representation earns a limited role

The auxiliary conflict experiment may show that explicit representation of objective incompatibility provides a robust functional advantage.

If it survives:

$$\begin{gathered}\text{scalarised controls},\\ \text{multi-objective controls},\\ \text{model-based controls},\\ \text{novel trade-off tests},\end{gathered} \tag{868}$$

conflict representation may deserve a limited place in the architecture of adaptive control.

The conclusion would remain:

$$\text{functional mechanism retained}, \tag{869}$$

not:

$$\text{consciousness mechanism established}. \tag{870}$$

18.20 Outcome 20: conflict representation is reduced

Alternatively, the conflict experiment may show no independent explanatory value.

If ordinary multi-objective or model-based control reproduces all decisive effects, the appropriate conclusion is:

$$\boxed{\text{explicit conflict representation: Reduce.}} \tag{871}$$

That result would support removing conflict negotiation from the active core of the programme while retaining the historical record of why it was tested.

18.21 No possible outcome establishes phenomenal consciousness

None of the outcome classes above changes the status of Open Problem 1.

Even the strongest result involving:

$$\begin{gathered}\text{world modelling,}\\ \text{counterfactual planning,}\\ \text{self-monitoring,}\\ \text{reflective control,}\\ \text{conflict representation}\end{gathered} \tag{872}$$

would remain a functional or causal result.

Therefore:

$$\boxed{\begin{gathered}\text{no possible Paper III outcome}\\ \text{is automatically evidence of phenomenal consciousness.}\end{gathered}} \tag{873}$$

Open Problem 1 remains open unless an independent bridge is discovered.

18.22 Possible effects on Paper I

Paper III can change the Minimal Principles architecture in several ways.

A result may cause a Paper I claim to be:

$$\begin{gathered}\text{retained,}\\ \text{restricted,}\\ \text{reduced,}\\ \text{suspended,}\\ \text{rejected.}\end{gathered} \tag{874}$$

It may also reveal:

$$\begin{gathered}\text{a missing transition,}\\ \text{a bypass route,}\\ \text{a branching dependency,}\\ \text{an unnecessary mechanism.}\end{gathered} \tag{875}$$

The dependency map should be revised to reflect those outcomes rather than preserved for continuity.

18.23 Possible effects on Paper II

Results may also change the Mathematics of Constraint programme.

Paper III could reveal that a candidate quantity is:

$$\begin{gathered}\text{causally informative},\\ \text{merely descriptive},\\ \text{redundant},\\ \text{representation-dependent},\\ \text{unstable across timescales.}\end{gathered} \tag{876}$$

Paper II should then revise the corresponding typed profile or candidate criterion.

The mathematics remains provisional until the experimental interpretation survives.

18.24 The final dependency map need not resemble the starting map

The starting architecture is a research instrument.

The final architecture is an empirical result.

Those two maps may differ substantially.

The final representation should therefore be reconstructed from the decision record rather than generated by simply colouring the original arrows as “passed” or “failed”.

Possible final structures include:

$$\begin{gathered}\text{linear chain},\\ \text{branching tree},\\ \text{convergent network},\\ \text{modular architecture},\\ \text{largely disconnected mechanisms.}\end{gathered} \tag{877}$$

The topology itself becomes an outcome.

18.25 Outcome-space summary table

The principal classes of result can be summarised as follows:

Table 22: Principal possible outcomes of Paper III.

Outcome class	**Experimental meaning**	**Architectural consequence**
Substantial hierarchy support	Most proposed dependencies survive adversarial tests	Retain a conditional artificial dependency architecture
Partial hierarchy	Some dependencies survive while others fail	Remove or revise failed arrows
Branching structure	Multiple higher-level routes arise from earlier capacities	Replace linear hierarchy with branching map
Convergent routes	Different mechanisms realise the same capacity	Reject unique-route necessity
Successful bypass	Target capacity exists without proposed prerequisite	Reject or restrict necessity claim
Reduction	Simpler mechanism explains decisive effect	Replace higher-level construct where appropriate
Conditional mechanism	Effect depends upon architecture, environment or timescale	Restrict scope explicitly
Descriptor without mechanism	Quantity predicts but intervention does not support causality	Retain descriptively; reject mechanistic interpretation
New hierarchy	Unexpected dependencies survive more strongly than proposed ones	Revise Paper I around discovered structure
No coherent hierarchy	Dependencies fail to organise into stable progression	Abandon hierarchical interpretation for tested capacities
Cross-scale relation	Artificial relation transports predictively to another domain	Promote to independent cross-domain testing
Cross-scale failure	Candidate relations do not transport	Reduce or abandon universal constraint claim

18.26 Success criteria for Paper III

Paper III should not define success as producing:

$$\text{a complete five-block hierarchy.} \tag{878}$$

The experiment succeeds if it produces justified decisions about the proposed dependencies.

Success therefore includes:

$$\begin{gathered}\text{confirming a mechanism,}\\ \text{restricting its scope,}\\ \text{discovering a bypass,}\\ \text{reducing a construct,}\\ \text{suspending an underdetermined claim,}\\ \text{rejecting an unsupported transition.}\end{gathered} \tag{879}$$

A clean falsification is as valuable as a clean confirmation.

18.27 Open-science interpretation of the outcome space

The full outcome space should be visible in the public record.

The archive should therefore preserve:

$$\begin{gathered}\text{successful hypotheses,}\\ \text{failed hypotheses,}\\ \text{successful bypasses,}\\ \text{failed bypasses,}\\ \text{successful rescues,}\\ \text{failed rescues,}\\ \text{inconclusive tests,}\\ \text{revised dependency maps.}\end{gathered} \tag{880}$$

Future researchers should be able to reconstruct how the architecture changed as evidence accumulated.

That historical record may itself become important if later technologies make possible tests that are unavailable when the first implementation is built.

18.28 No endpoint is assumed

Paper III should not assume that the final experiment produces a permanently settled architecture.

New systems may later:

$$\begin{gathered}\text{defeat an apparent necessity,}\\ \text{reveal a hidden dependency,}\\ \text{invalidate a measurement,}\\ \text{support a previously suspended mechanism.}\end{gathered} \tag{881}$$

The outcome space therefore remains revisable.

A retained claim is a claim that survived the current evidence, not one that has become immune to future challenge.

18.29 Provisional judgement

The most important possible outcome of Paper III is not confirmation of the starting hierarchy.

It is a more accurate dependency architecture.

That architecture may contain fewer mechanisms, more branches, alternative routes or deeper uncertainties than the original proposal.

The governing rule is:

$$\boxed{\begin{gathered}\text{the correct outcome is whatever structure}\\ \text{survives controlled attempts to break it;}\\ \text{if the hierarchy survives, retain it;}\\ \text{if it branches, redraw it;}\\ \text{if it reduces, simplify it;}\\ \text{if it fails, remove it.}\end{gathered}} \tag{882}$$

19 Conclusion

Paper III converts the proposed transition architecture from a conceptual sequence into an adversarial experimental programme. Its purpose is not to demonstrate that organisation, evolution, intelligence and reflective self-modelling form one predetermined hierarchy. It is to determine which candidate mechanisms and dependencies survive controlled attempts to separate them from simpler explanations, alternative architectures and successful bypasses.

The paper therefore begins from a deliberately weaker proposition:

$$\boxed{\begin{array}{c}\text{a proposed dependency is a hypothesis}\\ \text{until intervention, comparison and counterexample}\\ \text{show how much explanatory weight it can bear.}\end{array}} \tag{883}$$

This change is fundamental to the wider Theory of Everything programme. The architecture is no longer something the experiment is expected to reproduce. The architecture itself becomes the object of experiment.

19.1 From hierarchy construction to dependency discovery

The original artificial-hierarchy design risked combining several distinct questions into one developmental sequence. Persistent organisation could be mistaken for self-maintenance, reproduction for selection, memory for learning, prediction for world-model use, and self-report for reflective self-modelling.

The revised experiment separates these claims.

Paper III now contains five principal experimental blocks:

$$\begin{array}{c}\text{Block I: Dynamical Persistence and Maintenance,}\\ \text{Block II: Heredity, Reproduction and Population Change,}\\ \text{Block III: Retention, Feedback and Learning,}\\ \text{Block IV: Adaptive Agency and Structural Generalisation,}\\ \text{Block V: Reflective Self-Modelling.}\end{array} \tag{884}$$

An additional conflict-representation experiment remains outside the principal hierarchy because no current evidence warrants treating conflict negotiation as a necessary transition towards intelligence or consciousness.

The resulting structure is therefore:

$$\boxed{\text{a set of dependency tests rather than a ladder of assumed emergence.}} \tag{885}$$

19.2 Block I separates persistence from self-maintenance

The first block begins with a physical counterexample to overinterpretation. Driven reaction–diffusion systems can generate persistent organised patterns without establishing internal self-maintenance.

The experiment therefore distinguishes:

$$\begin{gathered}\text{passive persistence},\\ \text{external maintenance},\\ \text{internal feed-forward maintenance},\\ \text{reciprocal internal maintenance}.\end{gathered} \tag{886}$$

Ablation and rescue are used to ask whether nominated maintenance relations are causal rather than merely architectural.

The important transition is consequently not:

$$\text{persistence} = \text{self-maintenance}, \tag{887}$$

but:

$$\boxed{\text{persistence} \longrightarrow \text{a separate causal question about maintenance.}} \tag{888}$$

19.3 Block II establishes the population before invoking selection

The second block begins with a question that cannot be skipped:

$$\boxed{\text{What reproduces?}} \tag{889}$$

Individual identity, descendant formation, parent–offspring transmission and heritable variation must be established before population change is described as natural-selection-like.

Selection is then tested against:

$$\begin{gathered}\text{random reproduction},\\ \text{mutation without differential reproduction},\\ \text{random search},\\ \text{within-lifetime optimisation}.\end{gathered} \tag{890}$$

This preserves the distinction between the origin of a heritable population and the operation of selection once such a population exists.

19.4 Block III separates three mechanisms often conflated as adaptation

Retention, feedback and learning are treated factorially:

$$R \times F \times L. \tag{891}$$

Retention is tested by manipulating whether relevant history is available and causally used.

Feedback is tested by comparing closed-loop control with open-loop and yoked alternatives under predictable and unpredictable disturbance.

Learning is tested by changing the behavioural mapping and determining whether experience produces persistent modification beyond temporary corrective action.

The result may therefore show:

$$\begin{gathered}\text{retention is useful without being universally necessary,}\\ \text{feedback matters only under particular disturbance regimes,}\\ \text{learning changes behaviour without defining intelligence.}\end{gathered} \tag{892}$$

The experiment does not require these mechanisms to form a single progression.

19.5 Block IV places model-based intelligence at genuine risk

The fourth block asks whether structurally generalising agency requires explicit model-based control.

The decisive tasks move beyond ordinary held-out examples to:

$$\begin{gathered}\text{causal reversal,}\\ \text{novel composition,}\\ \text{delayed consequences,}\\ \text{hidden-state inference,}\\ \text{new problem families.}\end{gathered} \tag{893}$$

A predictive representation earns mechanistic credit only if intervention on its content alters planning in the predicted direction and correction restores the corresponding function.

At the same time, serious model-free and heuristic alternatives are allowed to succeed.

Paper III can therefore support:

$$\text{world-model use as one route to intelligent agency} \tag{894}$$

while rejecting:

$$\text{world-model use as a universal necessity.} \tag{895}$$

19.6 Block V separates self-modelling from self-description

The final principal block investigates representations concerning the system's own competence, uncertainty, internal reliability or limitations.

The central criterion is causal use.

A system that reports:

$$\text{“I am uncertain”} \tag{896}$$

has not thereby demonstrated reflective self-modelling.

Stronger evidence requires:

$$\boxed{\begin{array}{c}\text{self-related estimate}\\ \longrightarrow \text{predicted control change}\\ \longrightarrow \text{targeted misinformation effect}\\ \longrightarrow \text{correction and behavioural rescue.}\end{array}} \tag{897}$$

This gives reflective self-modelling an operational interpretation while leaving the question of phenomenal experience untouched.

19.7 Conflict representation remains auxiliary

The conflict-representation experiment tests whether explicit representation of incompatible objectives provides functional value beyond:

$$\begin{array}{c}\text{scalar reward optimisation,}\\ \text{ordinary multi-objective control,}\\ \text{model-based prediction.}\end{array} \tag{898}$$

A positive result would justify retaining conflict representation as a useful control mechanism under specified conditions.

A negative result would justify reducing it to simpler established mechanisms.

Neither result establishes Paradox–Negotiation Theory as a theory of phenomenal consciousness.

19.8 Measurement unity is not manufactured

Paper III also inherits an important result from Paper II: heterogeneous measurements should not be forced into one universal constraint vector or organisation score.

The experimental programme therefore uses a typed profile:

$$\mathfrak{C} = [\mathfrak{D}, \mathfrak{F}, \mathfrak{T}, \mathfrak{O}], \tag{899}$$

only where the relevant categories match the explanandum.

The same formal quantity can play different roles in different experiments, and different blocks

may legitimately require entirely different measurements.

Accordingly:

$$\boxed{\text{common experimental discipline} \neq \text{common scalar of organisation.}} \tag{900}$$

19.9 The hierarchy is protected against being engineered into existence

The revised controls address one of the deepest risks in artificial experimentation: later systems can appear "higher" simply because the researcher gives them more parameters, more compute, more information, more training or more directly encoded structure.

Paper III therefore requires:

$$\begin{gathered}\text{capacity matching,}\\ \text{compute matching,}\\ \text{information-access controls,}\\ \text{reward-leakage tests,}\\ \text{matched-damage ablations,}\\ \text{specific rescue controls,}\\ \text{serious bypass architectures,}\\ \text{held-out structural tests.}\end{gathered} \tag{901}$$

Later experimental blocks are not required to be strict supersets of earlier architectures.

This is essential because:

$$\boxed{\text{a hierarchy built into the experiment cannot later be discovered by it.}} \tag{902}$$

19.10 The final result is a set of judgements, not one verdict

Each principal claim receives one of:

$$\boxed{\text{Retain,} \quad \text{Restrict,} \quad \text{Reduce,} \quad \text{Suspend,} \quad \text{Reject.}} \tag{903}$$

Paper III therefore has no omnibus test asking whether:

$$\text{the hierarchy is true.} \tag{904}$$

A scientifically plausible final result may contain several different judgements:

$$\begin{gathered}\text{internal maintenance: Retain,}\\ \text{reciprocal closure necessity: Reject,}\\ \text{selection: Retain,}\\ \text{learning: Restrict,}\\ \text{explicit rollout necessity: Reduce,}\\ \text{self-model use: Suspend.}\end{gathered} \tag{905}$$

Such a result would be more informative than a forced binary conclusion.

19.11 Failure of one dependency does not terminate the programme

The former phase-gate architecture has been replaced by dependency logic.

If one arrow fails:

$$A \not\rightarrow B, \tag{906}$$

later questions may still remain experimentally meaningful.

Another route may exist:

$$X \longrightarrow B. \tag{907}$$

Likewise, capacity C may be constructible independently of the failed transition.

The experimental programme can therefore discover:

$$\begin{gathered}\text{branches,}\\ \text{convergent routes,}\\ \text{bypasses,}\\ \text{modular mechanisms,}\\ \text{partial independence.}\end{gathered} \tag{908}$$

The topology of the final dependency map is itself an empirical result.

19.12 Negative results are structurally informative

A rejected dependency removes unnecessary structure.

A successful bypass weakens a necessity claim.

A failed rescue challenges a proposed mechanism.

A representation-sensitive result limits universality.

An inconclusive intervention identifies an unresolved measurement problem.

None of these is merely an unsuccessful experiment.

Within a programme seeking explanatory compression:

$$\boxed{\text{removing an unnecessary mechanism} = \text{progress towards a better theory.}} \tag{909}$$

This is especially important for a Theory of Everything project, where the temptation to preserve an attractive unifying concept can otherwise exceed the evidence supporting it.

19.13 Open science becomes part of the evidential architecture

Paper III requires the complete experimental history to remain inspectable.

For each decisive claim, the intended chain is:

$$\boxed{\begin{array}{c}\text{preregistered hypothesis}\\ \longrightarrow \text{machine-readable experiment manifest}\\ \longrightarrow \text{versioned implementation}\\ \longrightarrow \text{raw trajectory}\\ \longrightarrow \text{reproducible analysis}\\ \longrightarrow \text{decision record.}\end{array}} \tag{910}$$

Failed hypotheses, successful bypasses, null results, invalid runs, model checkpoints, lineage histories and superseded dependency maps should remain in the archive.

Future researchers should be able to reconstruct not merely what the programme concluded, but how and why its architecture changed.

19.14 The three-paper research architecture is iterative

The relationship among Papers I–III can now be stated clearly.

Paper I provides:

$$\text{candidate dependency architecture.} \tag{911}$$

Paper II provides:

$$\text{mathematical and causal measurement discipline.} \tag{912}$$

Paper III provides:

$$\text{a protocol for controlled adversarial artificial testing.} \tag{913}$$

The legitimate research loop is:

$$\boxed{\text{propose} \longrightarrow \text{formalise} \longrightarrow \text{test} \longrightarrow \text{revise} \longrightarrow \text{test again.}} \tag{914}$$

It is iterative rather than circular because revision must be driven by new evidence or demonstrated logical correction.

19.15 No result in this paper closes the consciousness problem

The strongest epistemic boundary remains Open Problem 1.

Paper III may eventually demonstrate:

$$\begin{gathered}\text{memory,}\\ \text{learning,}\\ \text{world-model use,}\\ \text{counterfactual planning,}\\ \text{self-monitoring,}\\ \text{reflective control,}\\ \text{conflict representation.}\end{gathered} \tag{915}$$

None currently provides a validated bridge to phenomenal consciousness.

Therefore:

$$\boxed{\begin{gathered}\text{Paper III stops at functional architecture;}\\ \text{phenomenal consciousness remains an open problem.}\end{gathered}} \tag{916}$$

This is not an omission to be repaired by adding another artificial stage. It is the conclusion required by the present evidence.

19.16 The programme permits its own deepest assumptions to fail

Several ideas that motivated earlier versions of the wider programme now occupy weaker or quarantined positions.

A universal constraint vector has not survived adversarial analysis.

Conflict negotiation has not been established as the origin of phenomenal consciousness.

World-model use has not been assumed to be necessary for intelligence.

Reciprocal organisational closure has not been granted universal necessity.

Information has not been promoted to a fundamental substance by the artificial experiments.

This is a feature of the programme rather than a defect.

A serious Theory of Everything cannot be built by accumulating every interesting hypothesis encountered along the way.

It must also contain a mechanism for subtraction.

19.17 What a successful Paper III would contribute

If implemented fully, Paper III could produce several kinds of result that would materially improve the wider research architecture.

It could identify:

$$\begin{gathered}\text{which mechanisms are causally effective,}\\ \text{which proposed necessities admit bypasses,}\\ \text{which quantities remain merely descriptive,}\\ \text{which transitions branch or converge,}\\ \text{which claims depend upon environment or representation,}\\ \text{which parts of the hierarchy should be removed.}\end{gathered} \tag{917}$$

The strongest contribution would therefore not be a demonstration that the starting architecture was correct.

It would be:

$$\boxed{\text{a dependency map that has survived genuine experimental risk.}} \tag{918}$$

19.18 From minimal implementation to long-term programme

The first implementation need not attempt the entire architecture.

A small Block I experiment combining:

$$\begin{gathered}\text{driven-pattern baseline,}\\ \text{internal maintenance network,}\\ \text{matched controls,}\\ \text{targeted ablation,}\\ \text{specific rescue}\end{gathered} \tag{919}$$

would already place Transition Problem 1 at empirical risk.

Later blocks can then be added as independent experimental programmes.

The preferred development strategy remains:

$$\boxed{\begin{array}{c}\text{small interpretable system}\\ \longrightarrow \text{causal discrimination}\\ \longrightarrow \text{replication}\\ \longrightarrow \text{scaling.}\end{array}} \tag{920}$$

This keeps theoretical clarity ahead of computational complexity.

19.19 The wider objective

The wider Theory of Everything project seeks connections across:

$$\begin{array}{c}\text{physics,}\\ \text{organisation,}\\ \text{life,}\\ \text{evolution,}\\ \text{intelligence,}\\ \text{consciousness.}\end{array} \tag{921}$$

Paper III does not claim to unify those domains.

Its role is more foundational.

Before cross-scale unity can be accepted, the proposed bridges themselves must survive precise definition and controlled challenge.

The appropriate path towards unification is therefore not:

$$\text{find one idea and apply it everywhere.} \tag{922}$$

It is:

$$\boxed{\begin{array}{c}\text{identify candidate relations}\\ \longrightarrow \text{define them precisely}\\ \longrightarrow \text{test them adversarially}\\ \longrightarrow \text{retain only what survives}\\ \longrightarrow \text{test whether the surviving relations transport across scales.}\end{array}} \tag{923}$$

19.20 Final conclusion

Paper III therefore proposes an artificial experimental programme not to manufacture a Theory of Everything, but to remove from the candidate dependency architecture whatever the controlled evidence does not require within the tested scope.

The final dependency map may be simpler than the starting one.

It may be more complicated.

It may branch.

Several supposedly distinct mechanisms may collapse into one.

One proposed transition may disappear entirely.

A new relation may emerge that was not anticipated at all.

Any of these outcomes is acceptable if it follows from reproducible evidence.

The final principle of the paper is therefore:

$$\boxed{\begin{array}{c}\text{do not ask the experiment to preserve the theory;}\\ \text{ask the theory to survive the experiment.}\end{array}} \tag{924}$$

If the resulting architecture is eventually worthy of contributing to a Theory of Everything, it should be because its components remained after every serious effort was made to show that they did not belong.

References

[1] J. Pearl, *Causality: Models, Reasoning, and Inference*, 2nd ed., Cambridge University Press, Cambridge, 2009.

[2] A. K. Seth and T. Bayne, "Theories of Consciousness," *Nature Reviews Neuroscience*, 23, 439–452, 2022. https://doi.org/10.1038/s41583-022-00587-4

[3] Cogitate Consortium, O. Ferrante, U. Gorska-Klimowska, S. Henin, et al., "Adversarial Testing of Global Neuronal Workspace and Integrated Information Theories of Consciousness," *Nature*, 642, 133–142, 2025. https://doi.org/10.1038/s41586-025-08888-1

[4] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor, "The Preregistration Revolution," *Proceedings of the National Academy of Sciences*, 115(11), 2600–2606, 2018. https://doi.org/10.1073/pnas.1708274114

[5] M. Montévil and M. Mossio, "Biological Organisation as Closure of Constraints," *Journal of Theoretical Biology*, 372, 179–191, 2015. https://doi.org/10.1016/j.jtbi.2015.02.029

[6] J. E. Pearson, "Complex Patterns in a Simple System," *Science*, 261(5118), 189–192, 1993. DOI: https://doi.org/10.1126/science.261.5118.189

[7] G. R. Price, "Selection and Covariance," *Nature*, 227, 520–521, 1970. https://doi.org/10.1038/227520a0

[8] W. R. Ashby, *An Introduction to Cybernetics*, Chapman & Hall, London, 1956.

[9] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap, "Mastering Diverse Control Tasks through World Models," *Nature*, 640, 647–653, 2025. https://doi.org/10.1038/s41586-025-08744-2

[10] G. W. Brier, "Verification of Forecasts Expressed in Terms of Probability," *Monthly Weather Review*, 78(1), 1–3, 1950. DOI: https://doi.org/10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2

[11] M. D. Wilkinson et al., "The FAIR Guiding Principles for Scientific Data Management and Stewardship," *Scientific Data*, 3, 160018, 2016. https://doi.org/10.1038/sdata.2016.18